

BASIC TECHNOLOGY YEAR 9 STUDENT WORKBOOK



Technical Employment Skills and Training
Ministry of Education
2019

PREFACE

This student workbook has been prepared for the Technical Drawing Section of the Year 9 Basic Technology Syllabus offered in Fiji secondary schools. It is the third edition of the student workbook and is designed to inculcate interest in students to take Technical Drawing at Year 9 level.

Suggestions for amendments is welcomed.

It is anticipated that both teachers and students will find its approach to technical drawing stimulating and rewarding and that it provides students with opportunities to pursue further in this field.

MINISTRY OF EDUCATION,
SUVA, FIJI.
04th October 2018.

ACKNOWLEDGEMENT

This Workbook for Year 9 Basic Technology has been produced by the Industrial Arts Section of the Technical Employment Skills and Training (TEST) Section of the Ministry of Education.

It has been written and compiled by a panel of writers comprising of the following Industrial Arts teachers / TEST Officers.

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The final formatting and editing has been done by Mr. Isoa Tauribau SEO Industrial Arts, TEST.

This publication has been made possible through the support and assistance provided by the A/Principal Education Officer, TEST; Mr. Pene Aropio and other Senior Staff of The Ministry of Education.

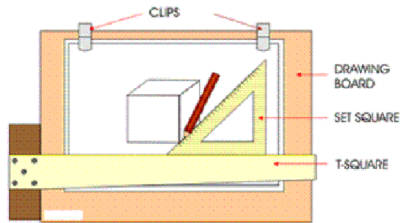
The TEST staff and family members of the writers are thanked above all for their patience and wholehearted support.

Every effort has been made to acknowledge all copyright.

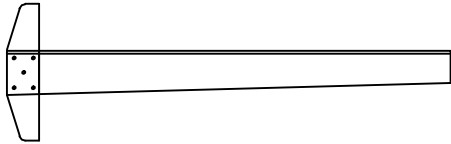
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TECHNICAL DRAWING IS A MEANS OF CLEARLY AND CONCISELY COMMUNICATING ALL THE INFORMATION NECESSARY TO TRANSFORM AN IDEA OR A CONCEPT INTO REALITY. THEREFORE TECHNICAL DRAWING OFTEN CONTAINS VARIOUS FORMS OF LINES, LETTERS AND FIGURES INSTEAD OF WORDS ONLY.

DRAWING INSTRUMENTS

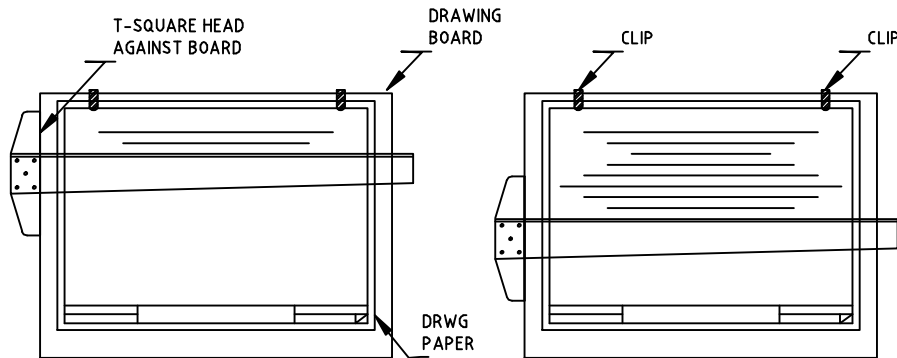


T-SQUARE

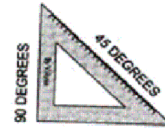


THE T-SQUARE IS USED FOR DRAWING HORIZONTAL LINES. THE T-SQUARE ACTS AS THE BASE FOR THE SET SQUARES WHILE DRAWING VERTICAL AND OBLIQUE LINES.

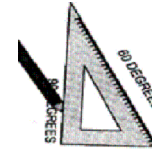
USING A T-SQUARE



SET SQUARES



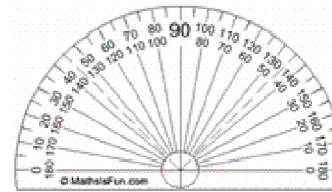
45° SET SQUARE



60° SET SQUARE

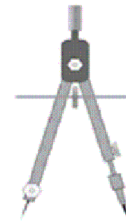
SET SQUARES ARE USED FOR DRAWING VERTICAL LINES AND LINES INCLINED AT 30°, 45° & 60° WHILE BEING HELD AGAINST THE EDGE OF THE T-SQUARE. IF THE TWO SET SQUARES ARE USED TOGETHER INCLINED LINES OF 15° AND 75° CAN ALSO BE OBTAINED.

PROTRACTOR



PROTRACTORS ARE USED FOR MEASURING AND OR MARKING ANGLES. THEY ARE GENERALLY USED IN GEOMETRICAL CONSTRUCTIONS, LAND SURVEYING AND COURSE PLOTTING.

COMPASS



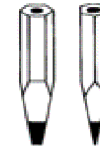
A COMPASS IS USED FOR DRAWING ARCS AND CIRCLES. ALWAYS HOLD THE COMPASS ON THE TOP WITH YOUR THUMB AND FORE FINGER. LEAN THE COMPASS SLIGHTLY FORWARD WHILE DRAWING ARCS AND CIRCLES.

PENCIL



Conical shape

A PENCIL IS THE MOST IMPORTANT TOOL FOR TECHNICAL DRAWING. GOOD QUALITY PENCILS ARE ESSENTIAL FOR NEAT AND ACCURATE WORK.



Chisel shape

DIVIDER

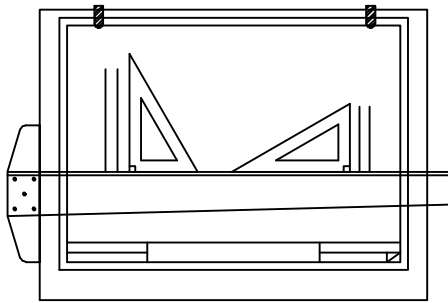


DIVIDERS ARE USED FOR STEPPING OFF OR MARKING EQUAL DISTANCES AND ALSO FOR TRANSFERRING MEASUREMENTS ACCURATELY.

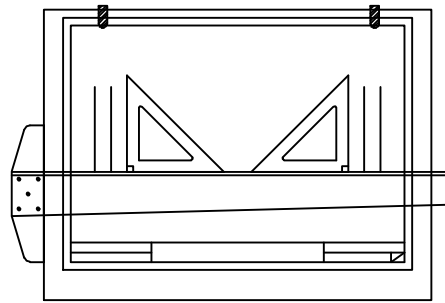
NAME: _____
FORM: _____

BASIC DRAWING INSTRUMENTS

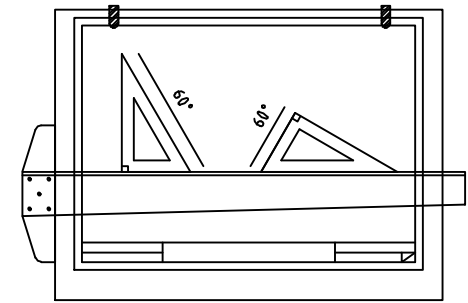
DATE: _____
SHEET NO: 1



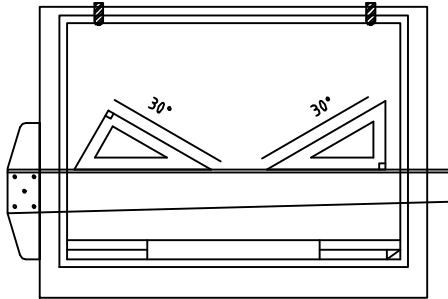
DRAWING VERTICLE LINES USING
THE 30-60° SET SQUARE



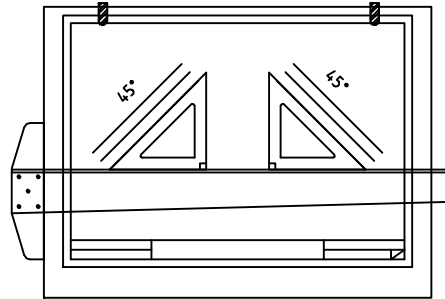
DRAWING VERTICLE LINES
USING THE 45° SET SQUARE



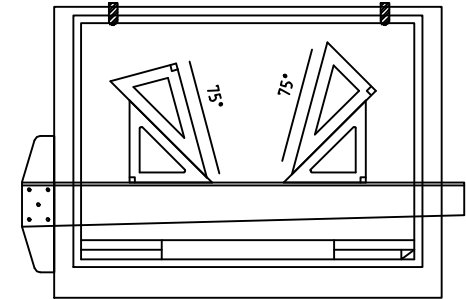
DRAWING 60° LINES USING
THE 30-60° SET SQUARE



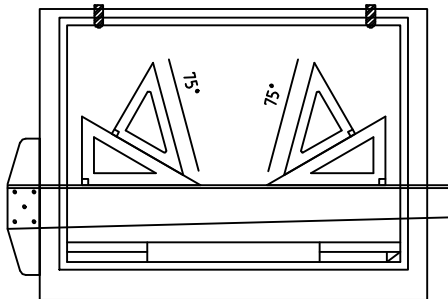
DRAWING 30° LINES USING
THE 30-60° SET SQUARE



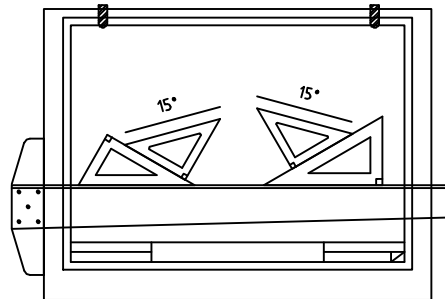
DRAWING 45° LINES USING
THE 45° SET SQUARE



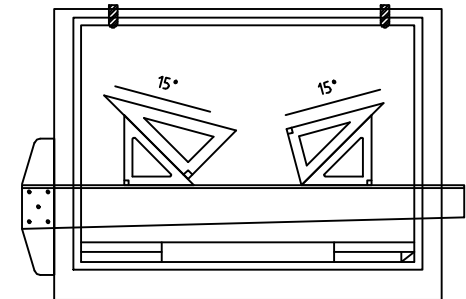
DRAWING 75° LINES USING
THE TWO SET SQUARES



DRAWING 75° LINES USING
THE TWO SET SQUARES



DRAWING 15° LINES USING
THE TWO SET SQUARES



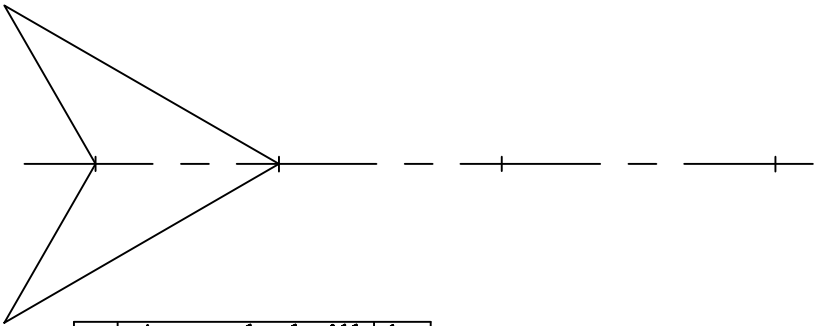
DRAWING 15° LINES USING
THE TWO SET SQUARES

NAME: _____
FORM: _____

USING THE SET SQUARES

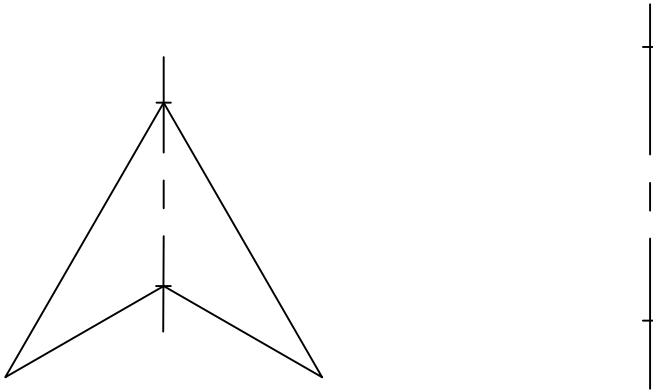
DATE: _____
SHEET NO: 2

GIVEN: THE SHAPE OF AN ARROW HEAD, CENTRE LINE & STARTING POINTS
REQUIRED: REDRAW THE ARROW HEAD BY PROJECTING PARALLEL LINES FROM THE GIVEN STARTING POINTS.



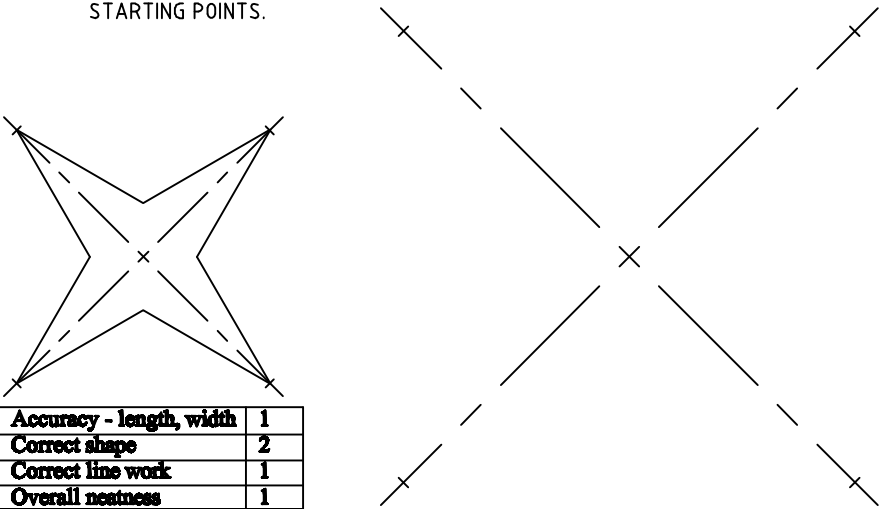
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE SHAPE OF AN ARROW HEAD, CENTRE LINE & STARTING POINTS
REQUIRED: REDRAW THE ARROW HEAD BY PROJECTING PARALLEL LINES FROM THE GIVEN STARTING POINTS.



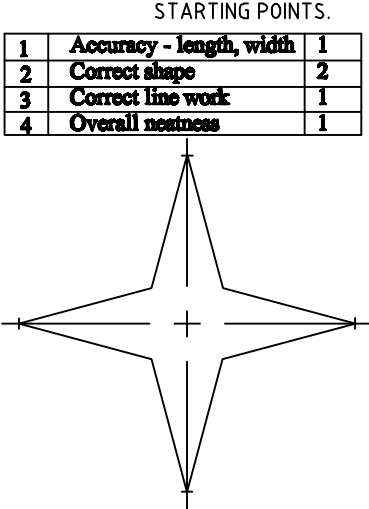
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE SHAPE OF A STAR, CENTRE LINES & STARTING POINTS
REQUIRED: REDRAW THE STAR SHAPE BY PROJECTING PARALLEL LINES FROM THE GIVEN STARTING POINTS.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE SHAPE OF A STAR, CENTRE LINES & STARTING POINTS
REQUIRED: REDRAW THE STAR SHAPE BY PROJECTING PARALLEL LINES FROM THE GIVEN STARTING POINTS.

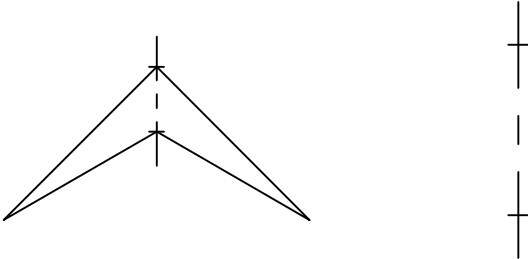


1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

NAME: _____
FORM: _____

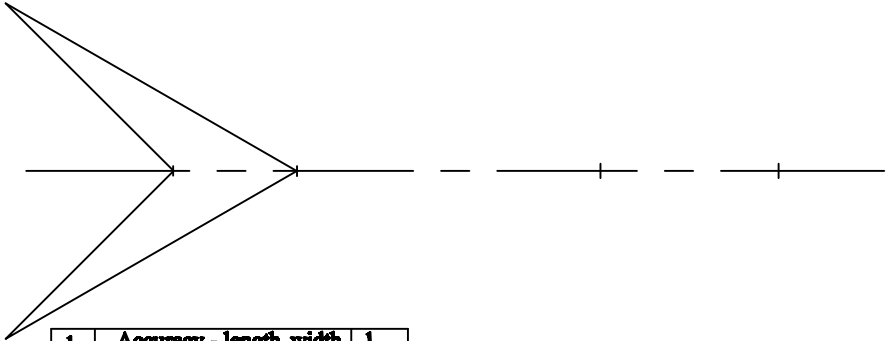
DATE: _____
SHEET NO: 3

GIVEN: THE SHAPE OF AN ARROW HEAD, CENTRE LINE & STARTING POINTS
REQUIRED: REDRAW THE ARROW HEAD BY PROJECTING PARALLEL LINES FROM THE GIVEN STARTING POINTS.



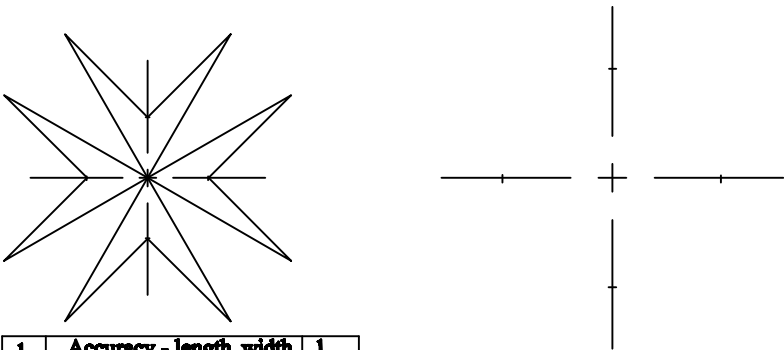
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE SHAPE OF AN ARROW HEAD, CENTRE LINE & STARTING POINTS
REQUIRED: REDRAW THE ARROW HEAD BY PROJECTING PARALLEL LINES FROM THE GIVEN STARTING POINTS.



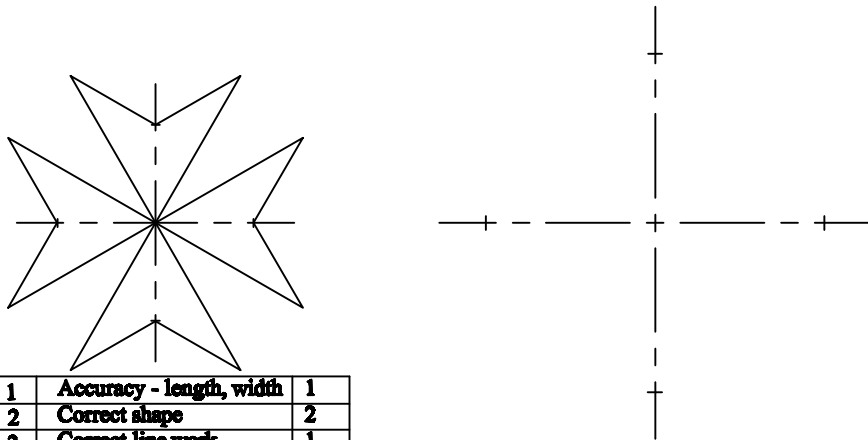
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE SHAPE OF A STAR, CENTRE LINES & STARTING POINTS
REQUIRED: REDRAW THE STAR SHAPE USING THE GIVEN STARTING POINTS.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE SHAPE OF A STAR, CENTRE LINES & STARTING POINTS
REQUIRED: REDRAW THE STAR SHAPE USING THE GIVEN STARTING POINTS.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

NAME: _____

FORM: _____

DATE: _____

SHEET NO: 4

LETTERING

RE-WRITE THE FOLLOWING IN VERTICLE AND SLOPING STYLE RESPECTIVELY

VERTICLE STYLE - UPPER CASE

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z & 1 2 3 4 5 6 7 8 9 0

VERTICLE STYLE - LOWER CASE

a b c d e f g h i j k l m n o p q r s t u v w x y z

SLOPING STYLE - UPPER CASE

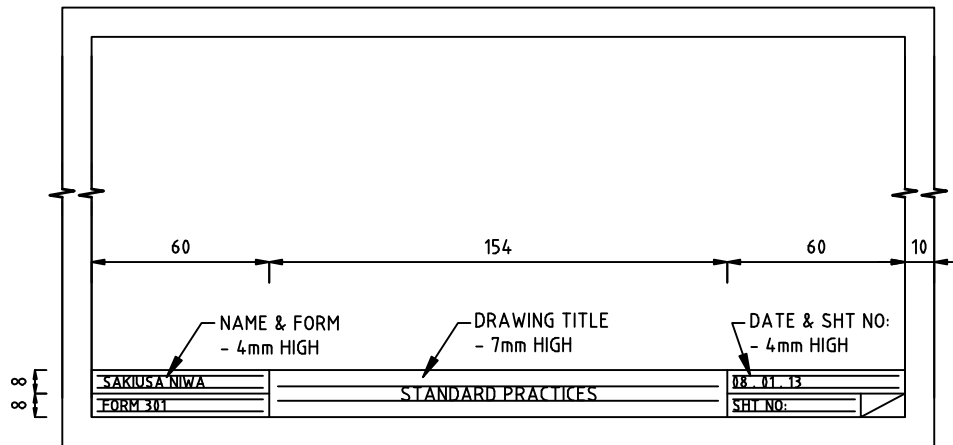
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z & 1 2 3 4 5 6 7 8 9 0

SLOPING STYLE - LOWER CASE

a b c d e f g h i j k l m n o p q r s t u v w x y z

PAPER LAYOUT & TITLE BLOCK

THE GIVEN PAPER LAYOUT AND TITLE BLOCK IS RECOMMENDED FOR USE. FOR THE NEXT 5 SHEETS DRAW THE BORDER LINES AND THE TITLE BLOCK AND COMPLETE THE TITLE BLOCK.



SYMBOLS AND CONVENTIONS

DRAW THE SYMBOLS AND CONVENTIONS AS IDENTIFIED IN THE SPACES PROVIDED

ONE WAY SWITCH

TWO WAY SWITCH

ILLUMINATING LAMP

SOCKET OUTLET

SOCKET OUTLET WITH SWITCH

WALL MOUNTED LAMP

FLUORESCENT LAMP

POINT OF ENTRY

DISTRIBUTION BOARD

SAWN TIMBER

DRESSED TIMBER

CONCRETE

EARTH

HARD CORE FILL

TIMBER BREAK

INSULATION

NAME: _____

FORM: _____

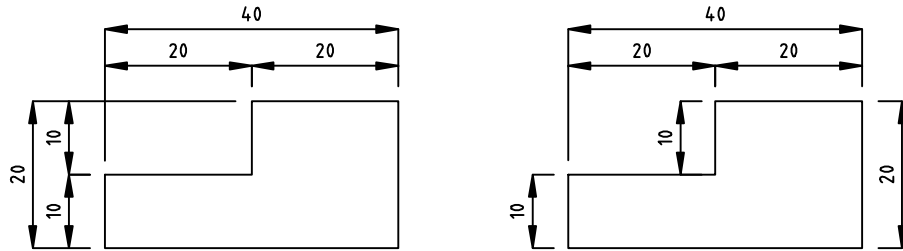
STANDARD PRACTICES

DATE: _____

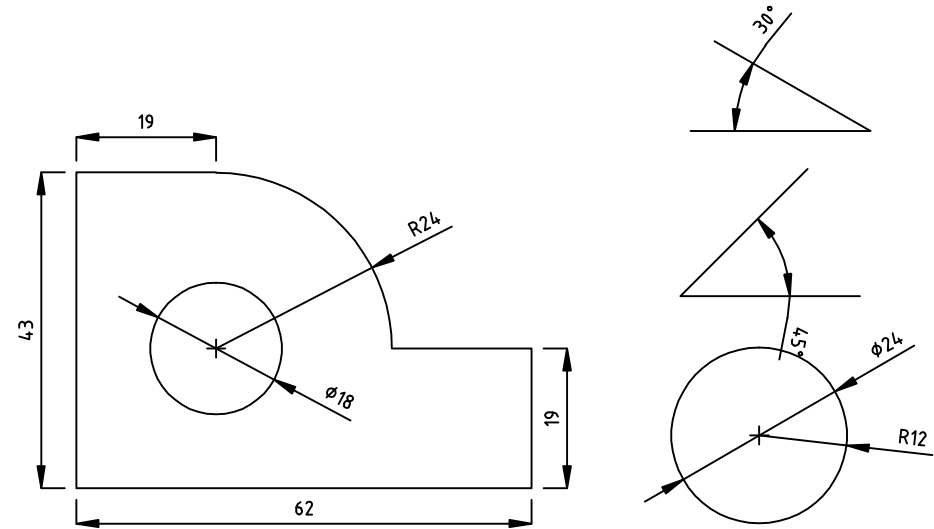
SHEET NO: 5

DIMENSION LINES ARE DRAWN USING CONTINUOUS THIN LINES AND SHOULD BE PLACED OUTSIDE THE OUTLINE OF THE OBJECT. TWO PROJECTION LINES INDICATE WHERE THE DIMENSION STARTS AND FINISHES. PROJECTION LINES DO NOT TOUCH THE OBJECT AND ARE DRAWN PERPENDICULAR TO THE SIDE BEING DIMENSIONED. ARROWHEADS ARE SMALL (3mm), SHARP, NEAT AND SHOULD TOUCH THE PROJECTION LINES.

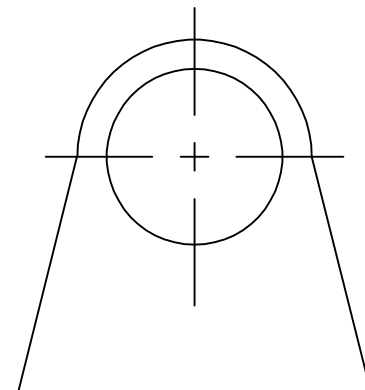
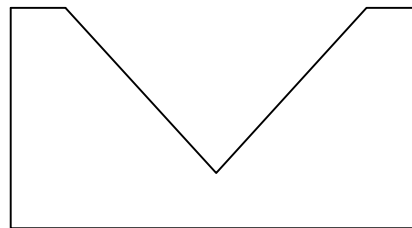
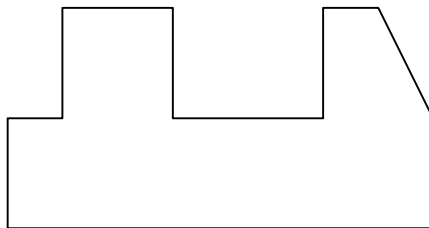
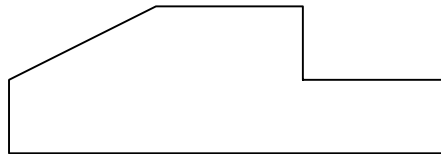
THE GENERAL PRACTICE IS TO HAVE ALL DIMENSIONS IN MILLIMETERS (Eg. 5mm). NUMBERS ARE TO BE PLACED ABOVE THE DIMENSION LINE, NOT TOUCHING THE LINES AND WHERE POSSIBLE PLACED CENTRALLY. DIMENSIONS LESS THAN 1mm SHOULD HAVE A LEADING ZERO. (Eg .35 SHOULD BE WRITTEN AS 0.35mm).



DIMENSIONING SIMPLE SHAPES AND AN ANGLES



DIMENSION THE SIMPLE SHAPES GIVEN BELOW



SKETCHING

"A PICTURE IS WORTH A THOUSAND WORDS". THE ABILITY TO MAKE FREEHAND SKETCHES IS VERY IMPORTANT FOR A DESIGNER. THE THOUGHT OR AN IDEA OF A PERSON CAN BE EASILY SEEN THROUGH SKETCHES. BEING ABLE TO PRODUCE ON THE SPOT SKETCHES QUICKLY, NEATLY AND CLEARLY IS A VERY VALUABLE SKILL.

IN TECHNICAL DRAWING IT IS IMPORTANT TO MAKE A SKETCH OF THE SOLUTION TO A PROBLEM. A SKETCHED SOLUTION HAS MANY ADVANTAGES. TO MAKE THE BEST USE OF THE DRAWING SPACE ON AN ACTIVITY SHEET A SKETCHED SOLUTION SHOULD BE USED FOR PLANNING OUT THE SPACES.

A SKETCH IS A QUICKLY EXECUTED FREEHAND DRAWING THAT IS NOT INTENDED AS A FINISHED WORK. IN GENERAL, A SKETCH IS A QUICK WAY TO RECORD AN IDEA FOR LATER USE.

FREEHAND SKETCHING

ALL FREEHAND SKETCHES ARE DRAWN WITH THE USE OF A PENCIL ONLY. SOMETIMES GRIDS ARE USED AS AN AID TO SKETCHING. THE SIZES OR PROPORTIONS ARE JUDGED BY THE EYES.

THE PURPOSE OF THE SKETCH AND THE NATURE OF OBJECT DETERMINES THE TYPE OF PROJECTION TO BE USED. THE TWO COMMON TYPES OF PROJECTIONS USED ARE ISOMETRIC AND PERSPECTIVE.



ORIGINAL PHOTO

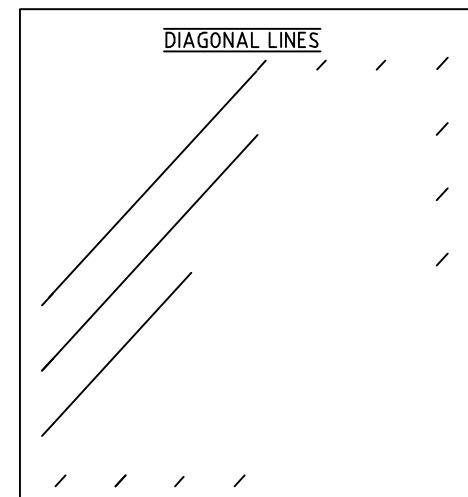
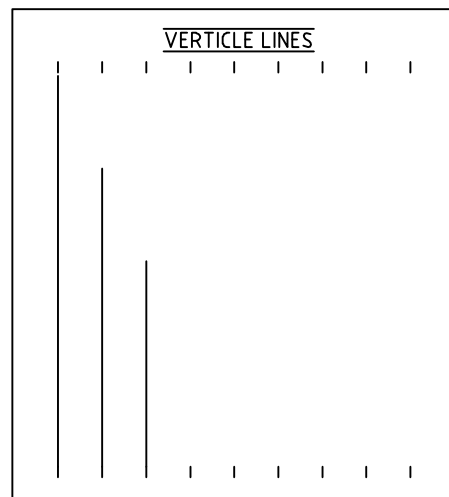
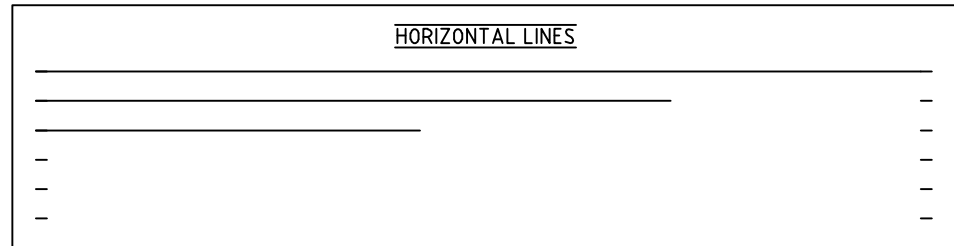


FREEHAND SKETCH

SKETCHING HORIZONTAL, VERTICLE AND DIAGONAL LINES

- MARK THE ENDS OF THE LINE WITH A SMALL DASH.
- HOLD PENCIL GENTLY IN NORMAL WRITING POSITION AT RIGHT ANGLES TO THE DIRECTION OF THE LINE.
- DRAW A COUPLE OF IMAGINARY LINES TO GET THE FEEL OF THE LINES.
- DRAW THE LINE STARTING FROM THE FIRST DASH WHILE KEEPING YOUR EYE ON THE FAR DASH.
- MOVE HAND AND ARM TOGETHER. KEEP HANDS FREE FROM THE PAPER.
- DO NOT ATTEMPT TO DRAW THE LINE IN ONE CONTINUOUS STROKE. SKETCH SEVERAL STROKES END TO END.
- MOVE YOUR ARM AFTER EACH STROKE. STROKES SHOULD NOT OVERLAP.
- FOR VERTICLE AND DIAGONAL LINES TURN THE PAPER SO THAT THE LINE LOOKS HORIZONTAL.

COMPLETE THE LINES GIVEN BELOW BETWEEN THE DASHES



RENDERING

RENDERING IS THE PROCESS OF SHADING DRAWINGS TO SHOW ITS SURFACE TEXTURE AND TO CONVEY AN IMPRESSION OF DEPTH THROUGH SHADE AND LIGHT.

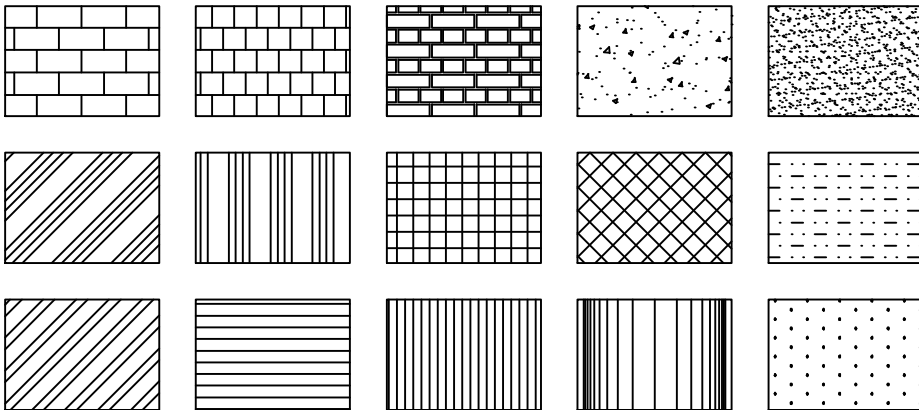
RENDERING OR SHADING IS USEFUL IN SHOWING MORE CLEARLY THE SHAPE AND TEXTURE OF OBJECTS. AN IMPRESSION OF TEXTURE CAN BE REALIZED WITH THE USE OF LINES, SHADES OR DOTS. QUITE A NUMBER OF METHODS OF SHADING CAN BE DONE USING A PENCIL. ITS USE IS LIMITED TO PICTORIAL SKETCHES. ALL RENDERING SHOULD BE SIMPLE AND MUST PRODUCE CLEAR AND EASILY READ SKETCHES.

THE SURFACE THAT IS CLOSEST TO THE LIGHT SOURCE ALWAYS HAS A LIGHTER TONE WHILE THE SURFACE THAT IS FURTHEST HAS A DARKER TONE. WITH SOME IMAGINATION ALMOST ANY SURFACE TEXTURE CAN BE OBTAINED.

RENDERING TIPS

- ALWAYS USE ENOUGH OF LIGHT SHADING AND NEVER USE HEAVY SHADING.
- WORK FROM LIGHT TO DARKER SURFACES.
- USE THE RIGHT AMOUNT OF LIGHT, SHADE AND TEXTURE RENDERING.
- PRACTICE AS MUCH AS POSSIBLE. "PRACTICE MAKES ONE PERFECT".

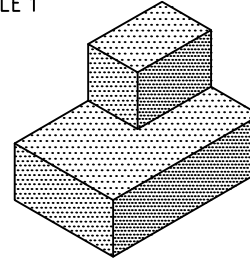
SOME OF THE MANY SURFACES THAT CAN BE USED FOR RENDERING ARE SHOWN BELOW



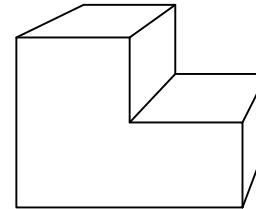
GIVEN: THE PICTORIAL DRAWINGS OF SHAPED BLOCKS. THE FIRST TWO SHAPES HAVE BEEN RENDERED.

REQUIRED: RENDER OR SHADE THE REMAINING SHAPES TO SHOW ITS TEXTURES CLEARLY

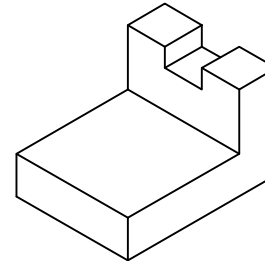
EXAMPLE 1



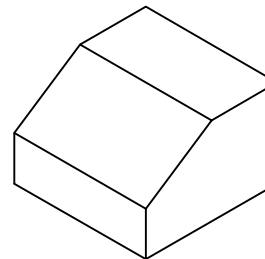
1.



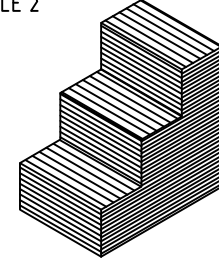
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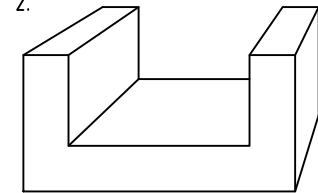
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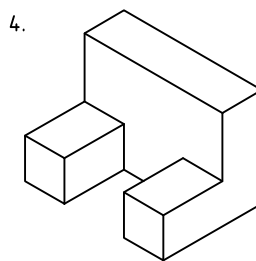
EXAMPLE 2



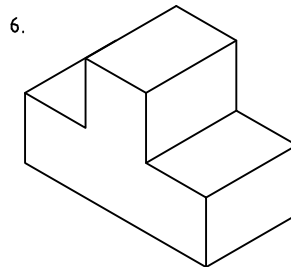
2.



4.

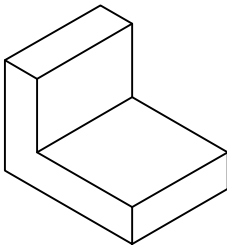


6.



GIVEN: THE PICTORIAL VIEW OF A SIMPLE SHAPED BLOCK.

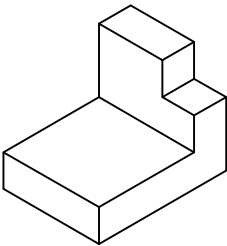
REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE BLOCK.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE OBJECT.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE PICTORIAL VIEW OF A SIMPLE SHAPED BLOCK.

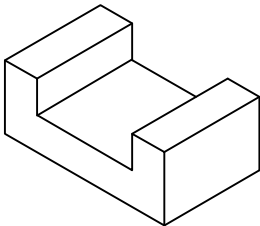
REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE BLOCK.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE OBJECT.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
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GIVEN: THE PICTORIAL VIEW OF A SIMPLE SHAPED BLOCK.

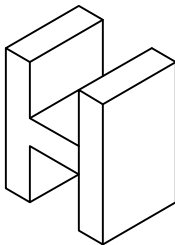
REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE BLOCK.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE OBJECT.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE PICTORIAL VIEW OF A SIMPLE SHAPED BLOCK.

REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE BLOCK.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE OBJECT.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: THE PICTURE OF A TEA TRAY

REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE TEA TRAY.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE TEA TRAY.



GIVEN: THE PICTURE OF A CABINET

REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE CABINET.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE CABINET.



GIVEN: PICTORIAL VIEW OF A WOODEN BENCH.

REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE WOODEN BENCH.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE WOODEN BENCH.



GIVEN: PICTURE OF A SIMPLE TABLE

REQUIRED: PRODUCE A FREEHAND PICTORIAL SKETCH OF THE TABLE.
BY RENDERING CLEARLY SHOW THE SHAPE AND TEXTURE OF THE TABLE.



LINES

IN TECHNICAL DRAWING VARIOUS TYPES OF LINES ARE USED TO SHOW THE DIFFERENT FEATURES OF A DRAWING THEREFORE IT IS IMPORTANT THAT THE APPROPRIATE LINE TYPES ARE USED TO REPRESENT THE GEOMETRY OF THE OBJECT.

TYPES OF LINES

VISIBLE OUTLINE (THICK CONTINUOUS)



Application

used to represent features that can be seen like outlines and border lines

CONSTRUCTION LINE (THIN CONTINUOUS)



used to represent features like dimension, hatching and section lines, leaders

HIDDEN LINE



used to represent features that cannot be seen in the current view

CENTRE LINE



used to show the centre of symmetrical objects

CUTTING PLANE LINE



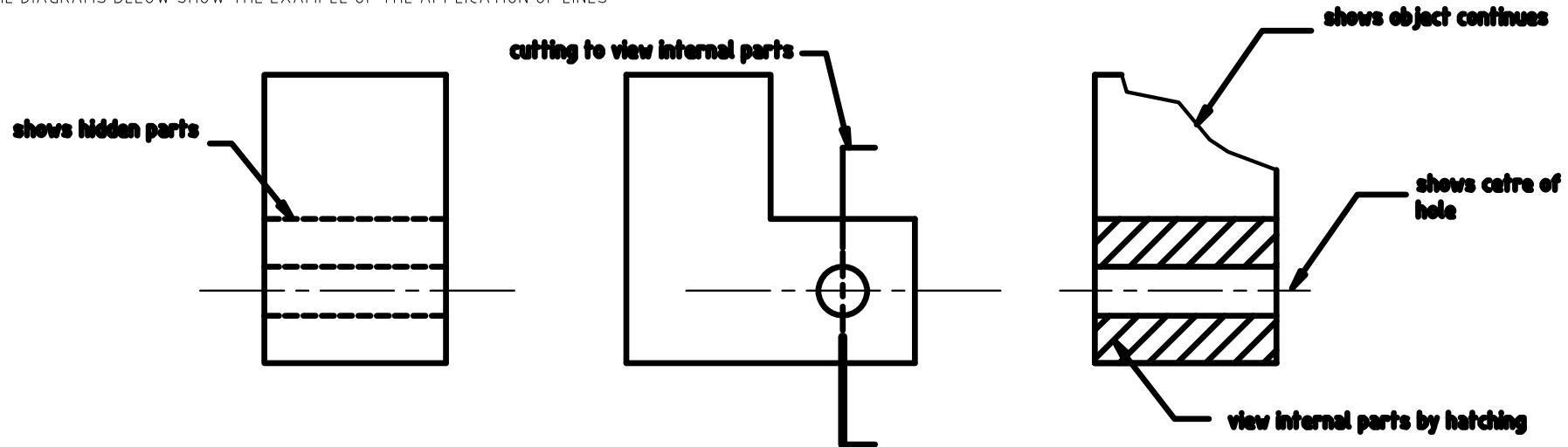
used to show the the part which have been cut to reveal internal features through sectioning

BREAK LINE



used to show where an object is broken to save space or to reveal internal features

THE DIAGRAMS BELOW SHOW THE EXAMPLE OF THE APPLICATION OF LINES



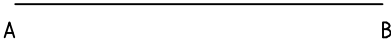
NAME: _____
FORM: _____

LINE TYPES

DATE: _____
SHEET NO: 11

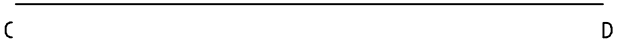
GIVEN: LINE AB
REQUIRED: BISECT THE LINE AB

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



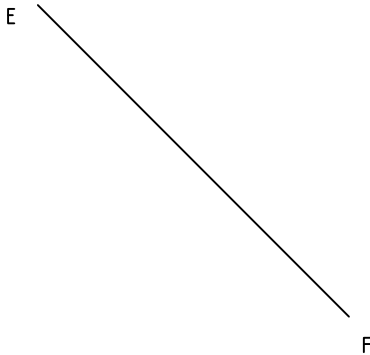
GIVEN: LINE CD
REQUIRED: QUADRISECT THE LINE CD

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



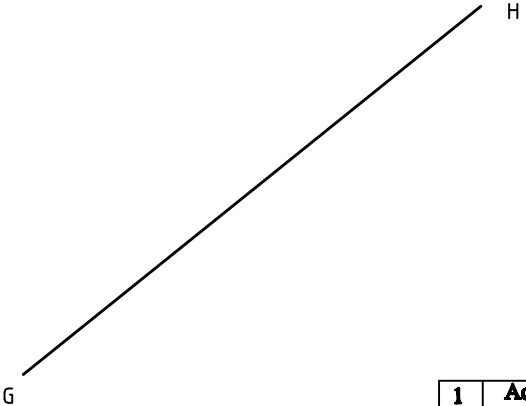
GIVEN: LINE EF
REQUIRED: BISECT THE LINE EF

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



GIVEN: LINE GH
REQUIRED: QUADRISECT THE LINE GH

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



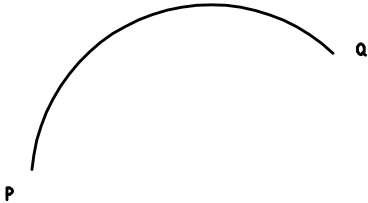
NAME: _____

FORM: _____

DATE: _____

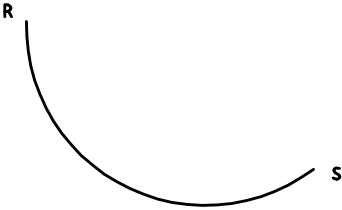
SHEET NO: 12

GIVEN: ARC PQ
REQUIRED: BISECT THE ARC PQ



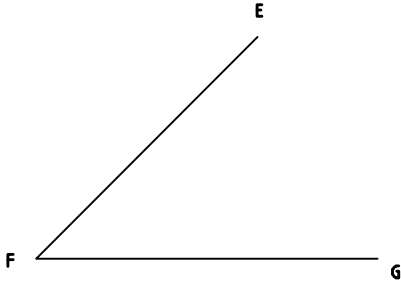
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: ARC RS
REQUIRED: QUADRISECT THE ARC RS.



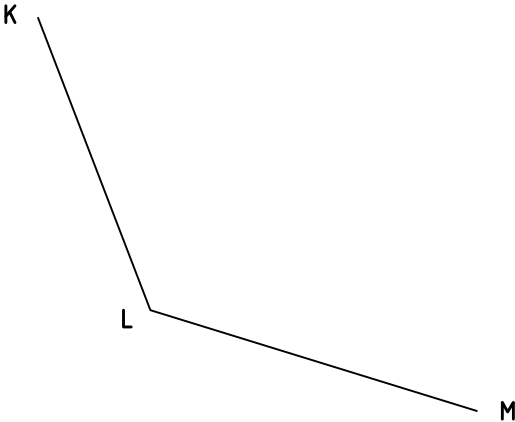
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: ANGLE EFG
REQUIRED: BISECT THE ANGLE



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: ANGLE KLM
REQUIRED: QUADRISECT THE ANGLE.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

NAME: _____

FORM: _____

DATE: _____

SHEET NO: 13

GIVEN: LINE AB
REQUIRED: DIVIDE THE LINE INTO 3 EQUAL PARTS.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

A _____ B

GIVEN: LINE CD
REQUIRED: DIVIDE THE LINE INTO 5 EQUAL PARTS.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

C _____ D

GIVEN: LINE EF
REQUIRED: DIVIDE THE LINE INTO 6 EQUAL PARTS.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

E _____ F

GIVEN: LINE GH
REQUIRED: DIVIDE THE LINE INTO 7 EQUAL PARTS.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

G _____ H

NAME: _____

FORM: _____

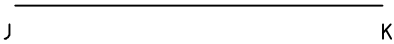
DATE: _____

SHEET NO: 14

GIVEN: LINE JK

REQUIRED: DIVIDE THE LINE JK INTO A RATIO OF 1:2

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



GIVEN: LINE LM

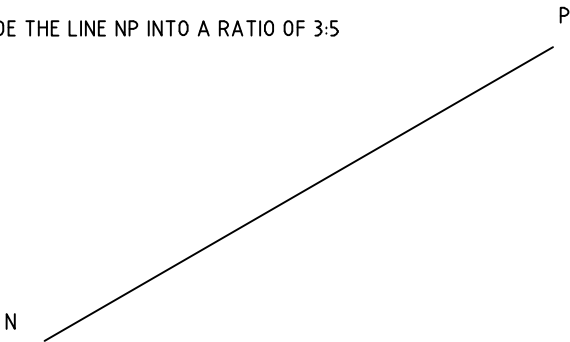
REQUIRED: DIVIDE THE LINE LM INTO A RATIO OF 2:3

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



GIVEN: LINE NP

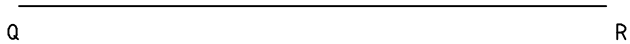
REQUIRED: DIVIDE THE LINE NP INTO A RATIO OF 3:5



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: LINE QR

REQUIRED: DIVIDE THE LINE QR INTO A RATIO OF 5:4



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

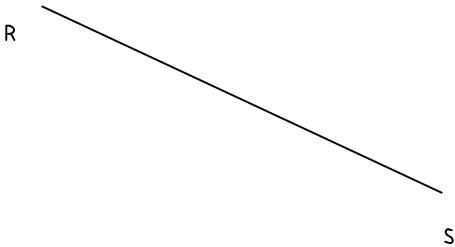
NAME: _____

FORM: _____

DATE: _____

SHEET NO: 15

GIVEN: LINE RS
REQUIRED: DIVIDE THE LINE RS INTO A RATIO OF 1:2:3



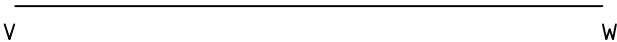
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: LINE TU
REQUIRED: DIVIDE THE LINE TU INTO A RATIO OF 2:3:4



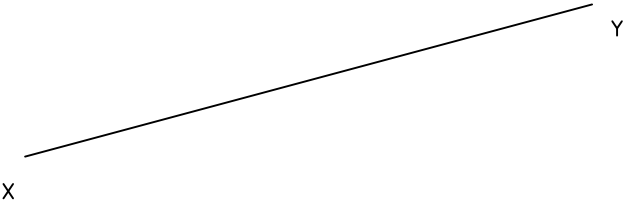
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: LINE VW
REQUIRED: DIVIDE THE LINE VW INTO A RATIO OF 2:5:3



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: LINE XY
REQUIRED: DIVIDE THE LINE XY INTO A RATIO OF 4:2:5



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

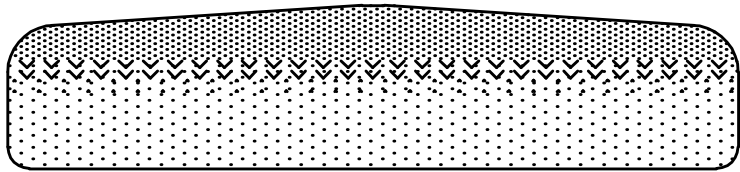
NAME: _____

FORM: _____

DATE: _____

SHEET NO: 16

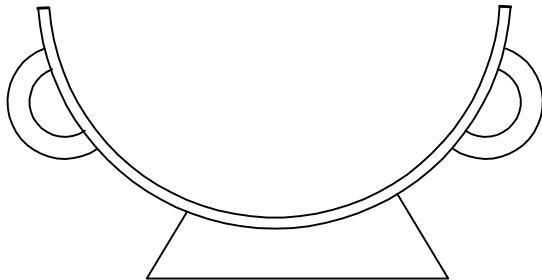
GIVEN: A LONG LOAF BREAD.
REQUIRED: EQUALLY SHARE THE BREAD BETWEEN TWO FRIENDS



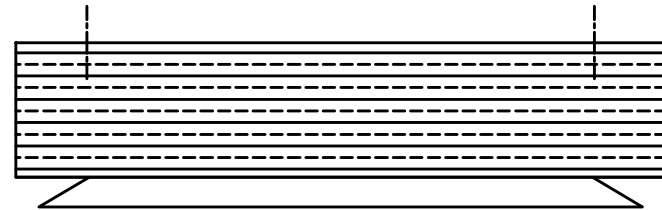
GIVEN: A MILK CHOCOLATE BAR
REQUIRED: SHARE THE CHOCOLATE EQUALLY AMONGST FIVE FAMILY MEMBERS.



GIVEN: A DOUBLE HANDLE PAPER PIN RACK
REQUIRED: PLACE A SEPARATOR SO THAT EQUAL SPACE IS AVAILABLE FOR STAPLER PINS AND OFFICE CLIPS



GIVEN: A WOODEN PENCIL RACK WITH THE CENTRES OF THE FIRST AND LAST HOLES
REQUIRED: MARK THE CENTRES FOR A TOTAL OF SEVEN PENS TO BE EQUALLY SPACED ON THE RACK

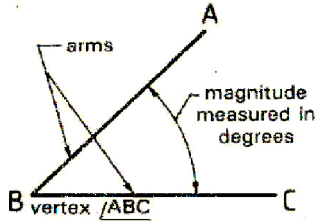


NAME: _____
FORM: _____

DATE: _____
SHEET NO: 17

ANGLES

WHEN TWO STRAIGHT LINES MEET AT A POINT THEY FORM AN ANGLE.
THE POINTS WHERE THEY MEET IS CALLED THE VERTEX.
ANGLES ARE USUALLY MEASURED IN DEGREES.



TYPES OF ANGLES

A RIGHT ANGLES TRIANGLE HAS A MAGNITUDE OF 90° .

AN ACUTE ANGLE HAS A MAGNITUDE LESS THAN 90° .

AN OBTUSE ANGLE HAS A MAGNITUDE GREATER THAN 90° BUT LESS THAN 180° .

A REFLEX ANGLE HAS A MAGNITUDE GREATER THAN 180° .

GIVEN: LINE GH

REQUIRED: CONSTRUCT AN ANGLE OF 120°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

G H

GIVEN: LINE AB

REQUIRED: CONSTRUCT AN ANGLE OF 60° AND BISECT IT.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

A B

GIVEN: LINE EF

REQUIRED: CONSTRUCT AN ANGLE OF 15°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

E F

GIVEN: LINE JK

REQUIRED: CONSTRUCT AN ANGLE OF 90°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

J K

GIVEN: LINE LM

REQUIRED: CONSTRUCT AN ANGLE OF 45°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

L M

NAME: _____

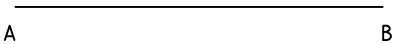
FORM: _____

DATE: _____

SHEET NO: 18

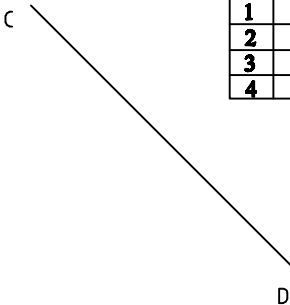
GIVEN: LINE AB
REQUIRED: CONSTRUCT AN ANGLE OF 75°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



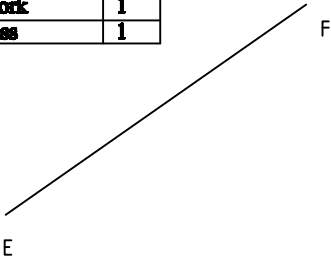
GIVEN: LINE CD
REQUIRED: CONSTRUCT AN ANGLE OF 150°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



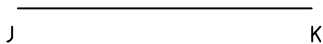
GIVEN: LINE EF
REQUIRED: CONSTRUCT AN ANGLE OF 135°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



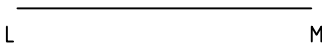
GIVEN: LINE JK
REQUIRED: CONSTRUCT AN ANGLE OF 67.5°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

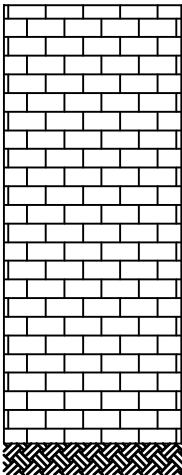


GIVEN: LINE LM
REQUIRED: CONSTRUCT AN ANGLE OF 37.5°

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



Given: A WALL AND A POINT T MARKED ON THE GROUND.
REQUIRED: PLACE A LADDER FROM POINT X TO LEAN AGAINST THE WALL AT AN ANGLE OF 75°



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



NAME: _____

FORM: _____

DATE: _____

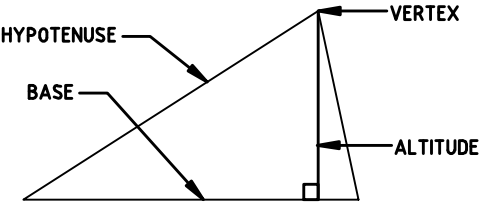
SHEET NO: 19

TRIANGLES

A TRIANGLE IS A PLANE FIGURE BOUNDED BY THREE STRAIGHT LINES. THE SUM OF THE THREE ANGLES IN A TRIANGLE IS EQUAL TO 180°.

PARTS OF A TRIANGLE

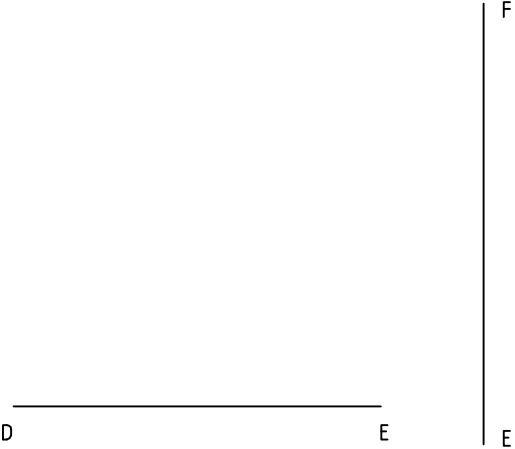
BASE: A TRIANGLE STANDS ON ONE OF ITS SIDES; TERMED AS THE BASE.
VERTEX: IS THE ANGLE OPPOSITE THE BASE.
ALTITUDE: IS A STRAIGHT LINE DRAWN FROM THE VERTEX PERPENDICULAR TO THE BASE.
HYPOTENUSE: IS THE LONGEST SIDE OF A TRIANGLE.



GIVEN: BASE LINE AB
REQUIRED: CONSTRUCT AN EQUILATERAL TRIANGLE ABC



GIVEN: BASE LINE DE & SIDE EF
REQUIRED: CONSTRUCT AN ISOCELES TRIANGLE DEF



GIVEN: BASE LINE GH AND ALTITUDE xJ
REQUIRED: CONSTRUCT AN ISOCELES TRIANGLE GHJ

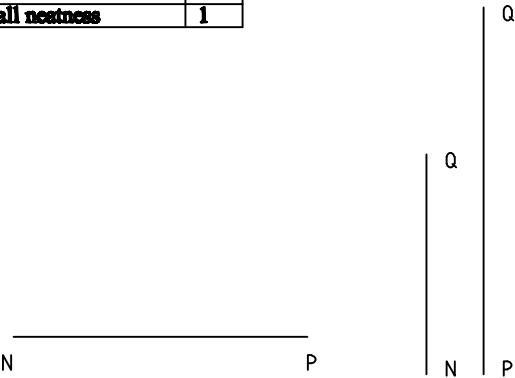
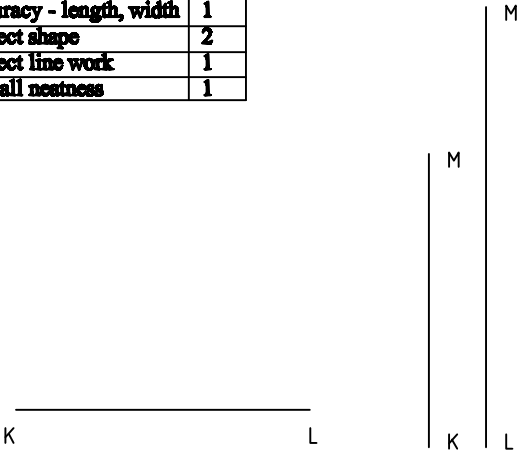
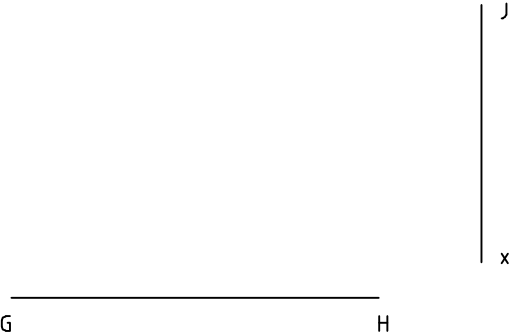
GIVEN: BASE LINE KL, SIDES KM & LM
REQUIRED: CONSTRUCT A SCALENE TRIANGLE KLM

GIVEN: BASE LINE NP, SIDES NQ & PQ
REQUIRED: CONSTRUCT A SCALENE TRIANGLE NPQ

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



NAME: _____
FORM: _____

DATE: _____
SHEET NO: 20

REQUIRED: CONSTRUCT AN EQUILATERAL TRIANGLE ABC

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

A B

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

D _____ E

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

$$\overline{GH}$$

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

K _____ L

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

N P

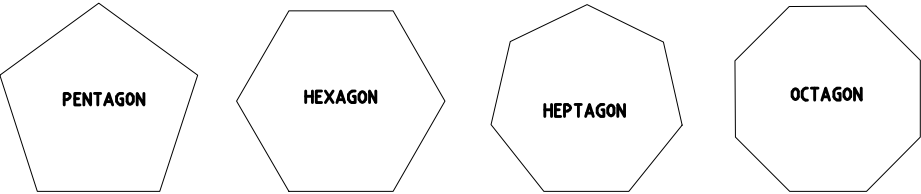
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

R S

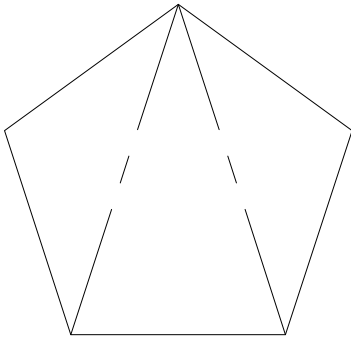
FORM: _____

SHEET NO: 21

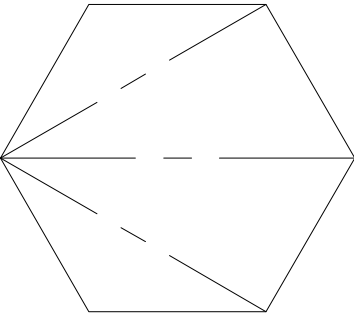
A POLYGON IS A PLANE FIGURE BOUNDED BY MORE THAN FOUR SIDES.
POLYGONS MAY BE REGULAR OR IRREGULAR.
REGULAR POLYGONS HAVE ALL THEIR SIDES AND ANGLES EQUAL.
DIAGONAL - IN ANY POLYGON THE LINE WHICH CONNECTS ANY TWO POINTS OR MORE ANGLES, NOT ADJACENT, IS CALLED A DIAGONAL.
CENTRE - THE CENTRE IS OBTAINED BY BISECTING ANY TWO ANGLES OF A REGULAR POLYGON.



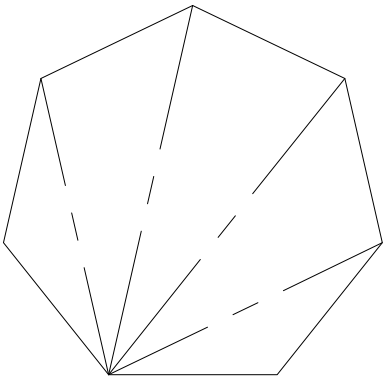
A REGULAR PENTAGON IS BOUNDED BY FIVE EQUAL SIDES. ALL THE INTERIOR ANGLES ADD UP TO 540° . ALL ANGLES ARE THE SAME -108° .



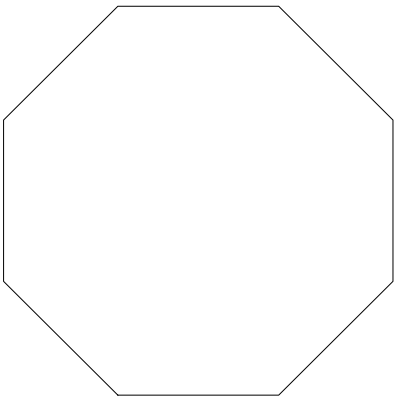
A REGULAR HEXAGON HAS SIX EQUAL SIDES. ALL THE INTERIOR ANGLES ADD UP TO 720° . ALL ANGLES ARE THE SAME -120° .



A REGULAR HEPTAGON IS BOUNDED BY SEVEN EQUAL SIDES. ALL THE INTERIOR ANGLES ADD UP TO 900° . ALL ANGLES ARE THE SAME.

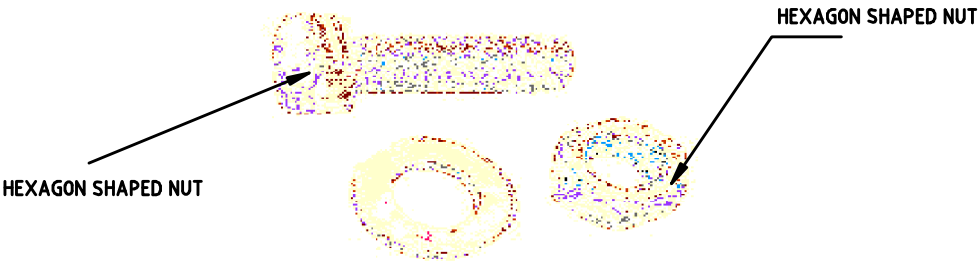


A REGULAR OCTAGON HAS EIGHT EQUAL SIDES. ALL THE INTERIOR ANGLES ADD UP TO 1080° . ALL ANGLES ARE THE SAME -135° .



SHOWN BELOW ARE EXAMPLES OF HEXAGONAL SHAPED BOLT AND NUT.

APPLICATION OF POLYGONS AS SEEN IN THE HEAD OF THE BOLT AND NUT

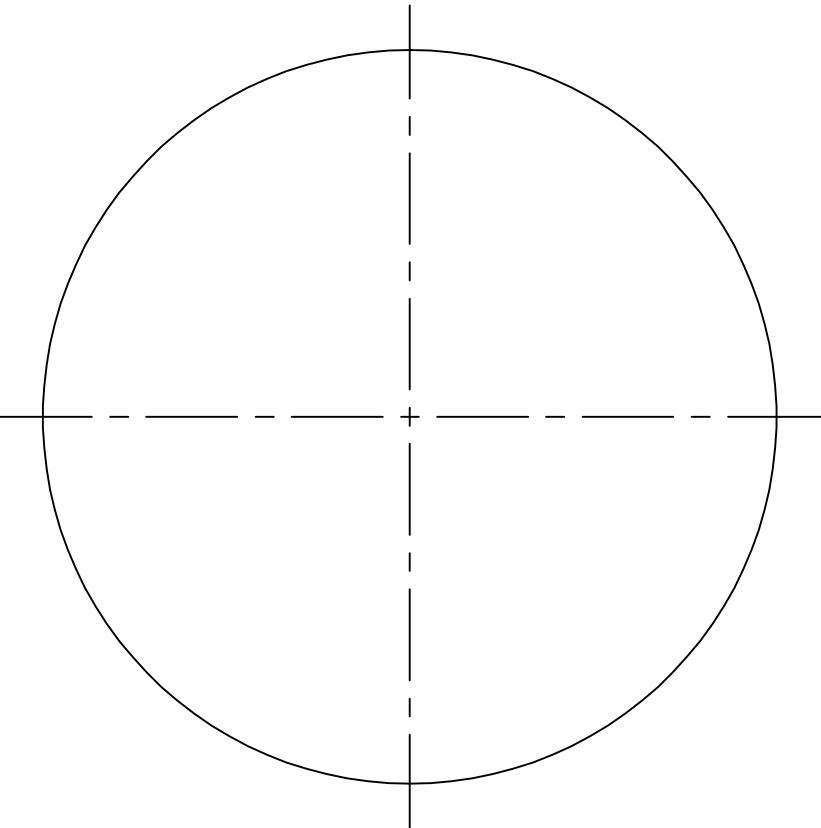


NAME: _____
FORM: _____

POLYGONS

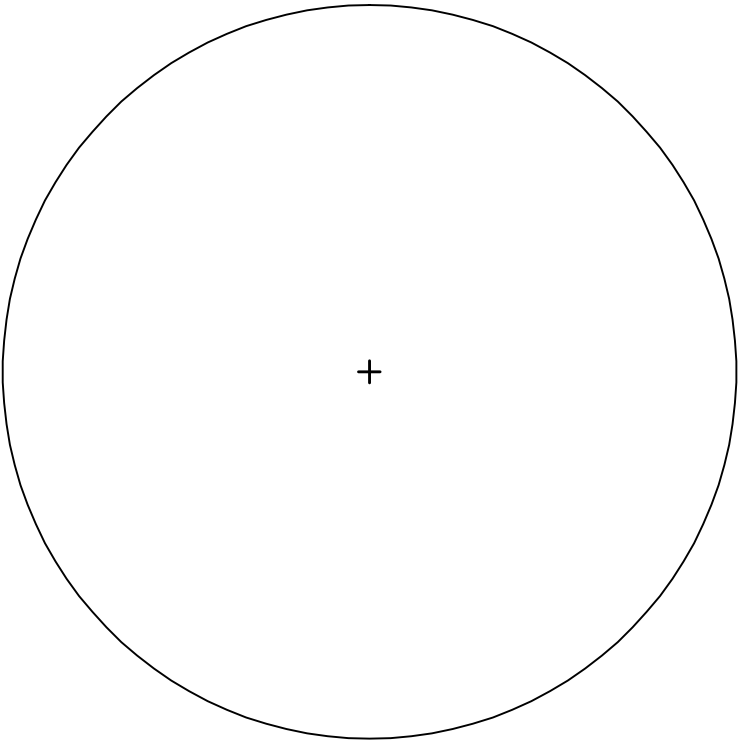
DATE: _____
SHEET NO: 23

GIVEN: A CIRCLE
REQUIRED: CONSTRUCT A REGULAR PENTAGON LMNOP WITHIN THE CIRCLE.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: A CIRCLE
REQUIRED: CONSTRUCT A REGULAR HEXAGON GHJKLM WITHIN THE CIRCLE.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

NAME: _____	_____	DATE: _____
FORM: _____	_____	SHEET NO: 24

REQUIRED: CONSTRUCT A REGULAR PENTAGON ABCDE.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

REQUIRED: CONSTRUCT A REGULAR HEXAGON FGHJKL.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

$\overline{FG}$

NAME: _____

FORM: _____

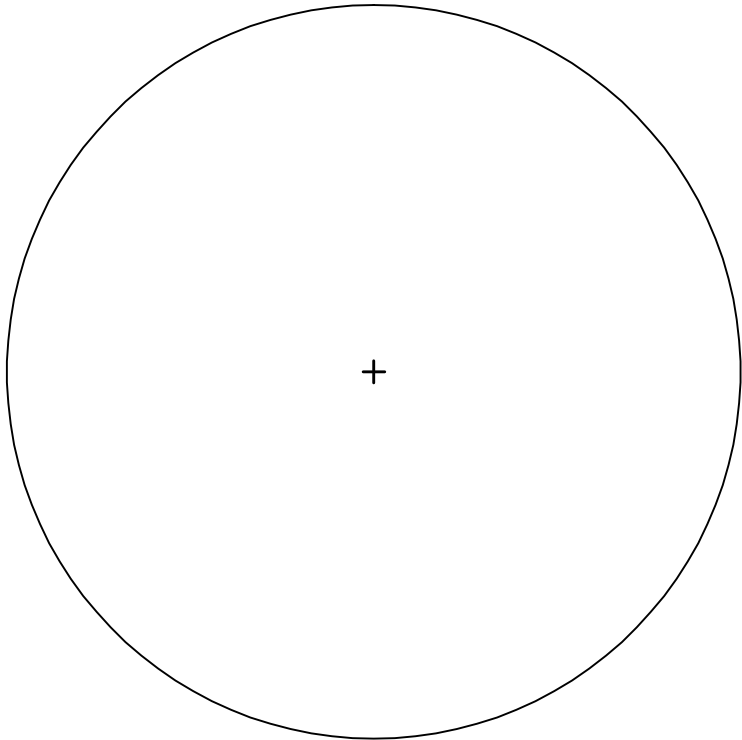
DATE: _____

SHEET NO: 25

GIVEN: A CIRCLE

REQUIRED: CONSTRUCT A REGULAR PENTAGON FGHJK WITHIN THE CIRCLE, USING THE GENERAL METHOD.

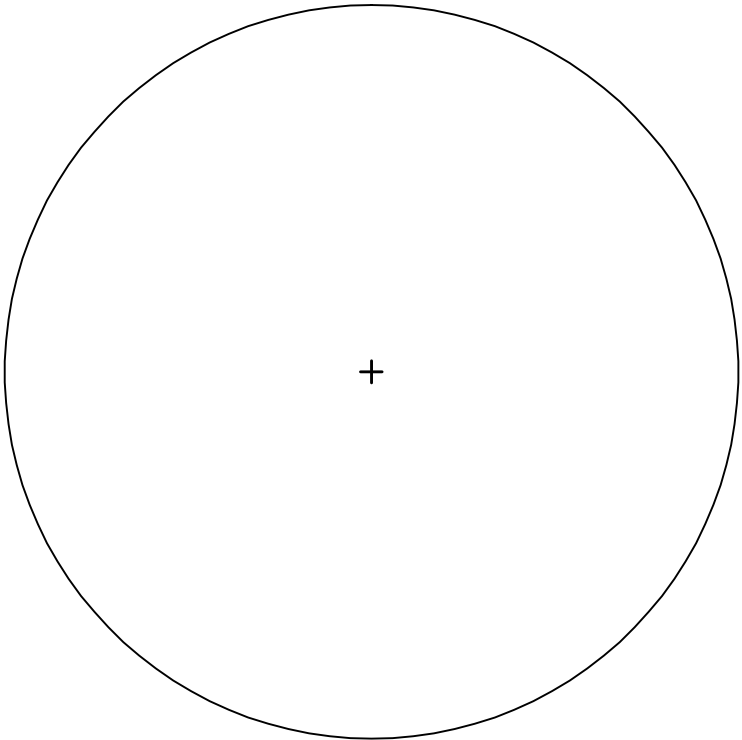
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



GIVEN: A CIRCLE

REQUIRED: CONSTRUCT A REGULAR HEXAGON GHJKLM WITHIN THE CIRCLE, USING THE GENERAL METHOD.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



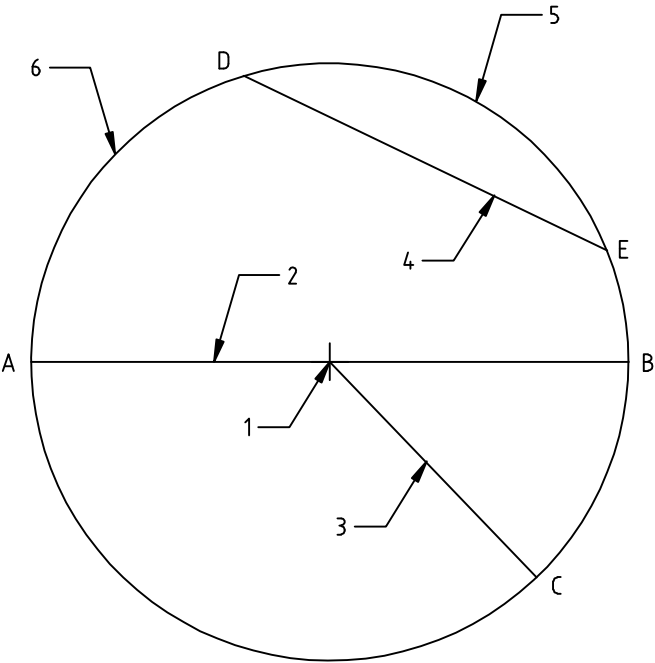
NAME: _____

FORM: _____

DATE: _____

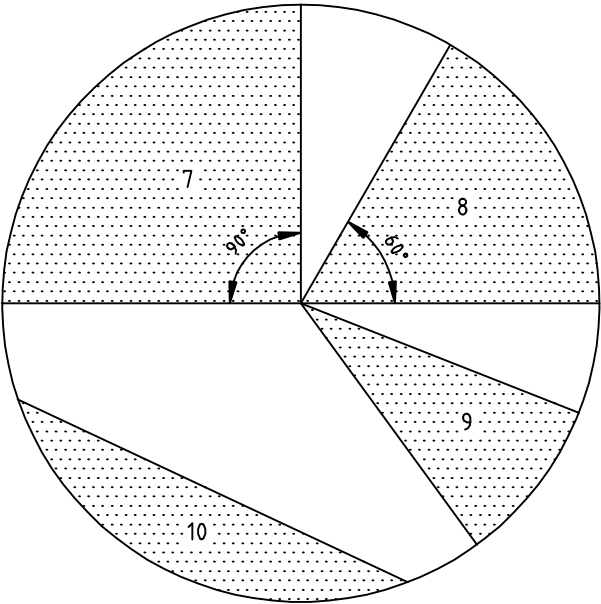
SHEET NO: 26

TABLE THE PARTS OF THE CIRCLE.



- 1 _____
- 2 _____
- 3 _____
- 4 _____
- 5 _____
- 6 _____

TABLE THE PARTS OF THE CIRCLE.



- 7 _____
- 8 _____
- 9 _____
- 10 _____

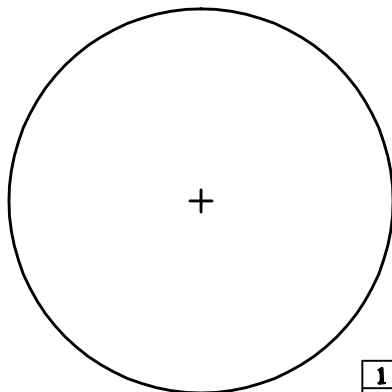
NAME: _____
FORM: _____

DATE: _____
SHEET NO: 27

GIVEN : A CIRCLE

REQUIRED: DIVIDE THE CIRCLE INTO TWELVE EQUAL PARTS.

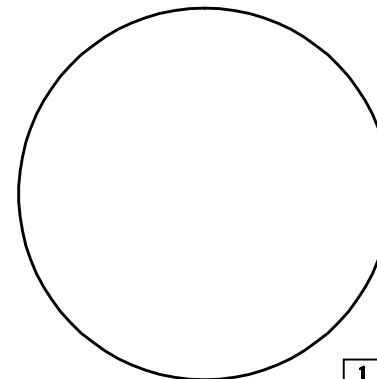
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN : A CIRCLE

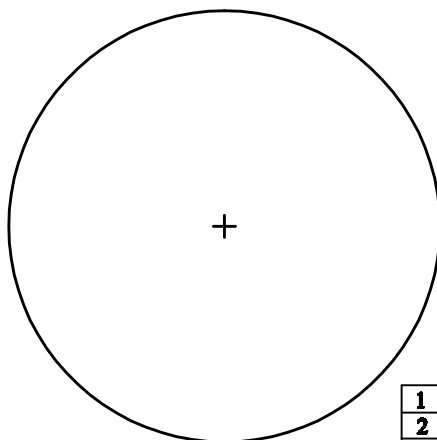
REQUIRED: FIND THE CENTRE OF THE CIRCLE AND DIVIDE IT INTO EIGHT EQUAL PARTS.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN : A CIRCLE

REQUIRED: DIVIDE THE CIRCLE INTO TEN EQUAL PARTS.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN : ARC RS

REQUIRED: FIND ITS CENTRE AND COMPLETE THE CIRCLE.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

NAME: _____

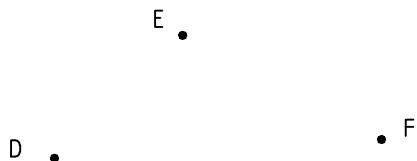
FORM: _____

DATE: _____

SHEET NO: 28

GIVEN: POINTS D, E AND F.

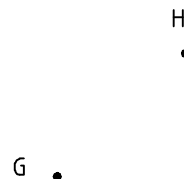
REQUIRED: DESCRIBE A CIRCLE TO PASS THROUGH THE THREE POINTS DEF



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: POINTS G AND H.

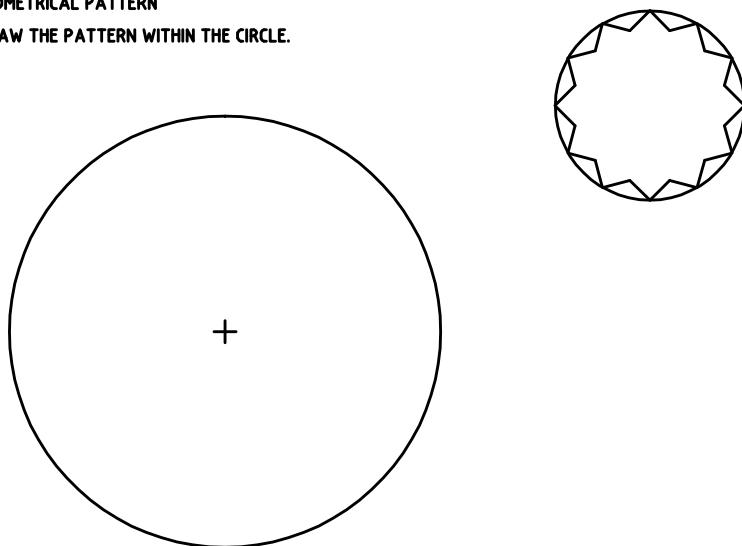
REQUIRED: DESCRIBE A CIRCLE OF 60mm DIAMETER TO PASS THROUGH THE TWO GIVEN POINTS.



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

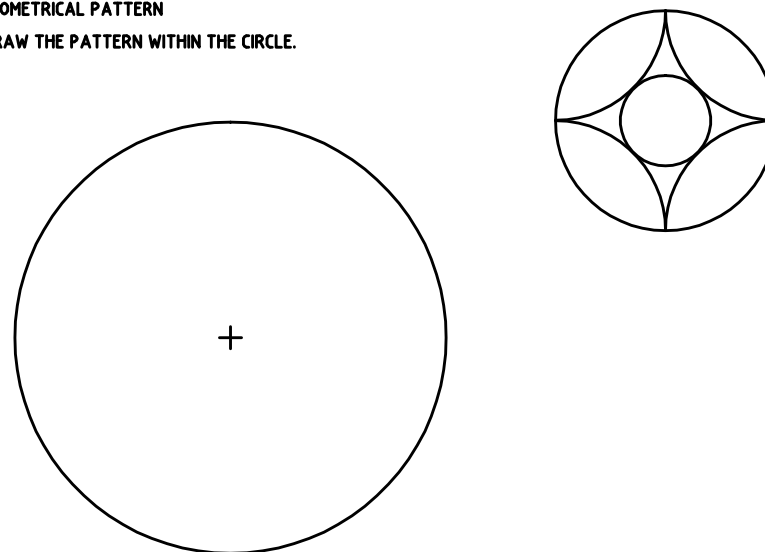
GIVEN : A GEOMETRICAL PATTERN

REQUIRED: REDRAW THE PATTERN WITHIN THE CIRCLE.



GIVEN : A GEOMETRICAL PATTERN

REQUIRED: REDRAW THE PATTERN WITHIN THE CIRCLE.



NAME: _____

FORM: _____

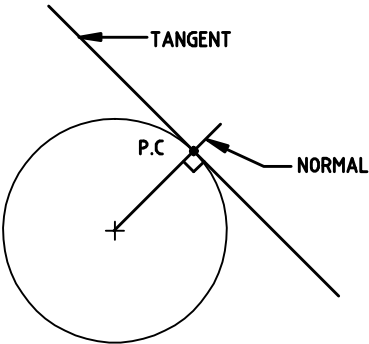
DATE: _____

SHEET NO: 29

TANGENTS TO CIRCLES

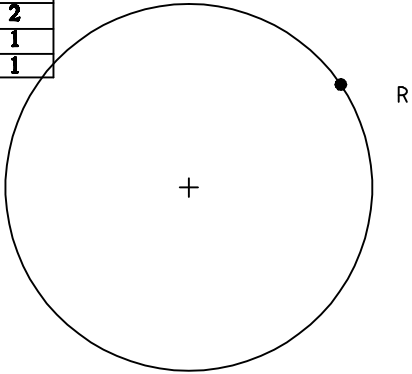
A TANGENT IS A STRAIGHT LINE THAT TOUCHES THE CIRCUMFERENCE OF A CIRCLE AT ONE POINT BUT DOES NOT CUT IT. THE POINT WHERE THE TANGENT TOUCHES THE CIRCLE IS CALLED THE POINT OF CONTACT (PC). A NORMAL IS A LINE THAT PASSES THROUGH THE POINT OF CONTACT AND IS PERPENDICULAR TO THE TANGENT.

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

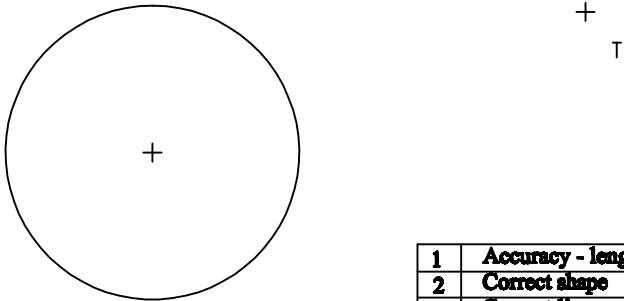


GIVEN: A CIRCLE AND A POINT R ON THE CIRCUMFERENCE OF THE CIRCLE.
REQUIRED: DRAW A TANGENT TO POINT R

1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

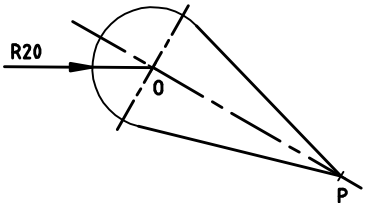


GIVEN: A CIRCLE AND A POINT T OUTSIDE THE CIRCLE.
REQUIRED: DRAW A TANGENT FROM POINT T TO THE GIVEN CIRCLE



1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1

GIVEN: A GEOMETRICAL PATTERN.
REQUIRED: REDRAW THE PATTERN.



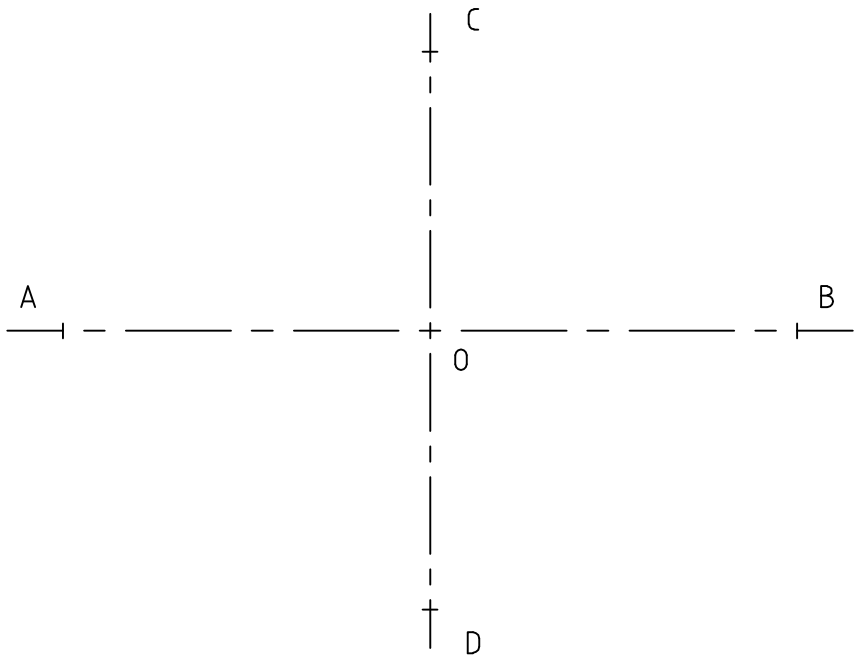
1	Accuracy - length, width	1
2	Correct shape	2
3	Correct line work	1
4	Overall neatness	1



NAME: _____
FORM: _____

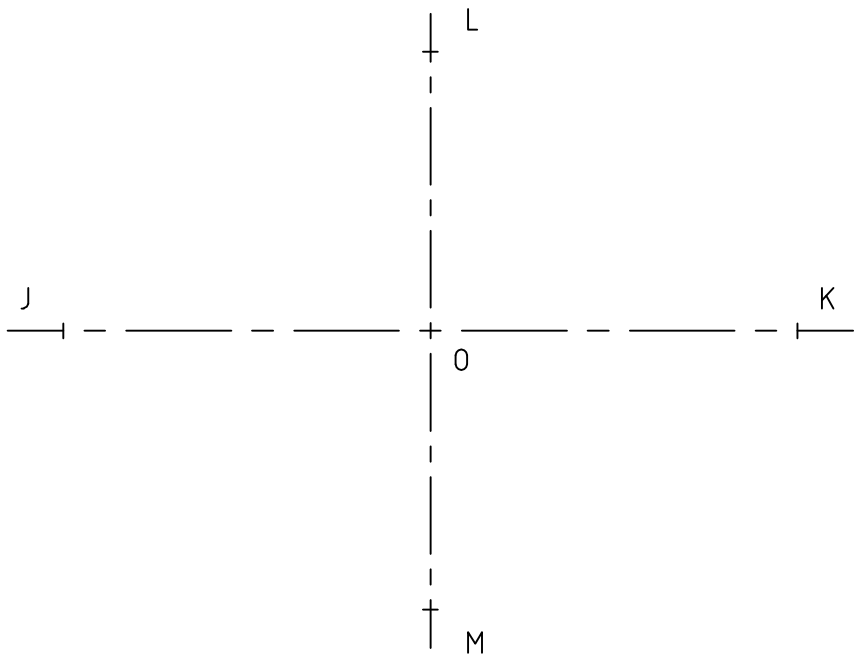
DATE: _____
SHEET NO: 30

GIVEN: A MAJOR AXIS AB & A MINOR AXIS CD
REQUIRED: CONSTRUCT AN ELLIPSE USING THE CONCENTRIC CIRCLES METHOD



1	Accuracy - Axis	3
2	Correct points	4
3	Correct line work	1
4	Overall neatness	2

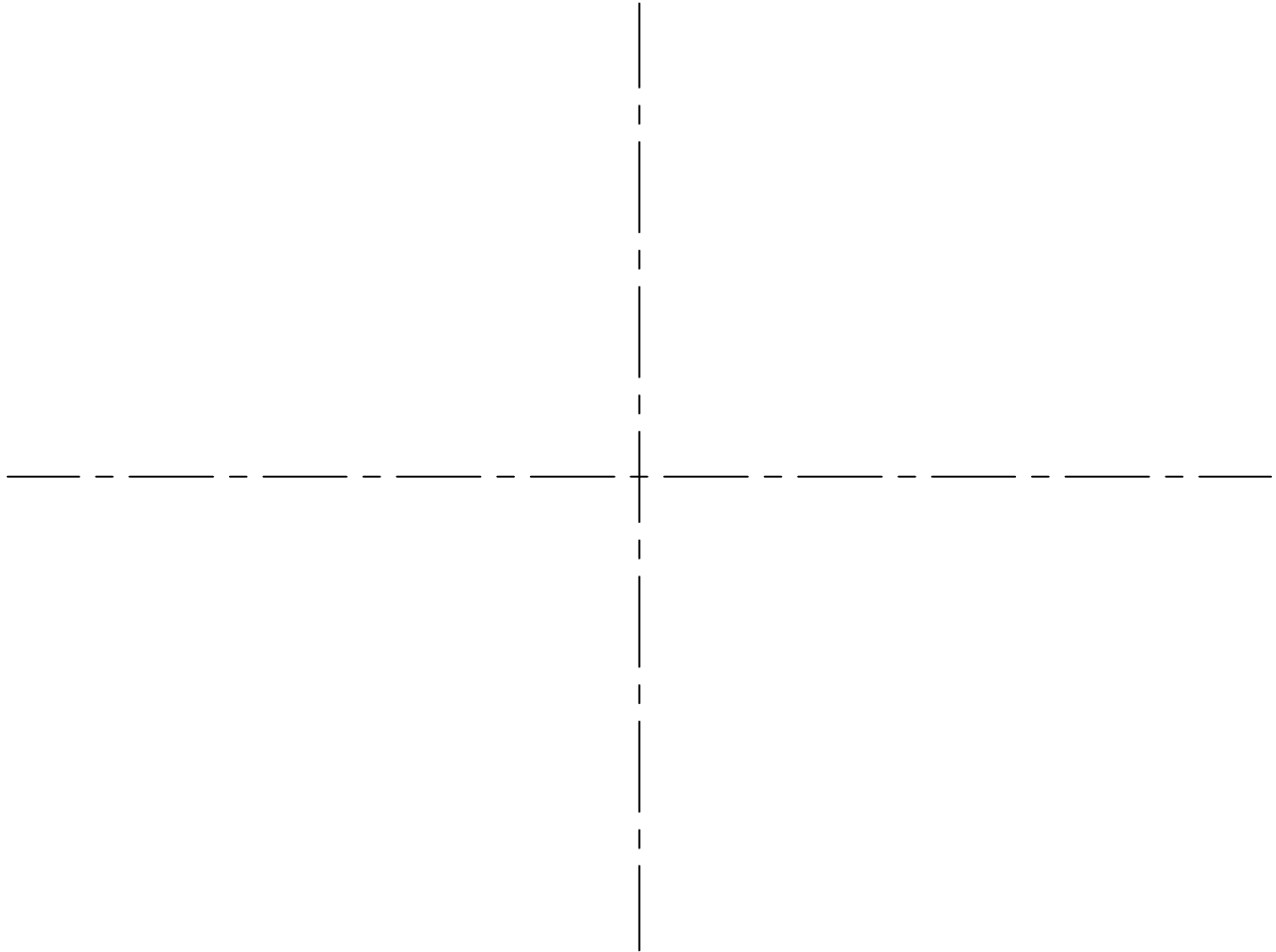
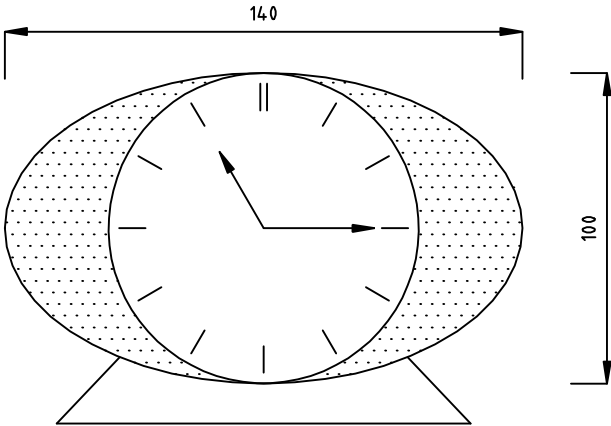
GIVEN: A MAJOR AXIS JK & A MINOR AXIS LM
REQUIRED: CONSTRUCT AN ELLIPSE USING THE INTERSECTING ARCS METHOD



1	Accuracy - Axis	3
2	Correct points	4
3	Correct line work	1
4	Overall neatness	2

NAME: _____	_____	DATE: _____
FORM: _____	_____	SHEET NO: 31

GIVEN: A TABLE CLOCK WITH AN ELLIPTICAL FRAME.
REQUIRED: CONSTRUCT THE ELLIPTICAL FRAME TO FULL SIZE.
NAME THE METHOD OF CONSTRUCTION USED. _____



1	Accuracy - Axis	3
2	Correct points	4
3	Correct line work	1
4	Overall neatness	2

NAME: _____	_____	DATE: _____
FORM: _____	_____	SHEET NO: 32

ORTHOGRAPHIC PROJECTION IS MOST COMMONLY USED FOR REPRESENTING DRAWINGS.

A SIMPLE SHAPED BLOCK FIG 8.1, WILL BE USED IN OUR UNDERSTANDING OF ORTHOGRAPHIC PROJECTION.

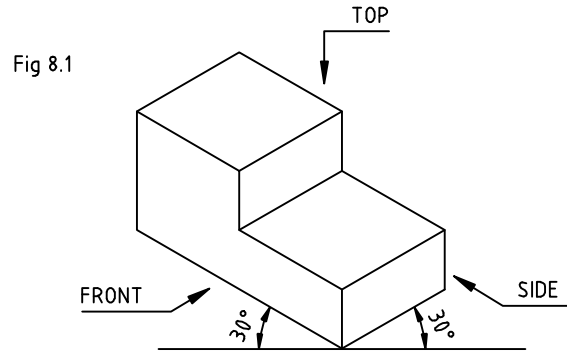
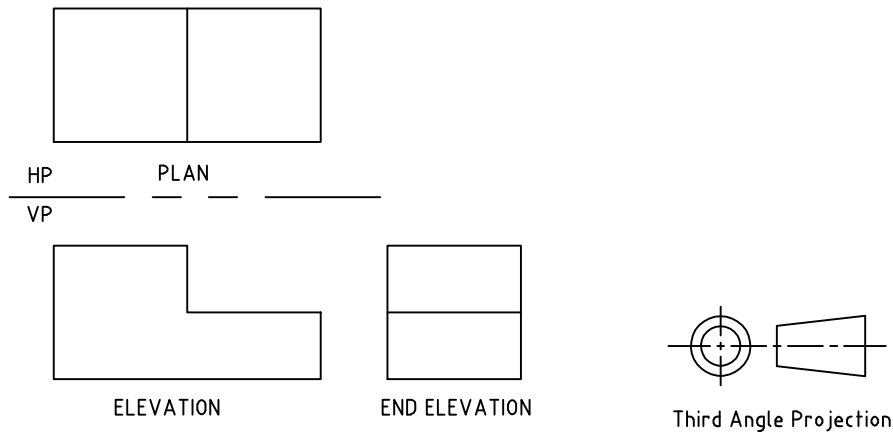


FIG 8.2 SHOWS THREE DRAWINGS, PLAN (VIEWED FROM TOP), ELEVATION (VIEWED FROM FRONT), END ELEVATION (VIEWED FROM SIDE) THAT REPRESENT THE SAME SIMPLE SHAPED BLOCK OF FIG 8.1, BUT SEEN FROM THREE DIFFERENT POSITIONS. EACH VIEW IS SEEN PERPENDICULAR TO OR AT 90° TO EACH OF THE SURFACES.

Fig 8.2



ORTHOGRAPHIC PROJECTION LOOKS AT THE REAL FORM OF OR THE TRUE SHAPE OF THE SIMPLE SHAPED BLOCK IN EACH VIEW. EACH VIEW IS IN LINE WITH THE NEXT EITHER SIDEWAYS OR ON THE TOP.

THIS IS THE INTERNATIONAL LANGUAGE OF DRAWINGS USED BY ENGINEERS, ARCHITECTS, DESIGNERS, DRAUGHTSPERSONS AND ALL OTHERS INVOLVED IN THE FIELD OF TECHNOLOGY AROUND THE WORLD.

PRINCIPAL PLANE OF PROJECTION

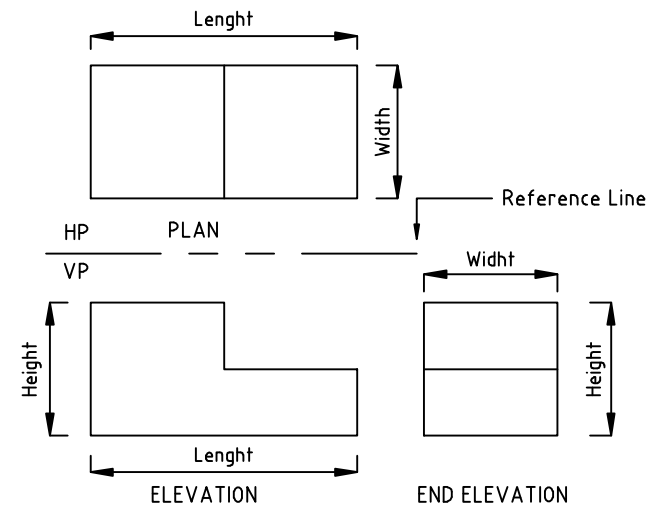
1) VERTICAL PLANE (VP) : IN VERTICAL PLANE THE ELEVATION AND END ELEVATION ARE DRAWN. THE ELEVATION IS VIEWED FROM FRONT AND END ELEVATION IS VIEWED FROM SIDE.

NOTE : IN THIRD ANGLE ORTHOGRAPHIC PROJECTION THE VERTICAL PLANE IS BELOW THE REFERENCE LINE.

2) HORIZONTAL PLANE (HP) : IN HORIZONTAL PLANE THE PLAN IS DRAWN. THE PLAN IS VIEWED FROM TOP.

NOTE : IN THIRD ANGLE ORTHOGRAPHIC PROJECTION THE HORIZONTAL PLANE IS ABOVE THE REFERENCE LINE.

Fig 8.3



NAME:	
YEAR:	

3rd ANGLE ORTHOGRAPHIC PROJECTION

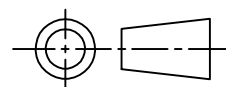
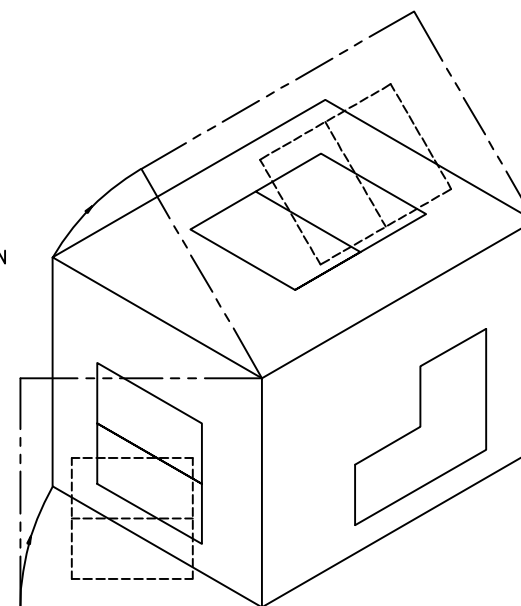
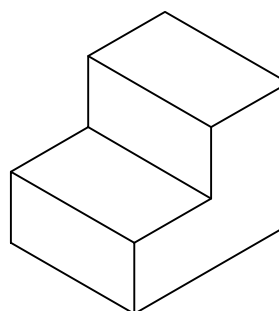
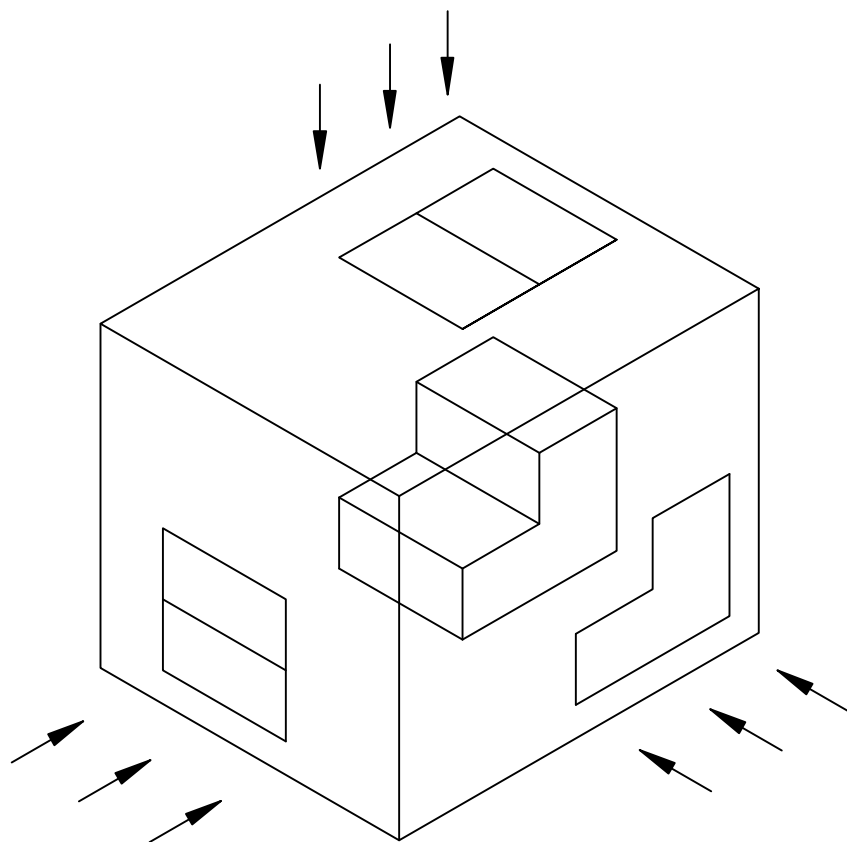
DATE:	
SHEET NO: 33	

A L - SHAPED BLOCK IS USED FOR UNDERSTANDING 3RD ANGLE PROJECTION WHERE THE OBJECT IS VIEWED AND DRAWN ON THE SAME SIDE.

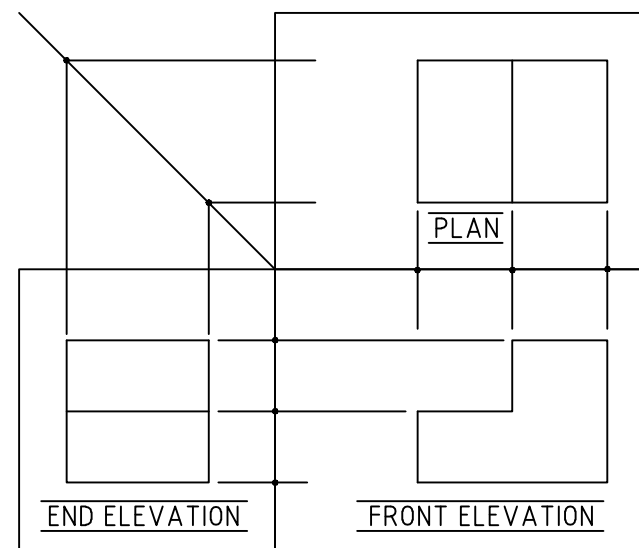
NOTE: IN ORTHOGRAPHIC PROJECTIONS THE LONGER SIDE (LENGTH) OF ANY OBJECT WILL BE TAKEN IN THE FRONT ELEVATION WHILE THE SHORTER SIDE (WIDTH) WILL BE TAKEN IN THE END ELEVATION.

NOTE:

WHEN THE BOX IS OPENED, THE PLAN AND THE END ELEVATION ARE IN LINE WITH THE FRONT ELEVATION. THE PLAN IS DRAWN IN LINE AND ABOVE THE FRONT ELEVATION AND THE END ELEVATION DRAWN IN LINE BUT ON THE SIDE OF THE FRONT ELEVATION WHERE IT IS VIEWED FROM.



THIRD ANGLE PROJECTION



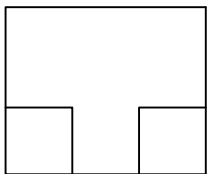
NAME:	
YEAR:	

3rd ANGLE ORTHOGRAPHIC PROJECTION

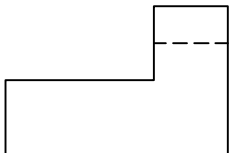
DATE:	
SHEET NO: 34	

EXERCISES : MATCHING

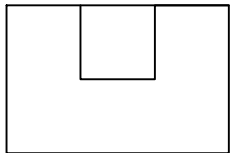
MATCH THE ELEVATIONS DRAWN BELOW WITH ITS CORRESPONDING PICTORIAL VIEWS
WRITE THE ANSWERS IN THE BOXES PROVIDED WITH EACH OF THE PICTORIAL VIEW.
NOTE: NO: 1 HAS BEEN DONE AS AN EXAMPLE.



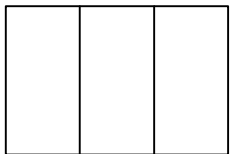
1



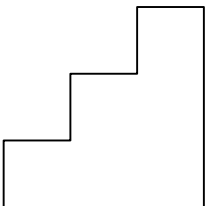
2



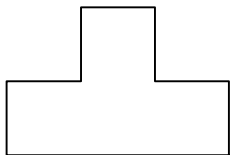
3



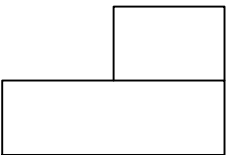
4



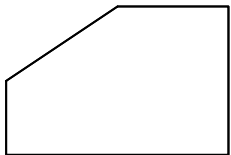
5



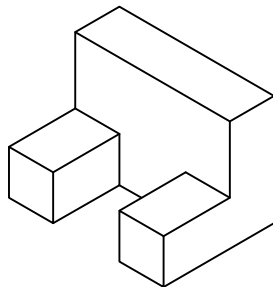
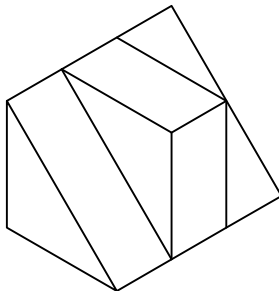
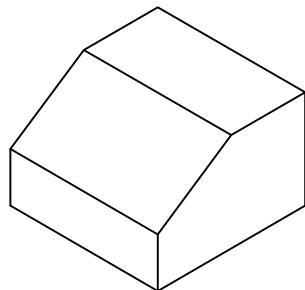
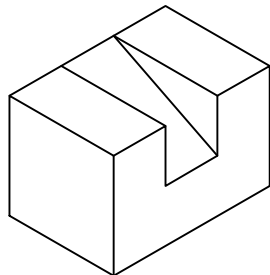
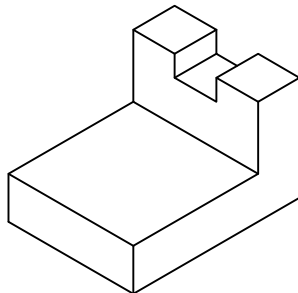
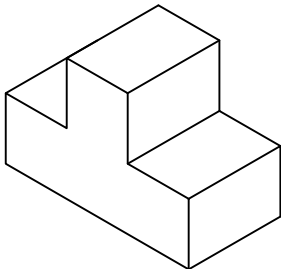
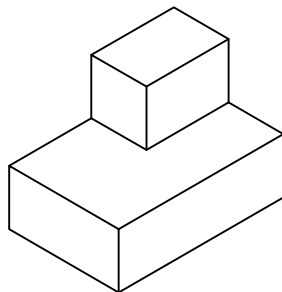
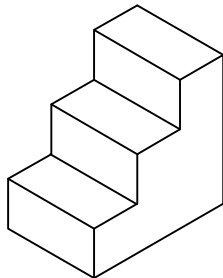
6



7



8



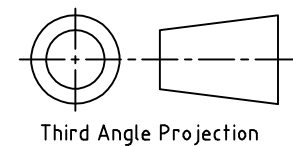
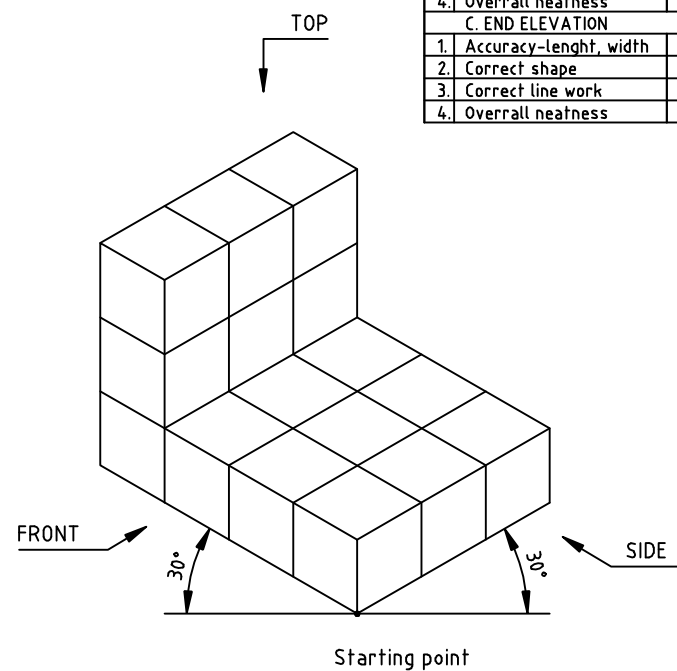
NAME: _____
YEAR: _____

3rd ANGLE ORTHOGRAPHIC PROJECTION

DATE: _____
SHEET NO: 35

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE PROJECTION.
USE THE GIVEN STARTING POINT AS GUIDES.
FILL IN THE TITTLE BLOCK

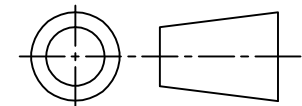
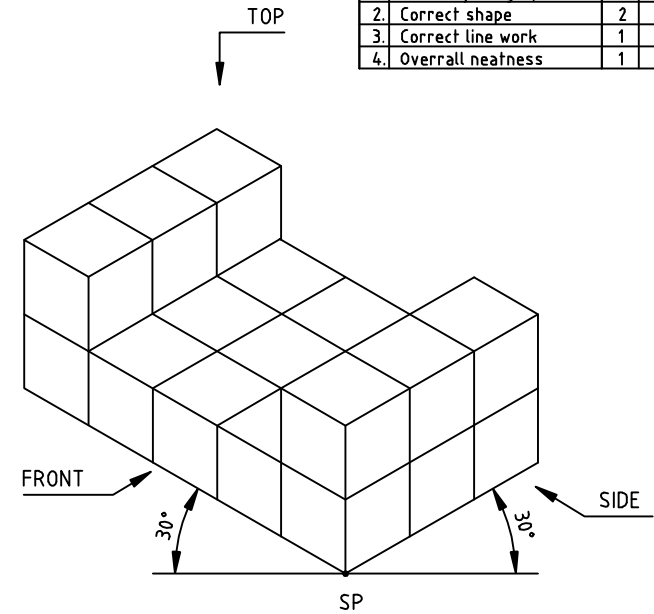
A. PLAN		
1.	Accuracy-length, width	1
2.	Correct shape	2
3.	Correct line work	1
4.	Overall neatness	1
B. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	2
3.	Correct line work	1
4.	Overall neatness	1
C. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	2
3.	Correct line work	1
4.	Overall neatness	1



NAME: _____		DATE: _____
YEAR: _____		SHEET NO: 36

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE PROJECTION.
USE THE GIVEN STARTING POINT AS GUIDES.
FILL IN THE TITTLE BLOCK

A. PLAN		
1.	Accuracy-length, width	1
2.	Correct shape	2
3.	Correct line work	1
4.	Overall neatness	1
B. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	2
3.	Correct line work	1
4.	Overall neatness	1
C. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	2
3.	Correct line work	1
4.	Overall neatness	1



Third Angle Projection

NAME: _____		DATE: _____
YEAR: _____		SHEET NO: 37

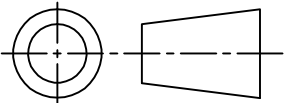
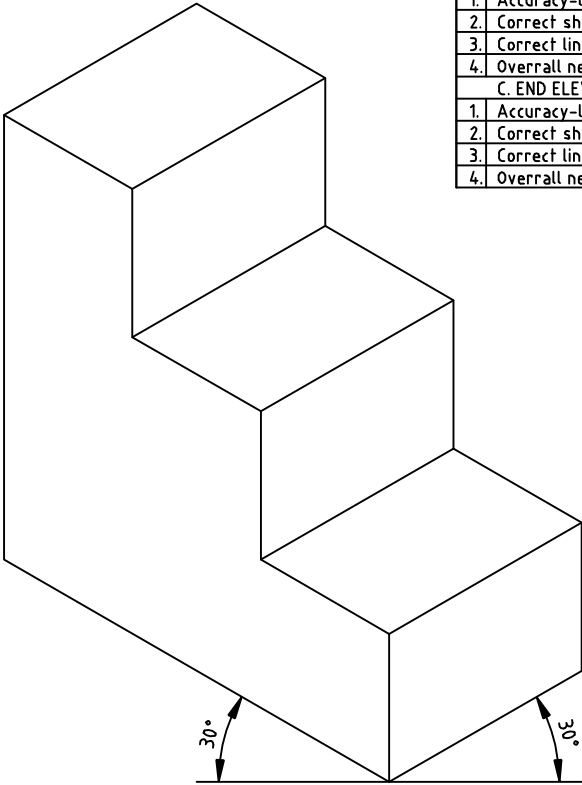
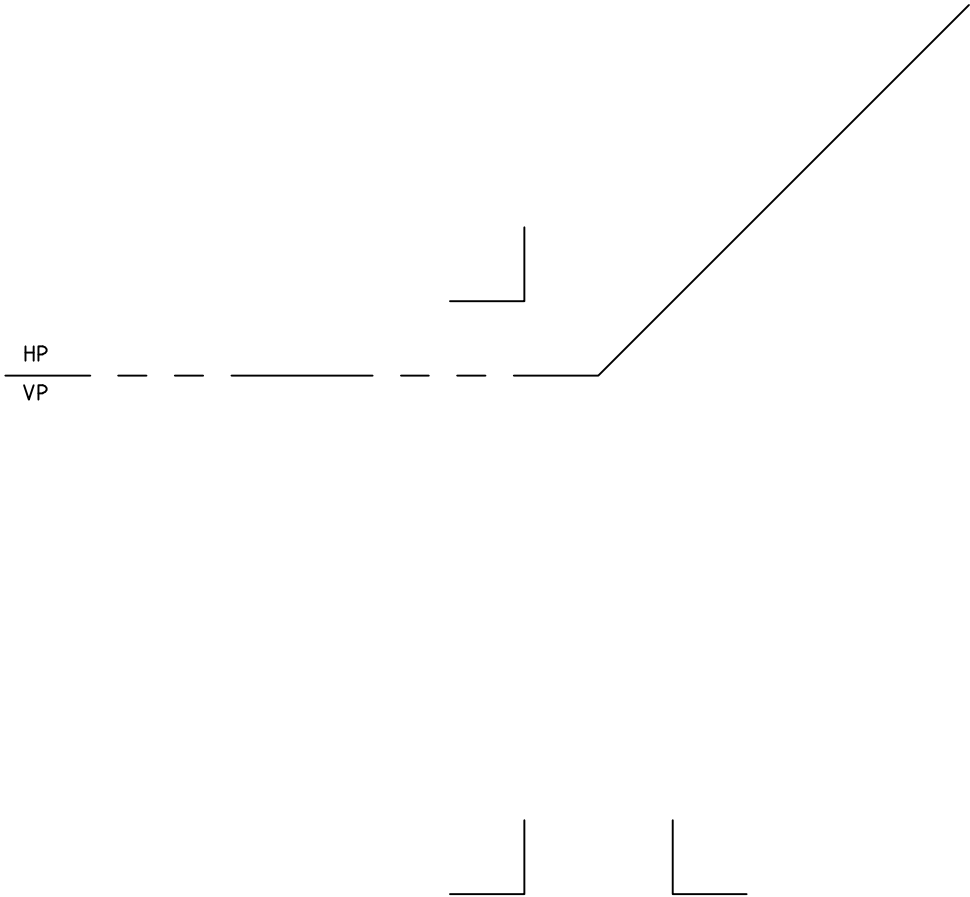
GIVEN: A SIMPLE SHAPED BLOCK DRAWN IN ISOMETRIC

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE ORTHOGRAPHIC PROJECTION USING THE GIVEN STARTING MARKS.

LABEL THE DRAWINGS AS FRONT ELEVATION, END ELEVATION OR PLAN.

NOTE : TAKE MEASUREMENTS FROM THE GIVEN ISOMETRIC DRAWING.

A. PLAN			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
B. ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
C. END ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	



Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 38	

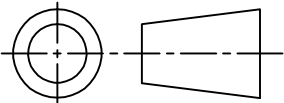
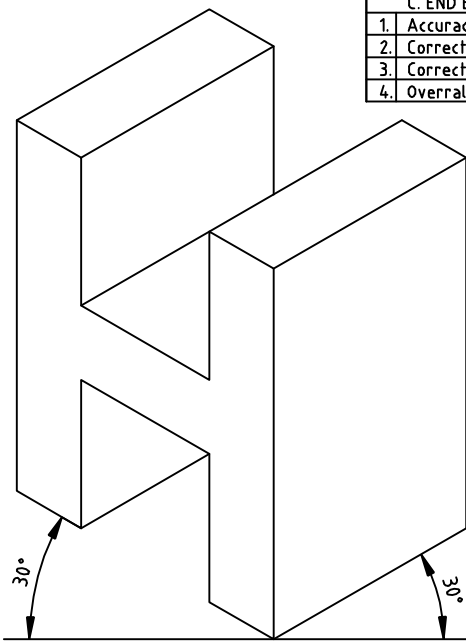
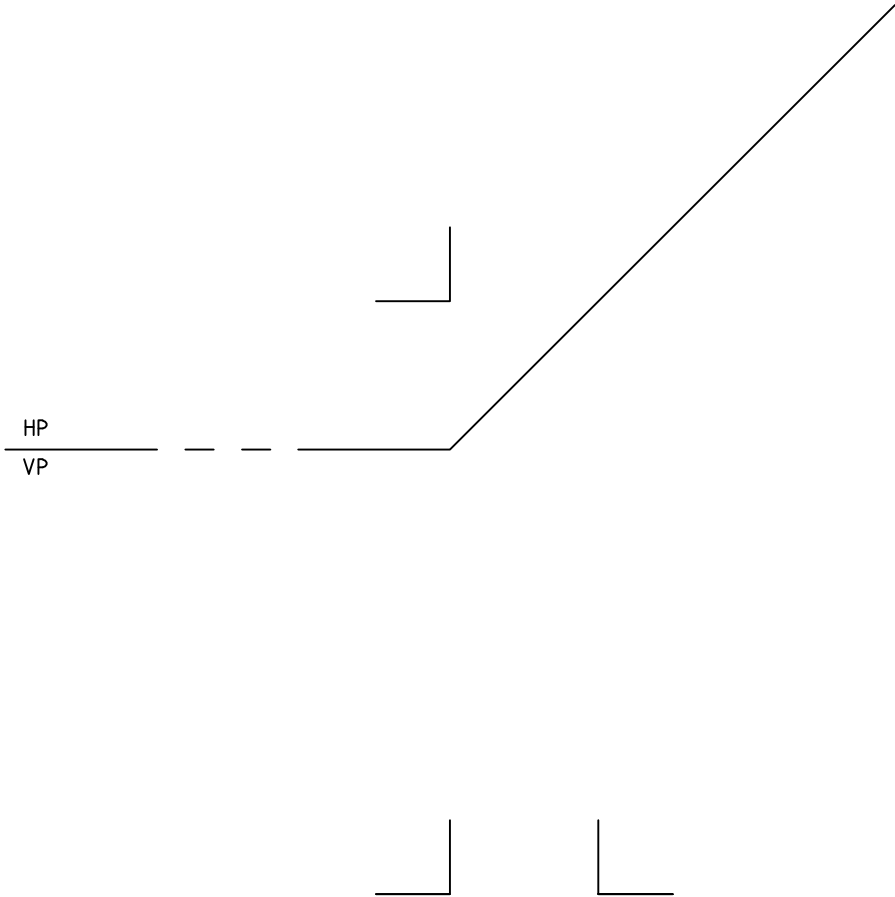
GIVEN: A SIMPLE SHAPED BLOCK DRAWN IN ISOMETRIC

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE ORTHOGRAPHIC PROJECTION USING THE GIVEN STARTING MARKS.

LABEL THE DRAWINGS AS FRONT ELEVATION, END ELEVATION OR PLAN.

NOTE : TAKE MEASUREMENTS FROM THE GIVEN ISOMETRIC DRAWING.

A. PLAN			
1.	Accuracy-lenght, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overrall neatness	1	
B. ELEVATION			
1.	Accuracy-lenght, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overrall neatness	1	
C. END ELEVATION			
1.	Accuracy-lenght, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overrall neatness	1	



Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 39	

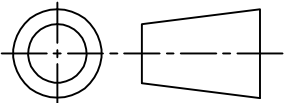
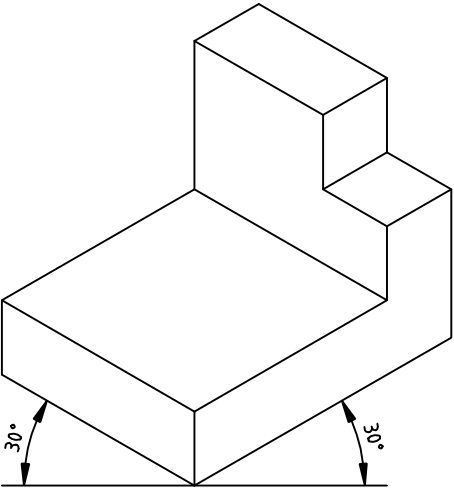
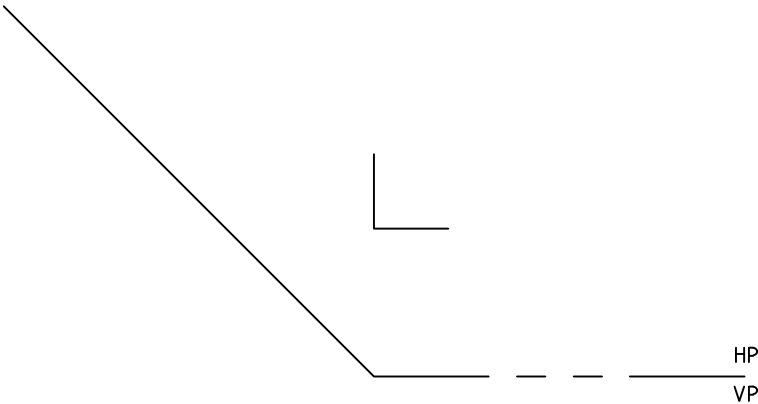
GIVEN: A SIMPLE SHAPED BLOCK DRAWN IN ISOMETRIC

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE ORTHOGRAPHIC PROJECTION USING THE GIVEN STARTING MARKS.

LABEL THE DRAWINGS AS FRONT ELEVATION, END ELEVATION OR PLAN.

NOTE : TAKE MEASUREMENTS FROM THE GIVEN ISOMETRIC DRAWING.

A. PLAN			
1.	Accuracy-lenght, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overrall neatness	1	
B. ELEVATION			
1.	Accuracy-lenght, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overrall neatness	1	
C. END ELEVATION			
1.	Accuracy-lenght, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overrall neatness	1	



Third Angle Projection

NAME: _____

YEAR: _____

DATE: _____

SHEET NO: 40

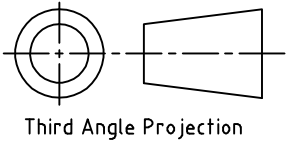
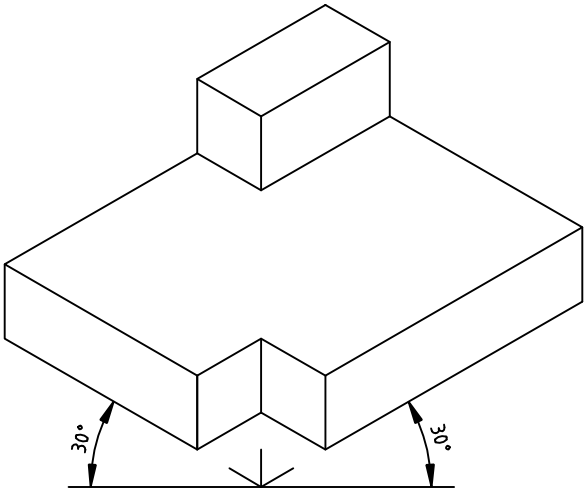
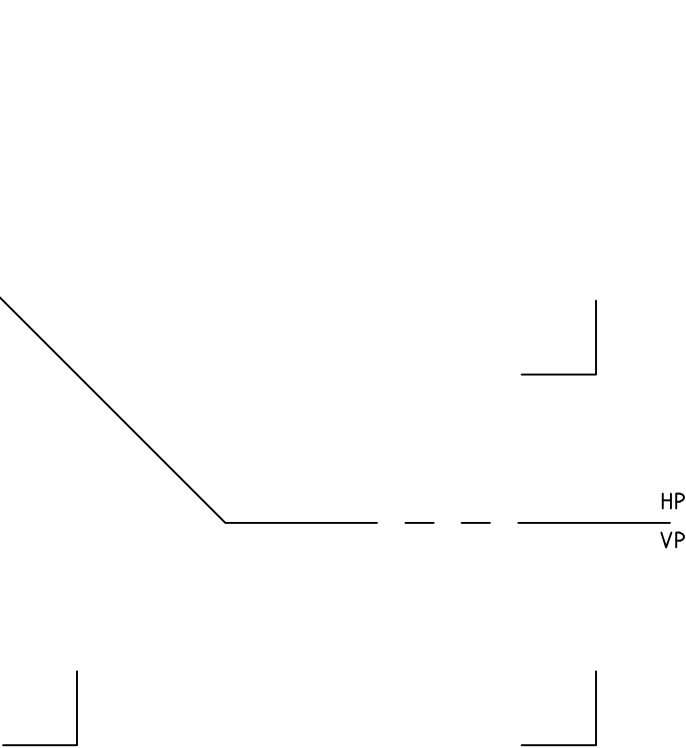
GIVEN: A SIMPLE SHAPED BLOCK DRAWN IN ISOMETRIC

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE ORTHOGRAPHIC PROJECTION USING THE GIVEN STARTING MARKS.

LABEL THE DRAWINGS AS FRONT ELEVATION, END ELEVATION OR PLAN.

NOTE : TAKE MEASUREMENTS FROM THE GIVEN ISOMETRIC DRAWING.

A. PLAN			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
B. ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
C. END ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	



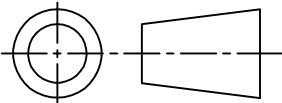
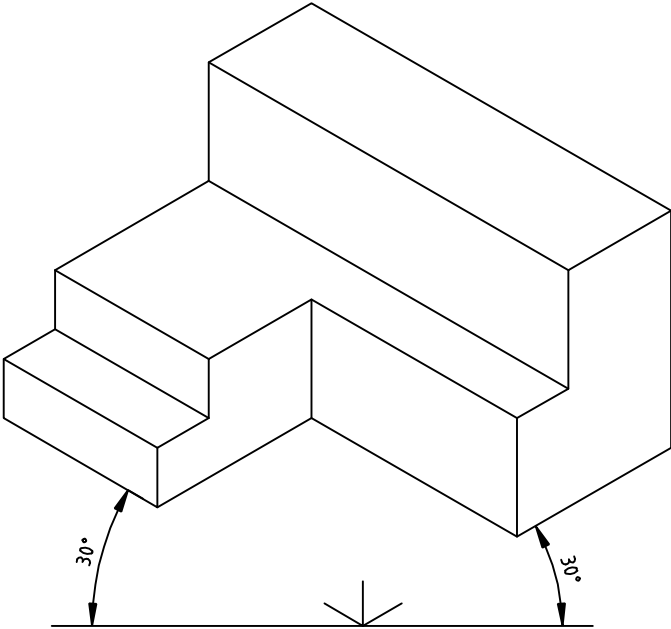
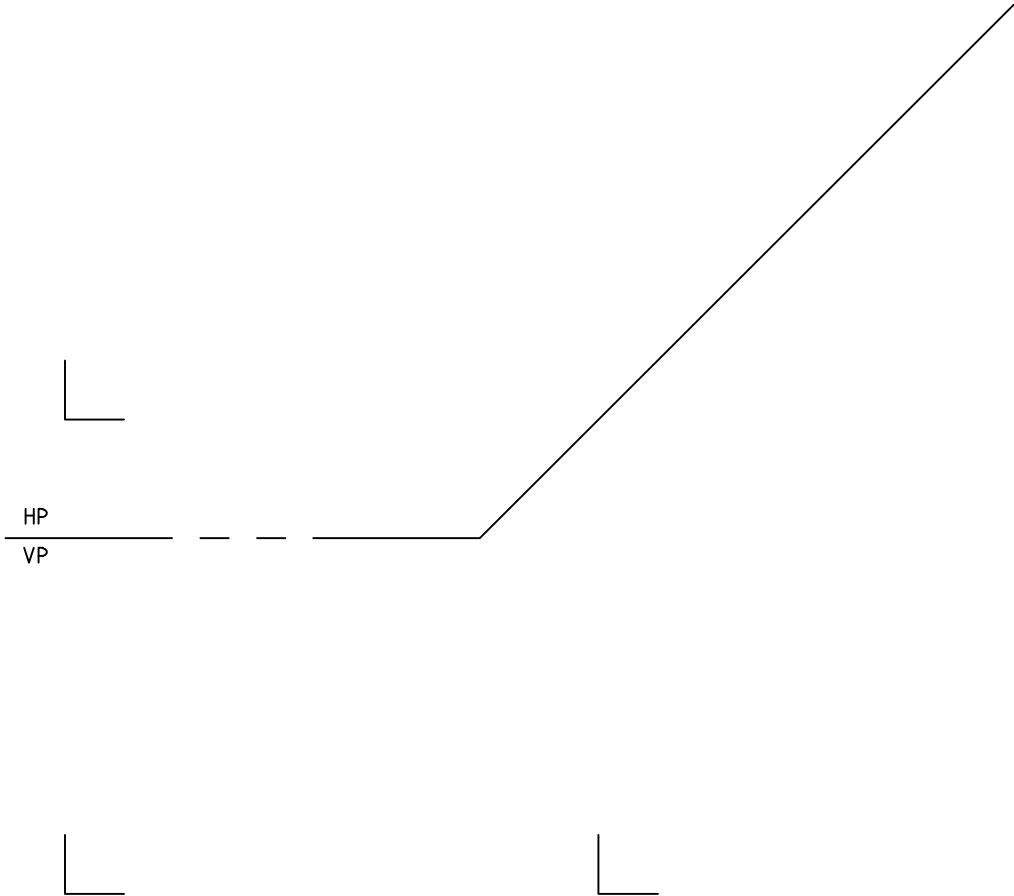
NAME:		DATE:	
YEAR:		SHEET NO: 41	

GIVEN: A SIMPLE SHAPED BLOCK DRAWN IN ISOMETRIC

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE ORTHOGRAPHIC PROJECTION USING THE GIVEN STARTING MARKS.

NOTE : TAKE MEASUREMENTS FROM THE GIVEN ISOMETRIC DRAWING.

A. PLAN			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
B. ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
C. END ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	



Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 42	

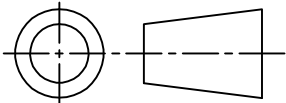
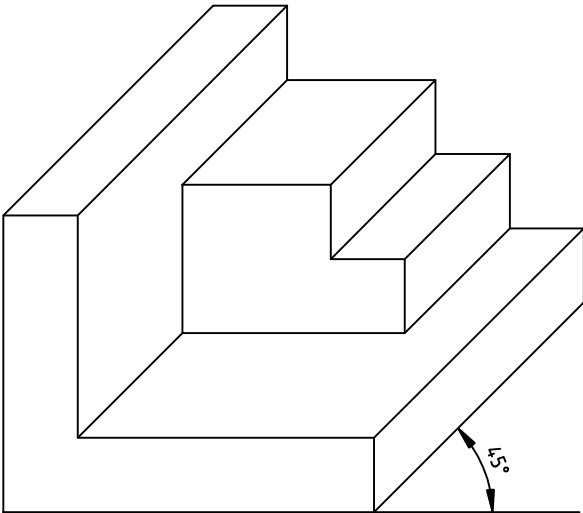
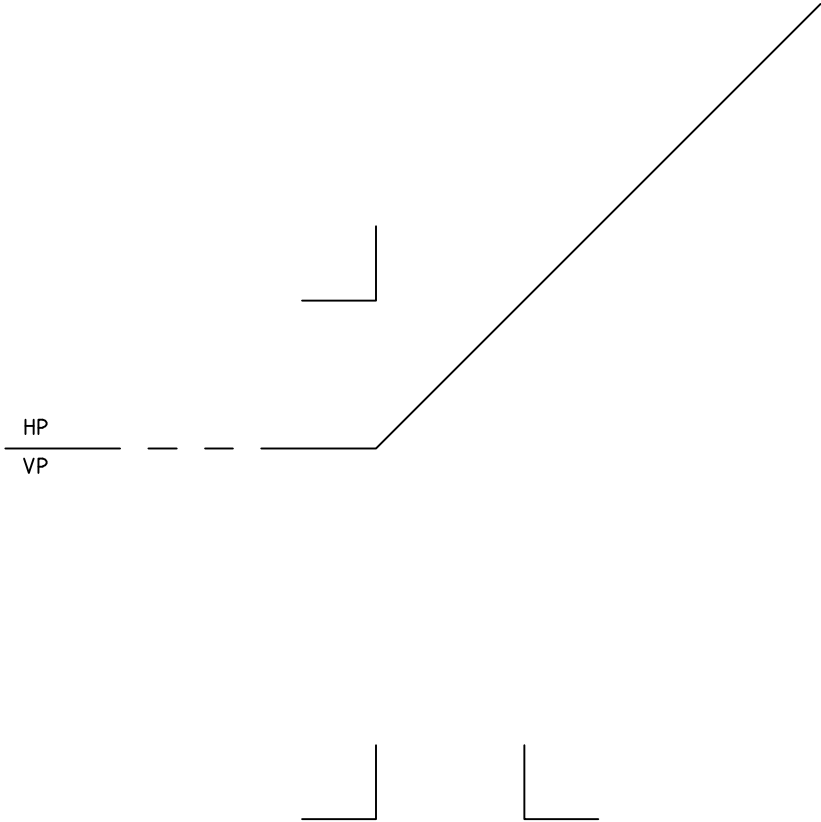
GIVEN: A SIMPLE SHAPED BLOCK DRAWN IN OBLIQUE

REQUIRED: DRAW THE SHAPED BLOCK IN 3rd ANGLE ORTHOGRAPHIC PROJECTION USING THE GIVEN STARTING MARKS.

LABEL THE DRAWINGS AS FRONT ELEVATION, END ELEVATION OR PLAN.

NOTE : TAKE MEASUREMENTS FROM THE GIVEN OBLIQUE DRAWING.

A. PLAN			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
B. ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	
C. END ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	2	
3.	Correct line work	1	
4.	Overall neatness	1	

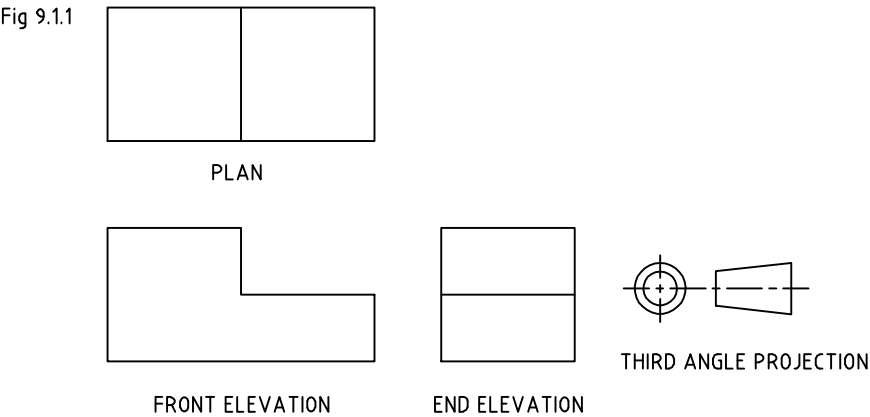


Third Angle Projection

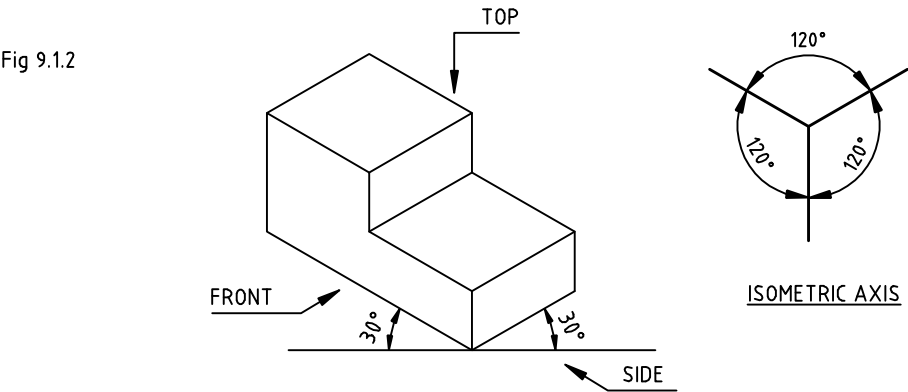
NAME:		DATE:	
YEAR:		SHEET NO: 43	

ISOMETRIC DRAWING IS THE MOST NATURAL LOOKING PICTORIAL DRAWING AND IT IS MUCH EASY TO PRODUCE. THE SHAPES ARE DRAWN INSIDE THE ISOMETRIC BOX.

THE ORTHOGRAPHIC PROJECTION FIGURE 9.1.1 OF A SIMPLE SHAPED BLOCK IS USED IN UNDERSTANDING AND DRAWING AN ISOMETRIC VIEW.



THE ISOMETRIC VIEW FIGURE 9.1.2 IS DRAWN FROM THE ORTHOGRAPHIC VIEWS OF THE SIMPLE SHAPED BLOCK, FIGURE 9.1.1.



ISOMETRIC DRAWING HELPS IN UNDERSTANDING THE SHAPE OF A SHAPED BLOCK.

ISOMETRIC DRAWINGS ARE DRAWN USING THE 30°-60°-90° SET SQUARE.

THE LENGTH AND WIDTH ON THE ISOMETRIC DRAWING IS DRAWN AT 30° AND THE HEIGHT IS DRAWN VERTICALLY AT 90° TO THE HORIZONTAL.

THE ISOMETRIC BOX FIGURE 9.1.3 IS USED AS A GUIDE TO DRAW THE ACTUAL SHAPE. IT IS DRAWN USING CONSTRUCTION LINES.

THE ISOMETRIC BOX FIGURE 9.1.4 ILLUSTRATES THE ORDER IN WHICH THE 9 LINES ARE DRAWN. NOTE THAT THREE OF THE LINES REPRESENT THE LENGTH, THREE REPRESENT THE WIDTH AND THREE REPRESENT THE HEIGHT OF THE SHAPED BLOCK.

Fig 9.1.3

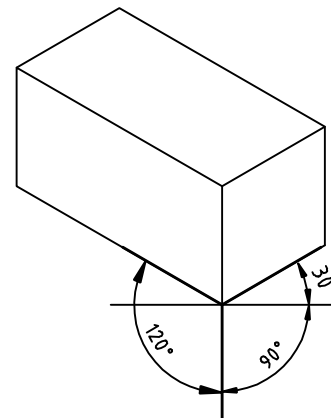
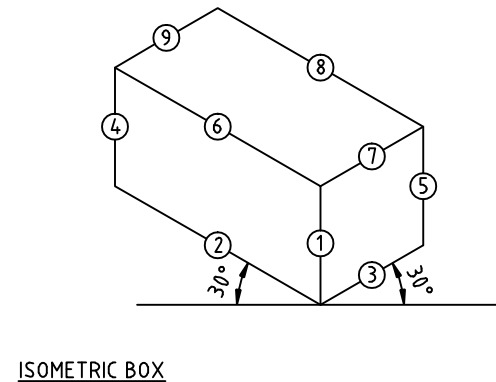
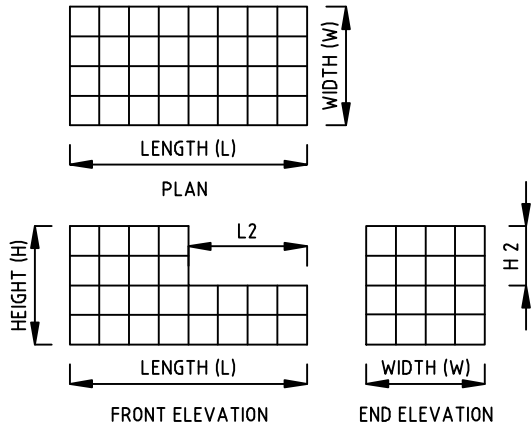


Fig 9.1.4



CONVERTING ORTHOGRAPHIC DRAWING TO ISOMETRIC DRAWING

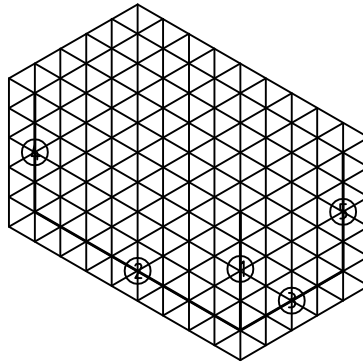
THE FRONT ELEVATION OF AN ORTHOGRAPHIC DRAWING SHOWS THE LENGTH & HEIGHT OF THE SHAPED BLOCK, THE END ELEVATION SHOWS THE WIDTH & HEIGHT AND THE PLAN SHOWS THE LENGTH & WIDTH.



CONSTRUCTION OF THE ISOMETRIC BOX

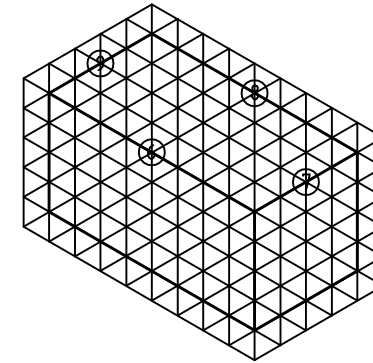
FIRST THE HEIGHT LINE IS DRAWN VERTICALLY. THEN THE LENGTH AND THE WIDTH OF THE ISOMETRIC BOX ARE DRAWN.

THE OTHER 2 HEIGHT LINES ARE ADDED AS WELL.



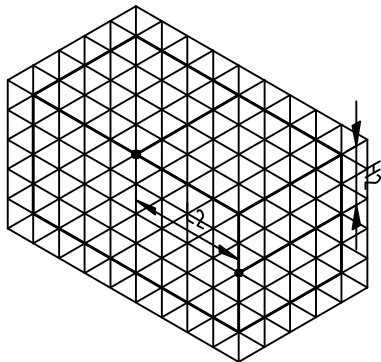
CONSTRUCTION OF THE ISOMETRIC BOX

THE OTHER LENGTH AND WIDTH LINES ARE THEN DRAWN TO COMPLETE THE ISOMETRIC BOX.



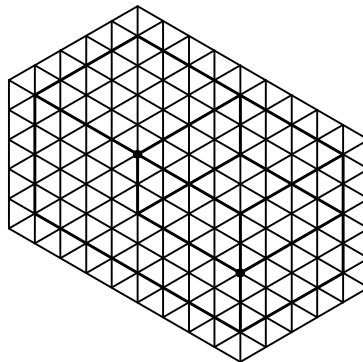
COMPLETING THE SHAPE

TO GET THE SHAPE OF THE OBJECT THE LENGTH 2 (L2) AND HEIGHT 2 (H2) ARE TAKEN FROM THE FRONT AND END ELEVATION AND DRAWN BY COUNTING THE NUMBER OF GRIDS.



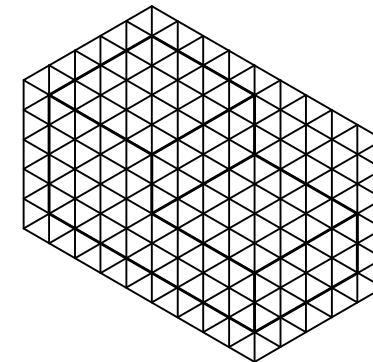
COMPLETING THE SHAPE

THE REMAINING LINES ARE DRAWN TO COMPLETE THE SHAPE.



COMPLETING THE SHAPE

THE SHAPE IS COMPLETED BY DRAWING THE OUTLINES.



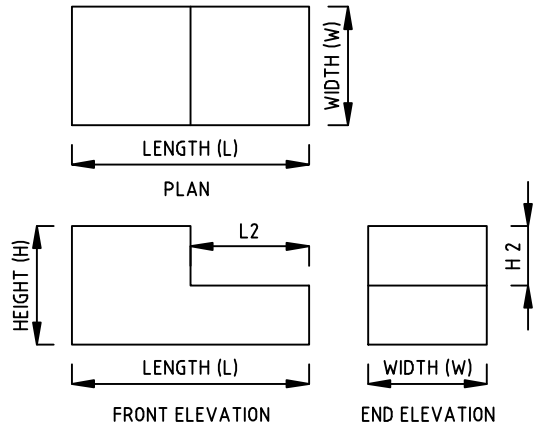
NAME: _____
YEAR: _____

ISOMETRIC DRAWING

DATE: _____
SHEET NO: 45

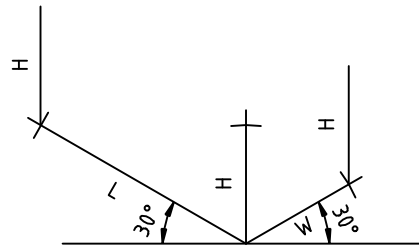
CONVERTING ORTHOGRAPHIC DRAWING TO ISOMETRIC DRAWING

THE FRONT ELEVATION OF AN ORTHOGRAPHIC DRAWING SHOWS THE LENGTH & HEIGHT OF THE SHAPED BLOCK, THE END ELEVATION SHOWS THE WIDTH & HEIGHT AND THE PLAN SHOWS THE LENGTH & WIDTH.



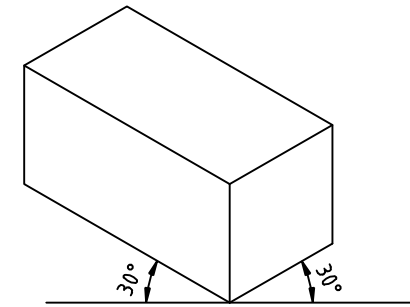
1) CONSTRUCTION OF THE ISOMETRIC BOX

DRAW THE ISOMETRIC AXIS BY DRAWING THE HEIGHT AT 90° , THE LENGTH AND WIDTH AT 30° TO THE BASE LINE. MARK THE LENGTH, WIDTH AND HEIGHT.



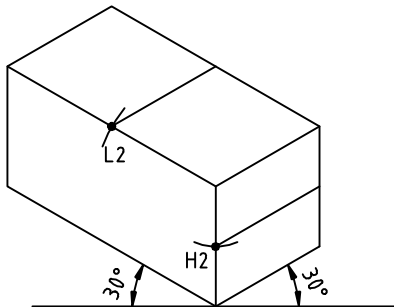
2) CONSTRUCTION OF THE ISOMETRIC BOX

COMPLETE THE ISOMETRIC BOX BY DRAWING LINES PARALLEL TO THE LENGTH, WIDTH AND HEIGHT.



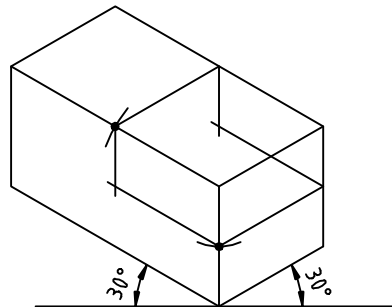
3) COMPLETING THE SHAPE

USING A COMPASS, MEASURE LENGTH (L_2) AND HEIGHT (H_2) AND TRANSFER TO THE ISOMETRIC BOX. USING THE $30/60^\circ$ SET SQUARE DRAW 30° LINES TO TOUCH THE OTHER SIDE.



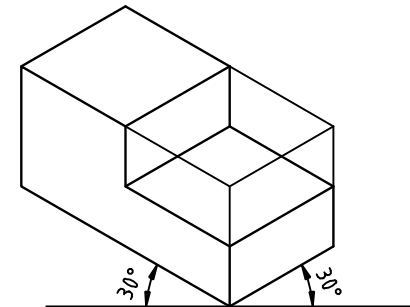
4) COMPLETING THE SHAPE

USING THE $30/60^\circ$ SET SQUARE DRAW THE REMAINING LINES PARALLEL TO THE LENGTH AND WIDTH SLIGHTLY LONGER THAN L_2 AND H_2 AND COMPLETE THE SHAPE.



5) COMPLETING THE SHAPE

THE SHAPE IS COMPLETED BY DRAWING THE OUTLINES.



NAME: _____
YEAR: _____

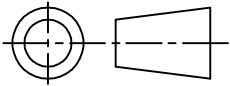
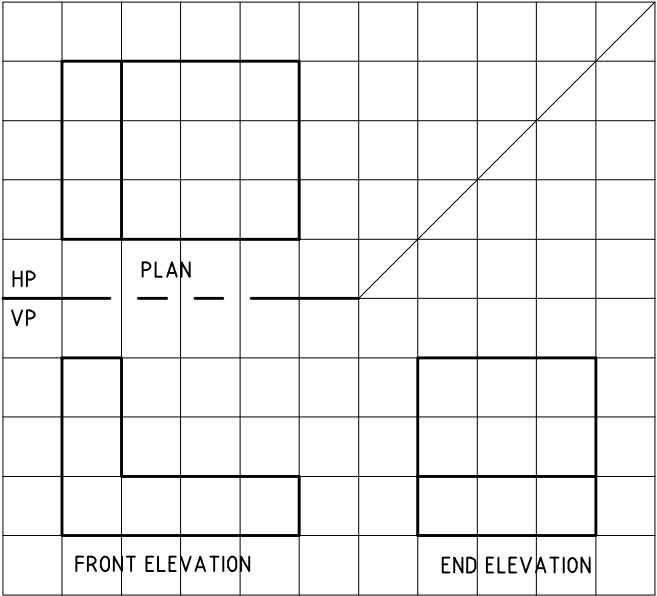
ISOMETRIC DRAWING

DATE: _____
SHEET NO: 46

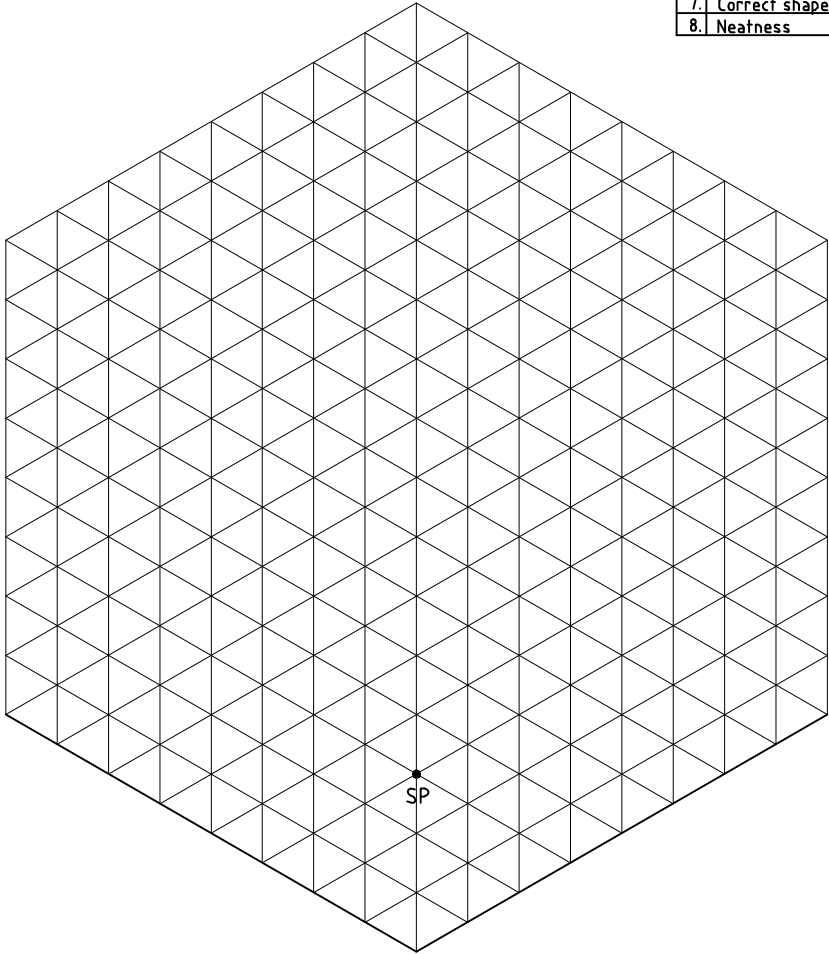
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND AN ISOMETRIC GRID

REQUIRED: DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK ON THE ISOMETRIC GRID.

1.	Guideines shown	1	
2.	Correctprinting	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection

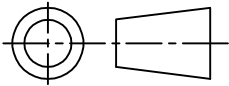
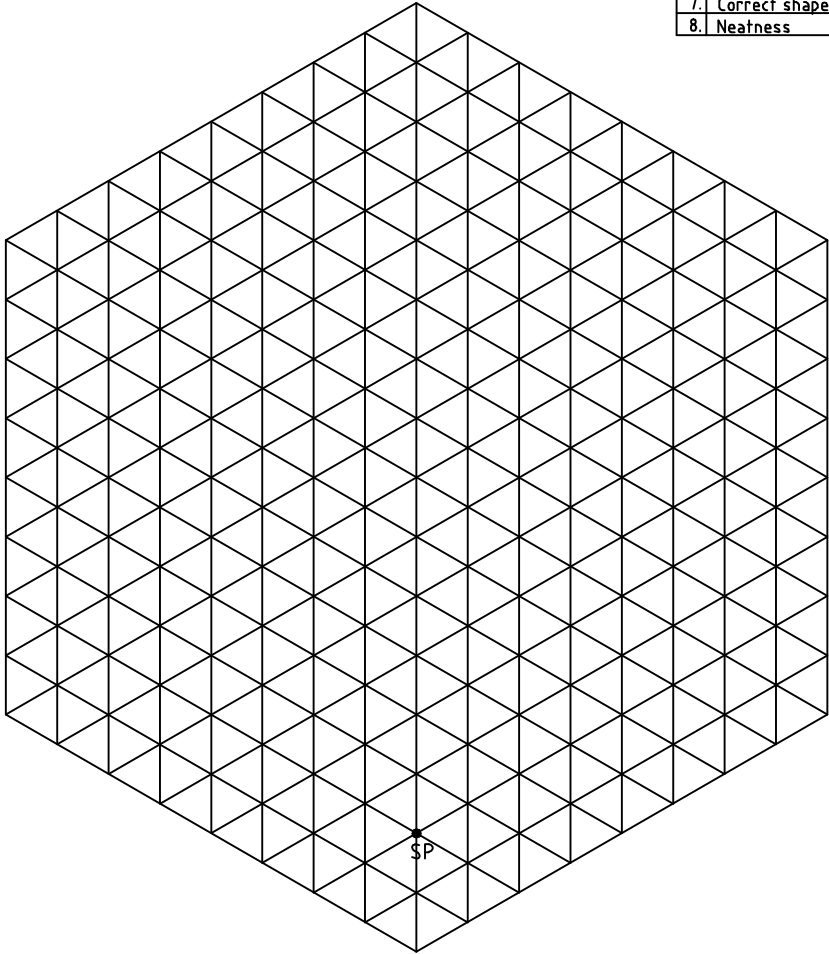
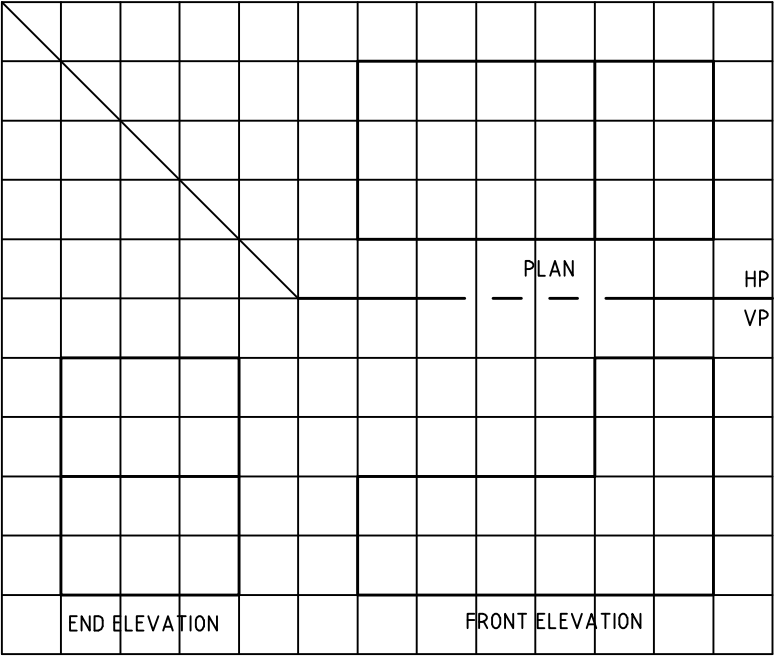


NAME:		DATE:	
YEAR:		SHEET NO: 47	

GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND AN ISOMETRIC GRID

REQUIRED: DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK ON THE ISOMETRIC GRID.
(USE THE GIVEN STARTING POINT)

1.	Guideines shown	1	
2.	Correctprinting	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



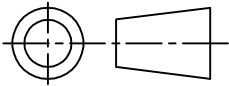
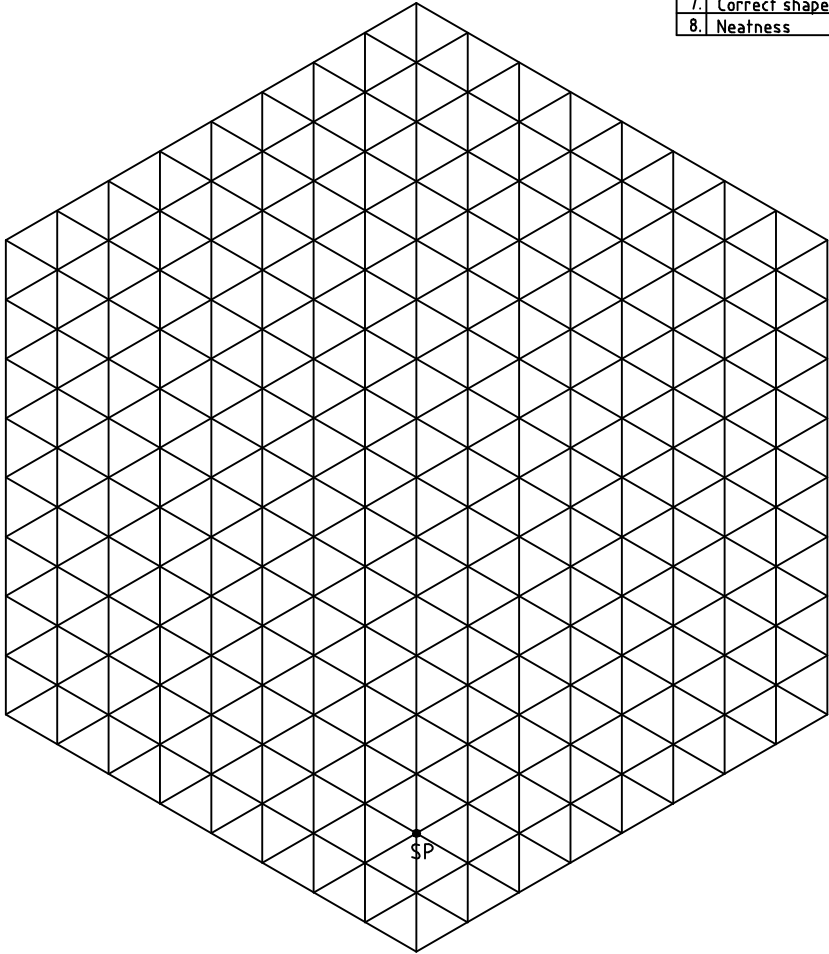
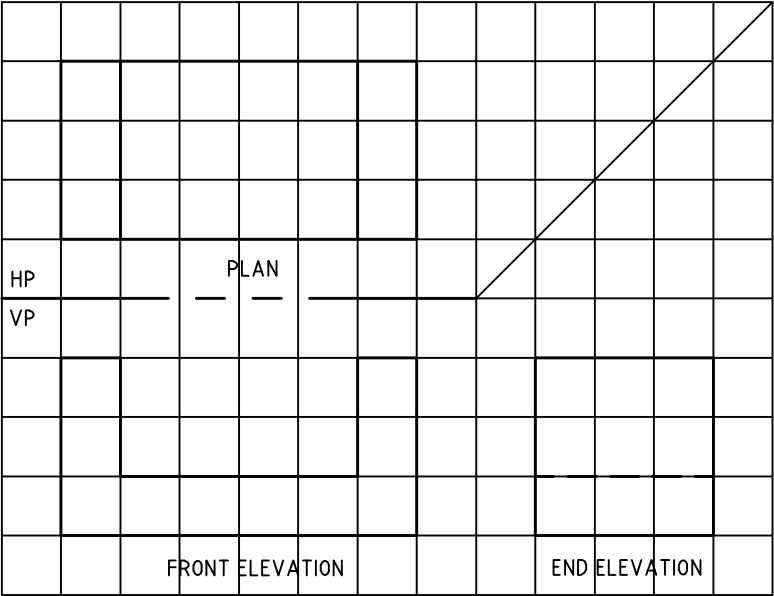
Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 48	

GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND AN ISOMETRIC GRID

REQUIRED: DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK ON THE ISOMETRIC GRID.
(USE THE GIVEN STARTING POINT)

1.	Guideines shown	1	
2.	Correctprinting	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 49	

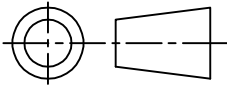
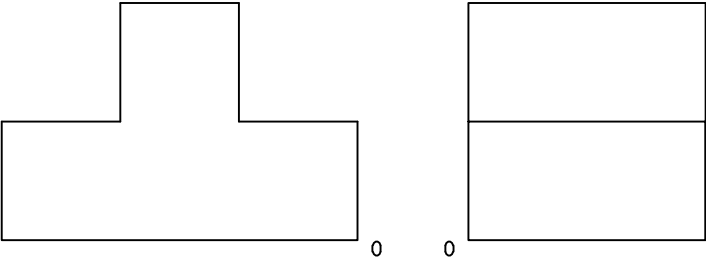
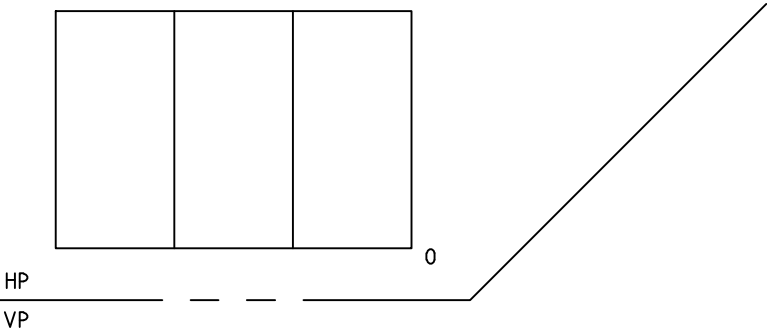
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION, ISOMETRIC AXIS AND THE STARTING POINT 'O'.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

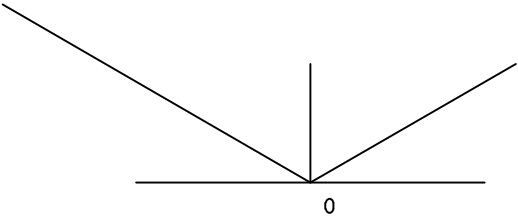
DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK USING THE GIVEN ISOMETRIC AXIS AND THE STARTING POINT 'O'.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEW

1.	Guideines shown	1	
2.	Correctprinting	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection



NAME: _____
YEAR: _____

DATE: _____
SHEET NO: 50

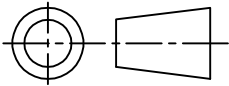
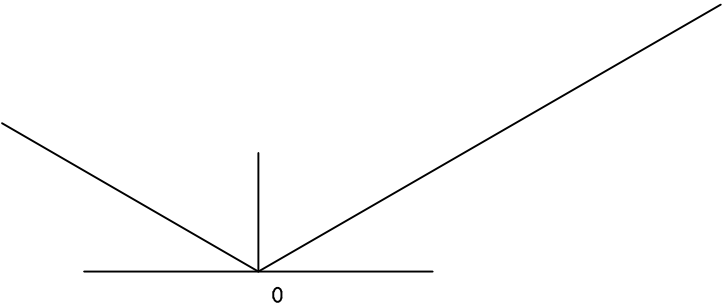
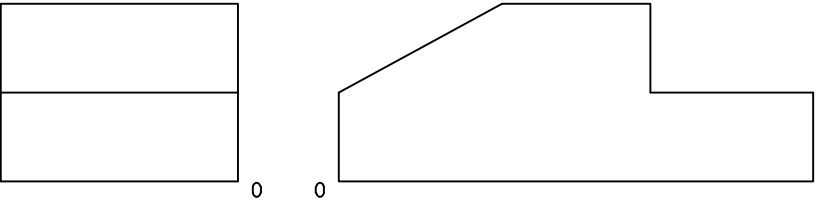
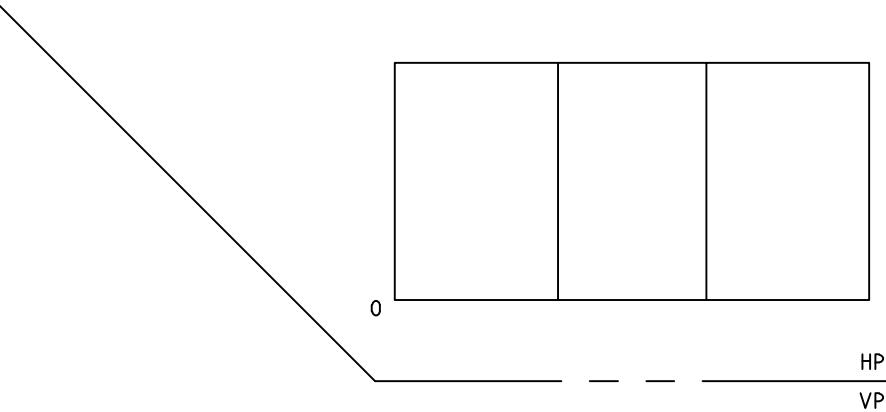
1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	

GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION, ISOMETRIC AXIS AND THE STARTING POINT 'O'.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK USING THE GIVEN ISOMETRIC AXIS AND THE STARTING POINT 'O'.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEW



Third Angle Projection

NAME: _____

YEAR: _____

DATE: _____

SHEET NO: 51 _____

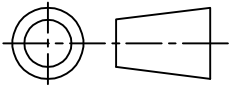
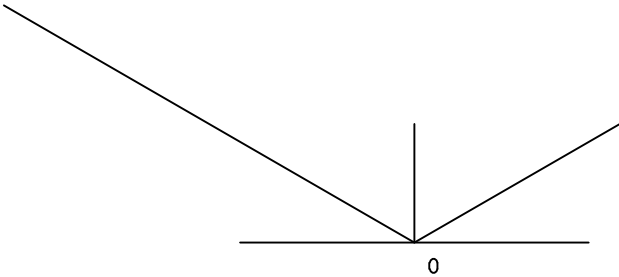
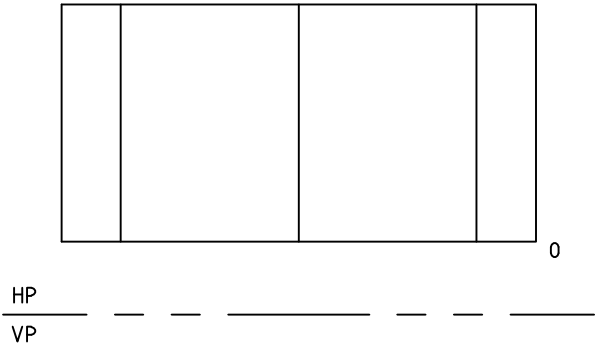
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION, ISOMETRIC AXIS AND THE STARTING POINT 'O'.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK USING THE GIVEN ISOMETRIC AXIS AND THE STARTING POINT 'O'.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEW

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 52	

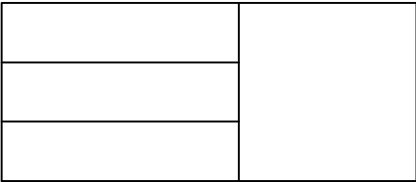
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND THE ISOMETRIC AXIS.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

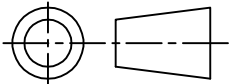
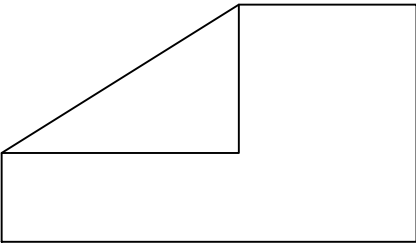
DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK USING THE GIVEN ISOMETRIC AXIS.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEW

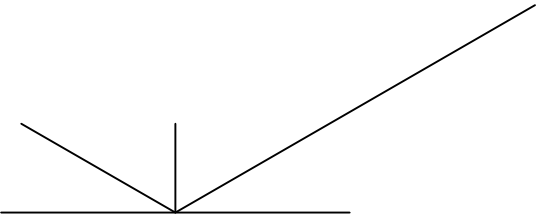
1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



HP
VP



Third Angle Projection



NAME:		DATE:	
YEAR:		SHEET NO: 53	

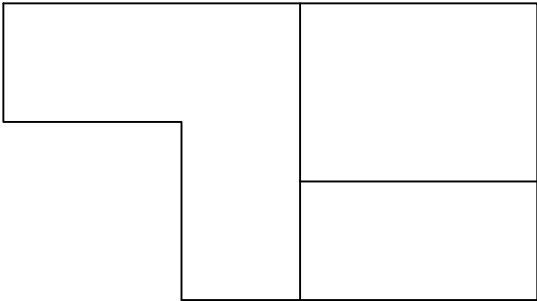
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND THE ISOMETRIC AXIS.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

DRAW THE ISOMETRIC VIEW OF THE SHAPED BLOCK USING THE GIVEN ISOMETRIC AXIS.

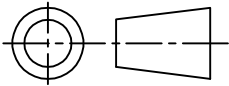
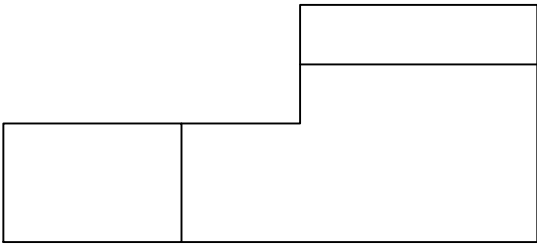
NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEW

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	

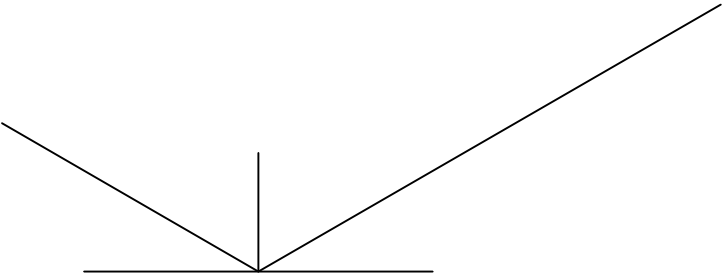


HP

VP



Third Angle Projection

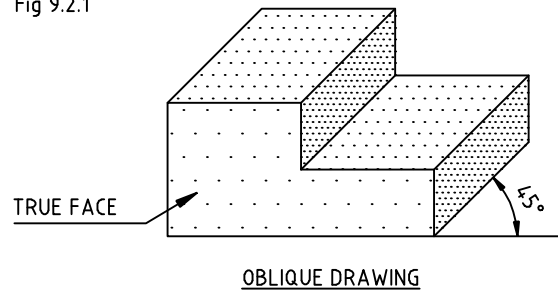


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OBLIQUE DRAWING IS ANOTHER FORM OF NATURAL LOOKING DRAWING PICTORIAL DRAWINGS. THE FRONT VIEW OF AN OBLIQUE DRAWING IS THE SAME AS THE FRONT ELEVATION IN THE ORTHOGRAPHIC DRAWING. BUT THE END ELEVATION IS DRAWN ON RECEDING AXIS WHICH AT AN ANGLE OF 45° .

A SHAPED BLOCK FIGURE 9.2.1 IS USED IN UNDERSTANDING AND DRAWING OBLIQUE VIEWS.

Fig 9.2.1



TYPES OF OBLIQUE DRAWING

CAVALIER OBLIQUE DRAWING, FIGURE 9.2.2 THE RECEDING AXIS DIMENSIONS ARE DRAWN TO FULL SIZE. CABINET OBLIQUE DRAWING FIGURE 9.2.3 IS THE RECEDING AXIS DIMENSIONS ARE DRAWN TO HALF FULL SIZE

Fig 9.2.2

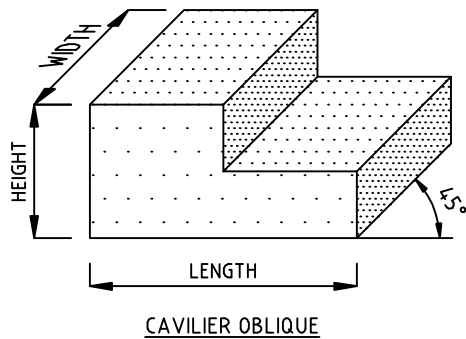
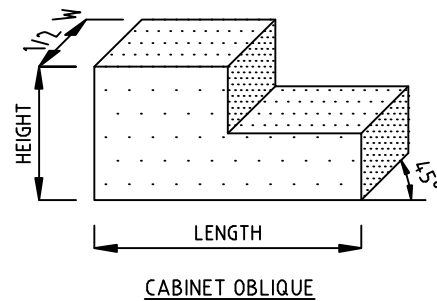


Fig 9.2.3



THE LENGTH IS DRAWN PARALLEL, THE HEIGHT VERTICALLY OR AT 90° AND THE WIDTH ON THE RECEDING LINES AT AN ANGLE OF 45° TO THE HORIZONTAL.

THE 45° - 90° SET SQUARE IS USED FOR DRAWING OBLIQUE VIEWS.

THE OBLIQUE BOX FIGURE 9.2.4 IS USED AS A GUIDE TO DRAW THE ACTUAL SHAPE. IT IS DRAWN USING CONSTRUCTION LINES.

THE OBLIQUE BOX FIGURE 9.2.5 ILLUSTRATES THE ORDER IN WHICH THE 9 LINES ARE DRAWN. NOTE THAT THREE OF THE LINES ARE REPRESENTING THE LENGTH, THREE REPRESENTING THE WIDTH AND THREE REPRESENTING THE HEIGHT OF THE SHAPED BLOCK.

Fig 9.2.4

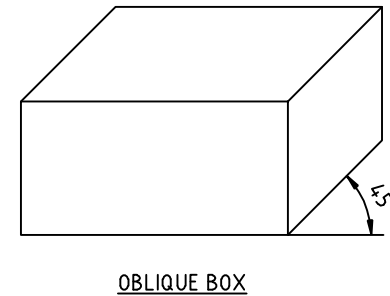
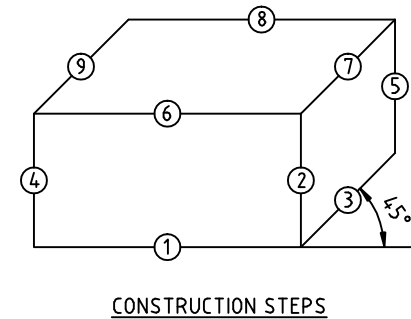


Fig 9.2.5



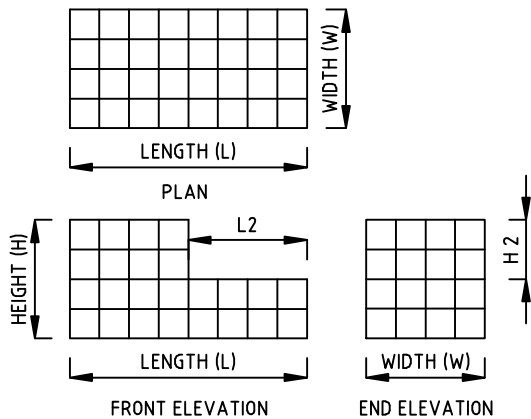
NAME:	
YEAR:	

OBLIQUE DRAWING	
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DATE:	
SHEET NO: 55	

CONVERTING ORTHOGRAPHIC DRAWING TO OBLIQUE DRAWING

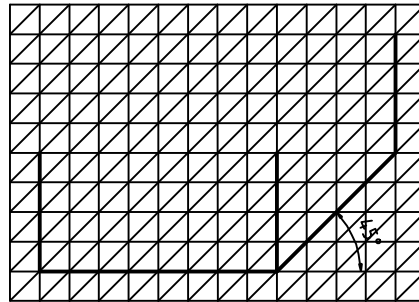
THE FRONT ELEVATION OF AN ORTHOGRAPHIC DRAWING SHOWS THE LENGTH & HEIGHT OF THE SHAPED BLOCK, THE END ELEVATION SHOWS THE WIDTH & HEIGHT AND THE PLAN SHOWS THE LENGTH & WIDTH.



1) CONSTRUCTION OF THE OBLIQUE BOX

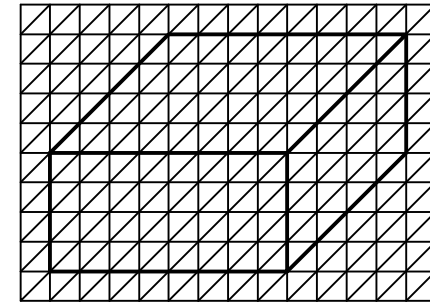
FIRST THE HEIGHT LINE IS DRAWN VERTICALLY, THEN THE LENGTH HORIZONTALLY AND THE WIDTH RECEDING AT AN ANGLE OF 45° TO THE HORIZONTAL. THE OTHER 2 HEIGHT LINES ARE ADDED AS WELL.

(ALL DIMENSIONS ARE TAKEN FROM ORTHOGRAPHIC DRAWING)



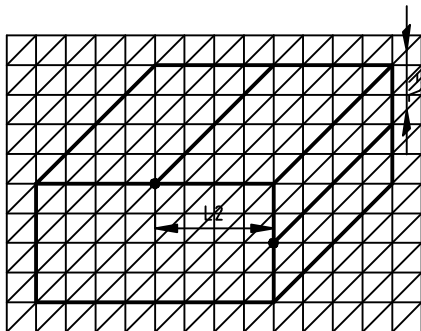
2) CONSTRUCTION OF THE OBLIQUE BOX

THE OTHER LENGTH AND WIDTH LINES ARE THEN DRAWN TO COMPLETE THE OBLIQUE BOX.



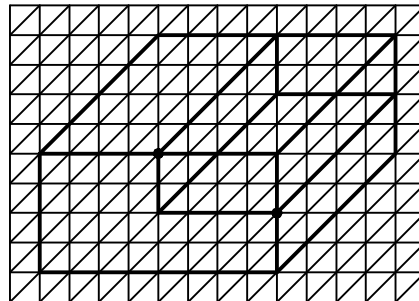
3) COMPLETING THE SHAPE

TO GET THE SHAPE OF THE OBJECT THE LENGTH 2 (L2) AND HEIGHT 2 (H2) ARE TAKEN FROM THE FRONT AND END ELEVATION AND DRAWN BY COUNTING THE NUMBER OF GRIDS.



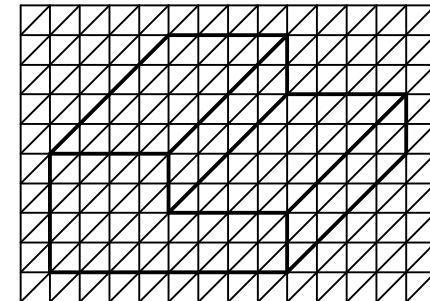
4) COMPLETING THE SHAPE

THE REMAINING LINES ARE DRAWN TO COMPLETE THE SHAPE.



5) COMPLETING THE SHAPE

THE SHAPE IS COMPLETED BY DRAWING THE OUTLINES.



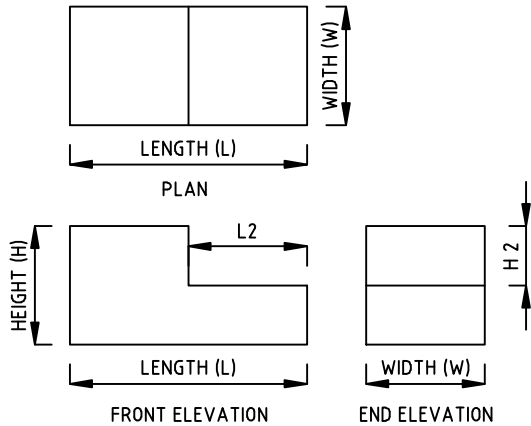
NAME: _____
YEAR: _____

OBLIQUE DRAWING

DATE: _____
SHEET NO: 56

CONVERTING ORTHOGRAPHIC DRAWING TO OBLIQUE DRAWING

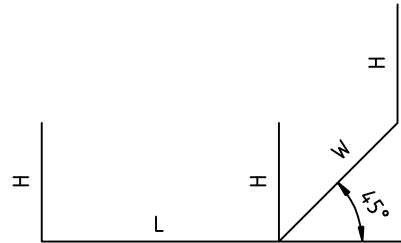
THE FRONT ELEVATION OF AN ORTHOGRAPHIC DRAWING SHOWS THE LENGTH & HEIGHT OF THE SHAPED BLOCK, THE END ELEVATION SHOWS THE WIDTH & HEIGHT AND THE PLAN SHOWS THE LENGTH & WIDTH.



1) CONSTRUCTION OF THE OBLIQUE BOX

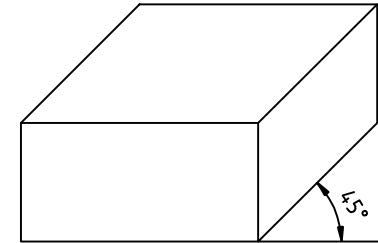
FIRST THE HEIGHT LINE IS DRAWN VERTICALLY, THEN THE LENGTH HORIZONTALLY AND THE WIDTH RECEDING AT AN ANGLE OF 45° TO THE HORIZONTAL.

THE OTHER 2 HEIGHT LINES ARE ADDED AS WELL.



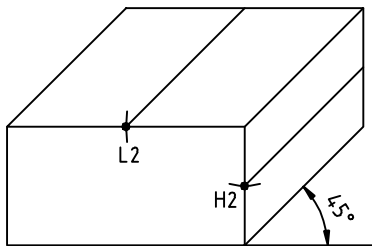
2) CONSTRUCTION OF THE OBLIQUE BOX

COMPLETE THE OBLIQUE BOX BY DRAWING LINES PARALLEL TO THE LENGTH, WIDTH AND HEIGHT.



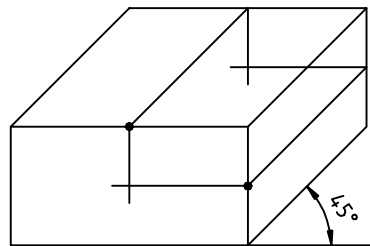
3) COMPLETING THE SHAPE

USING A COMPASS, MEASURE LENGTH (L2) AND HEIGHT (H2) AND TRANSFER TO THE OBLIQUE BOX. USING THE 45° SET SQUARE DRAW 45° LINES TO TOUCH THE OTHER SIDE.



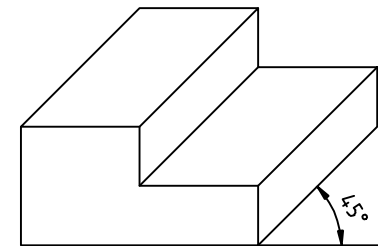
4) COMPLETING THE SHAPE

USING THE SET SQUARES DRAW THE REMAINING LINES PARALLEL TO THE LENGTH AND WIDTH SLIGHTLY LONGER THAN L2 AND H2 AND COMPLETE THE SHAPE.



5) COMPLETING THE SHAPE

THE SHAPE IS COMPLETED BY DRAWING THE OUTLINES.



NAME: _____
YEAR: _____

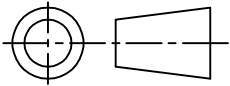
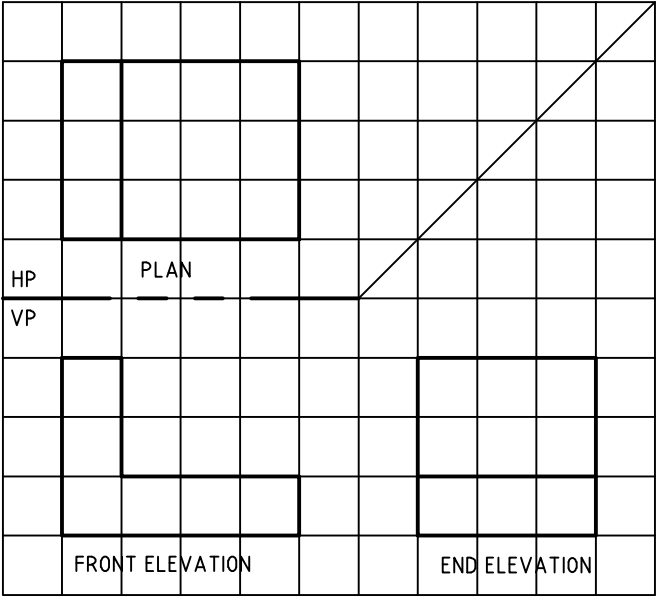
OBLIQUE DRAWING

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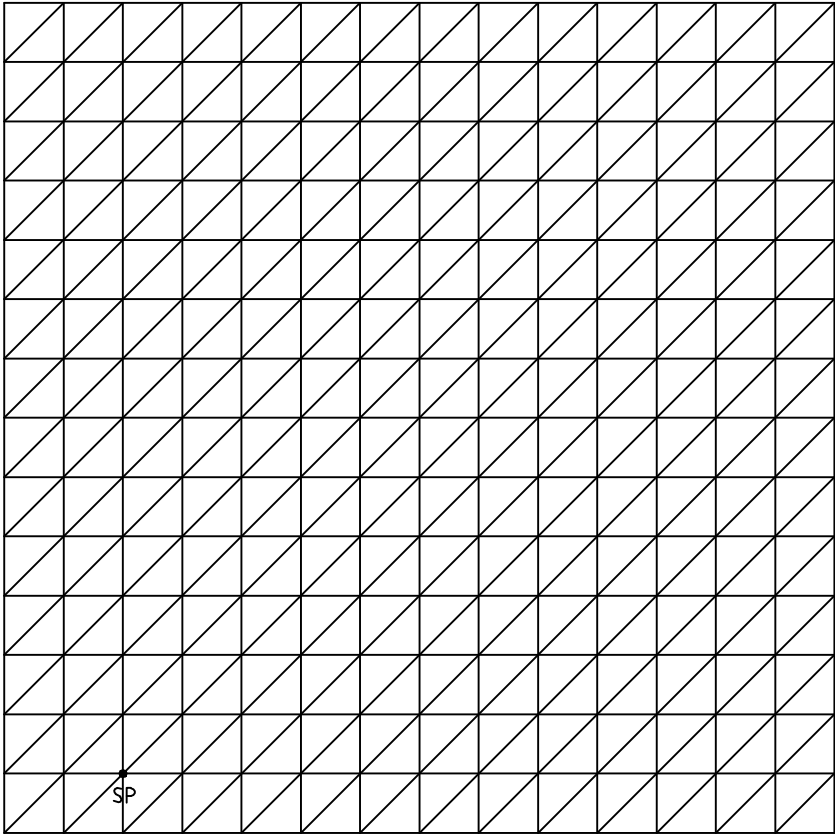
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND AN OBLIQUE GRID

REQUIRED: DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK ON THE OBLIQUE GRID.
(USE GIVEN THE STARTING POINT)

1.	Guideines shown	1	
2.	Correctprinting	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection

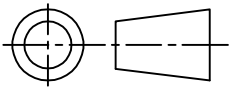
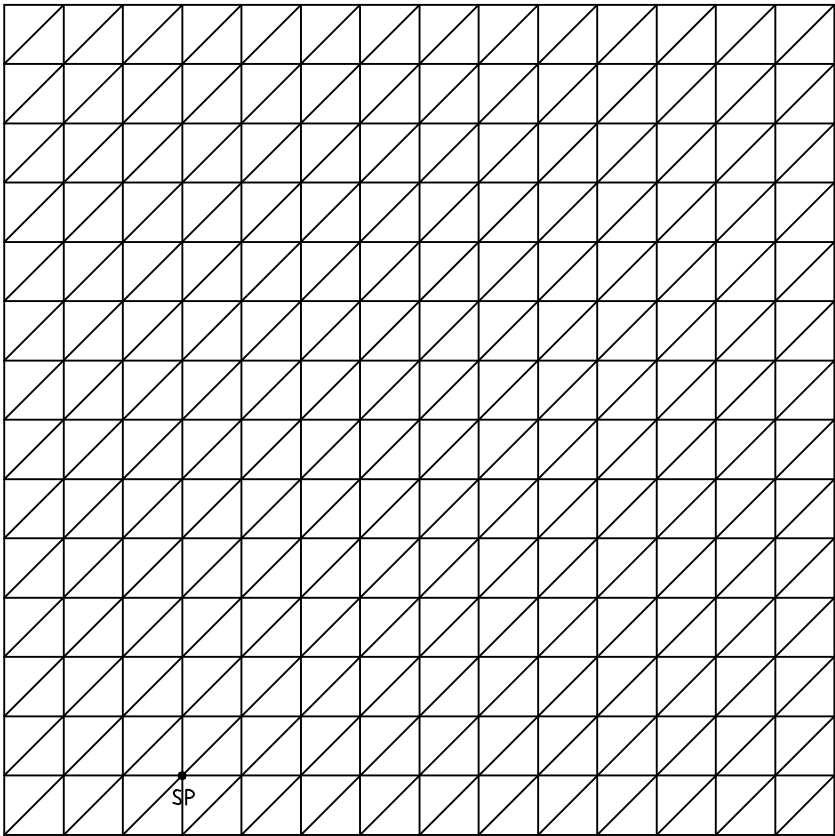
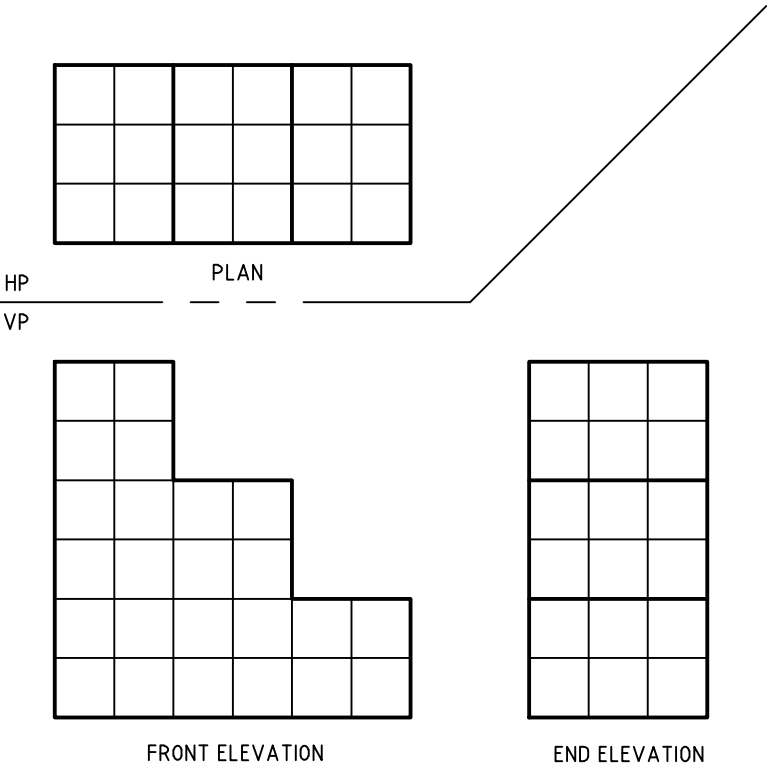


NAME:		DATE:	
YEAR:		SHEET NO: 58	

GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND AN OBLIQUE GRID

REQUIRED: DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK ON THE OBLIQUE GRID.

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



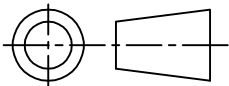
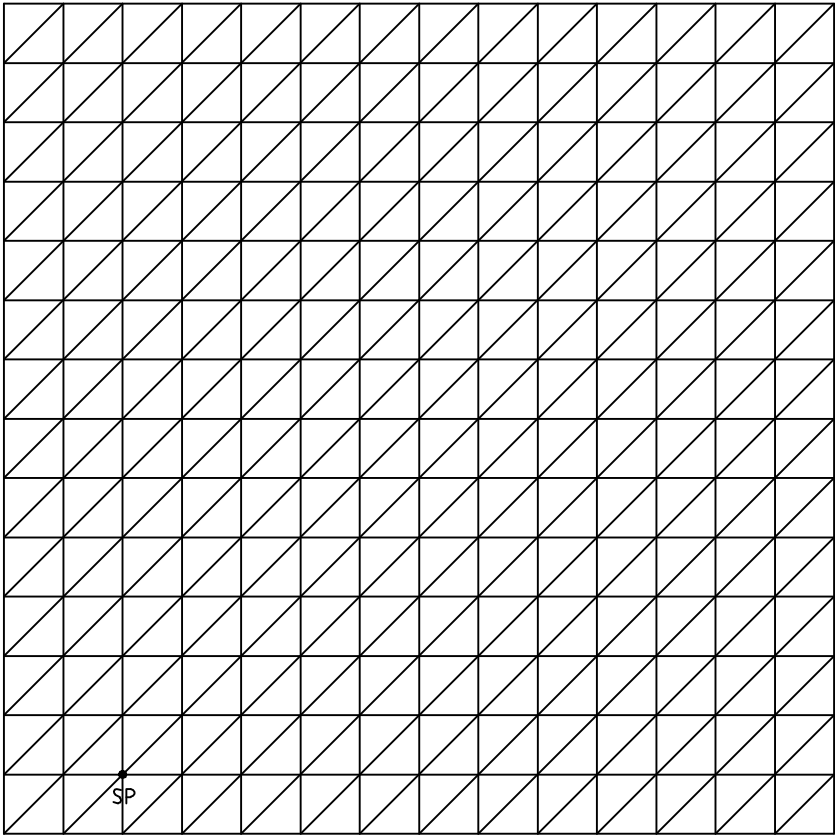
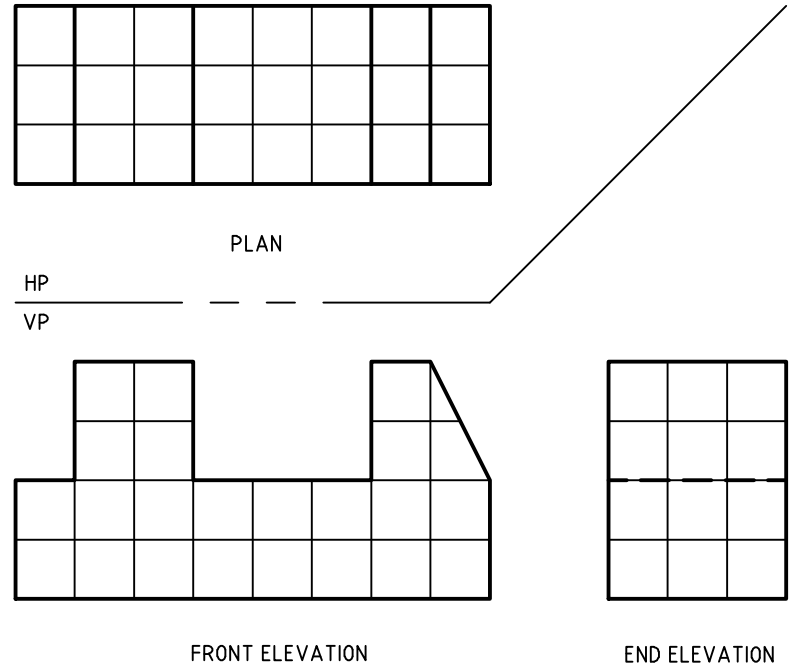
Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 59	

GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND AN OBLIQUE GRID

REQUIRED: DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK ON THE OBLIQUE GRID.

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 60	

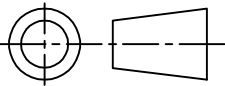
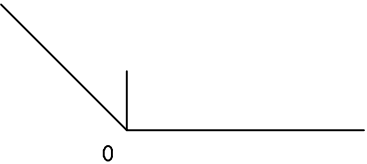
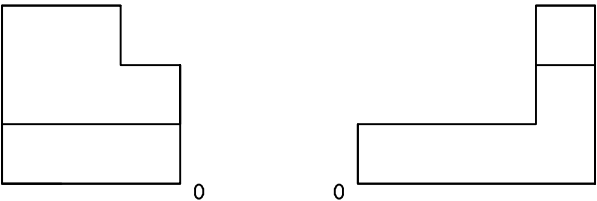
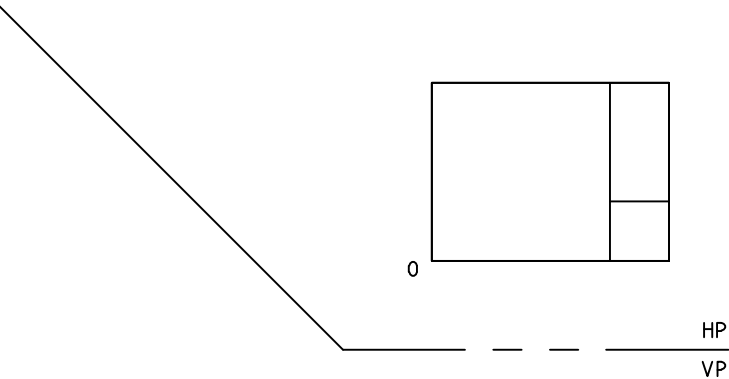
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION, OBLIQUE AXIS AND THE STARTING POINT 'O'.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK USING THE GIVEN OBLIQUE AXIS AND THE STARTING POINT 'O'.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEWS

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection

NAME: _____

YEAR: _____

DATE: _____

SHEET NO: 61

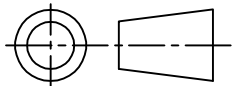
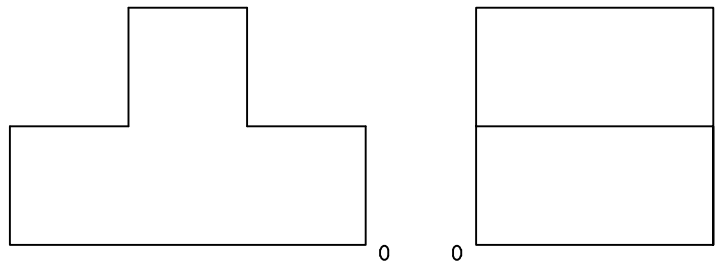
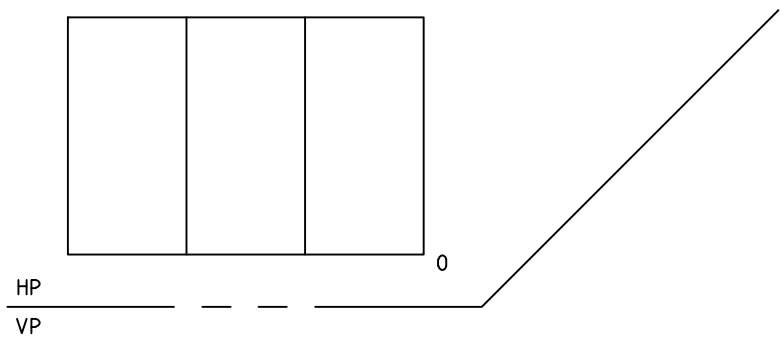
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION, OBLIQUE AXIS AND THE STARTING POINT 'O'.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

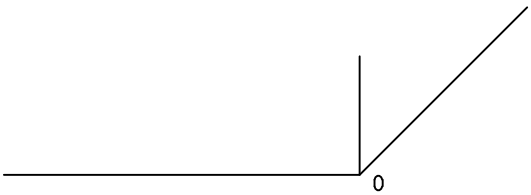
DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK USING THE GIVEN OBLIQUE AXIS AND THE STARTING POINT 'O'.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEWS

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection



NAME: _____

YEAR: _____

DATE: _____

SHEET NO: 62

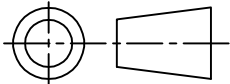
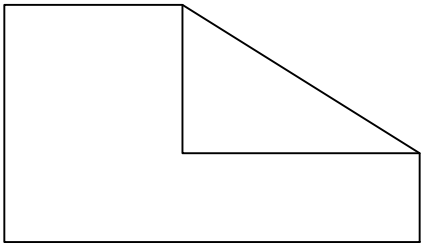
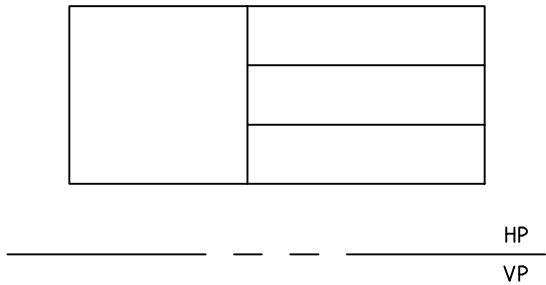
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND THE OBLIQUE AXIS.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK USING THE GIVEN OBLIQUE AXIS.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEWS

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection



NAME: _____
YEAR: _____

DATE: _____
SHEET NO: 63 _____

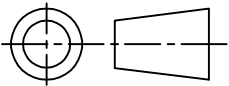
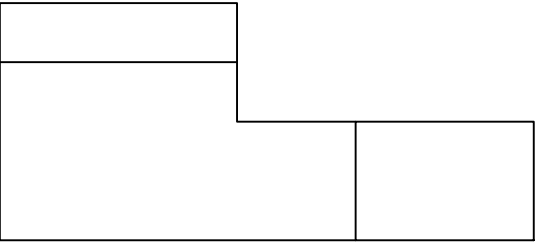
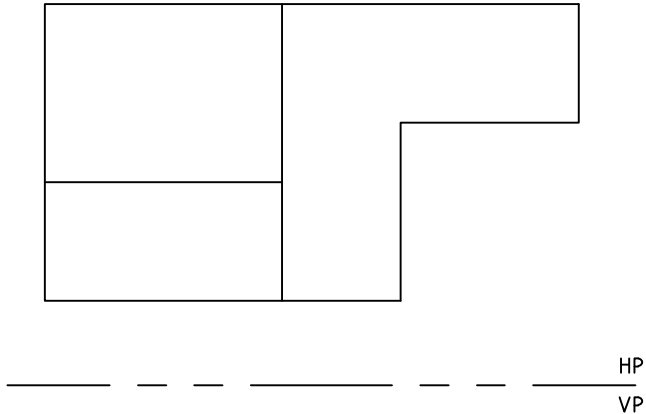
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND THE OBLIQUE AXIS.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK USING THE GIVEN OBLIQUE AXIS.

NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEWS

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



Third Angle Projection

NAME:		DATE:	
YEAR:		SHEET NO: 64	

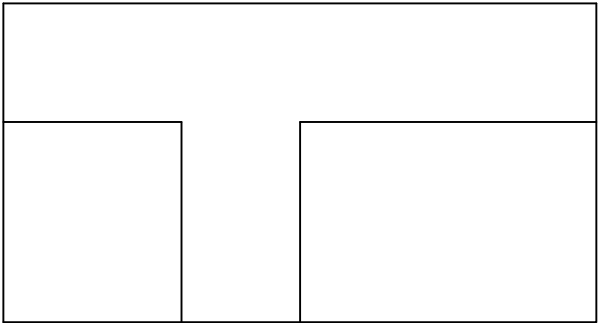
GIVEN: A SHAPED BLOCK DRAWN IN 3rd ANGLE PROJECTION AND THE OBLIQUE AXIS.

REQUIRED: LABEL THE GIVEN ORTHOGRAPHIC VIEWS

DRAW THE OBLIQUE VIEW OF THE SHAPED BLOCK USING THE GIVEN OBLIQUE AXIS.

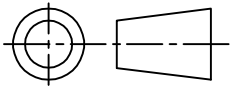
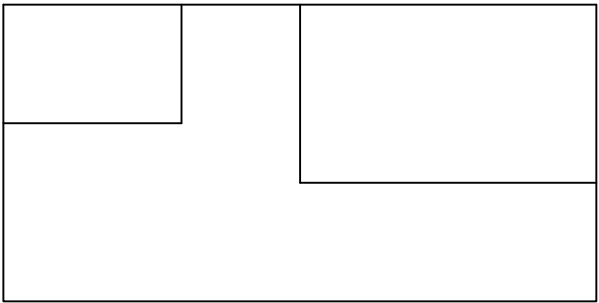
NOTE: TAKE MEASUREMENTS DIRECTLY FROM THE ORTHOGRAPHIC VIEWS

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



HP

VP



Third Angle Projection

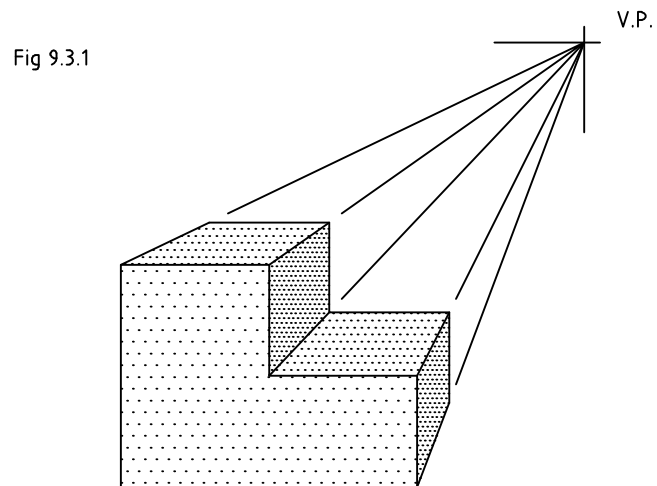


NAME: _____	_____	DATE: _____
YEAR: _____	_____	SHEET NO: 65

ONE POINT PERSPECTIVE DRAWING IS A FORM OF NATURAL LOOKING PICTORIAL DRAWING. OBJECTS DRAWN IN PERSPECTIVE APPEAR TO BE GETTING SMALLER THE FURTHER THEY ARE WHEN VIEWED. LINES THAT ARE PARALLEL TO EACH OTHER APPEAR TO CONVERGE TO A POINT ON THE HORIZON KNOWN AS THE VANISHING POINT (VP).

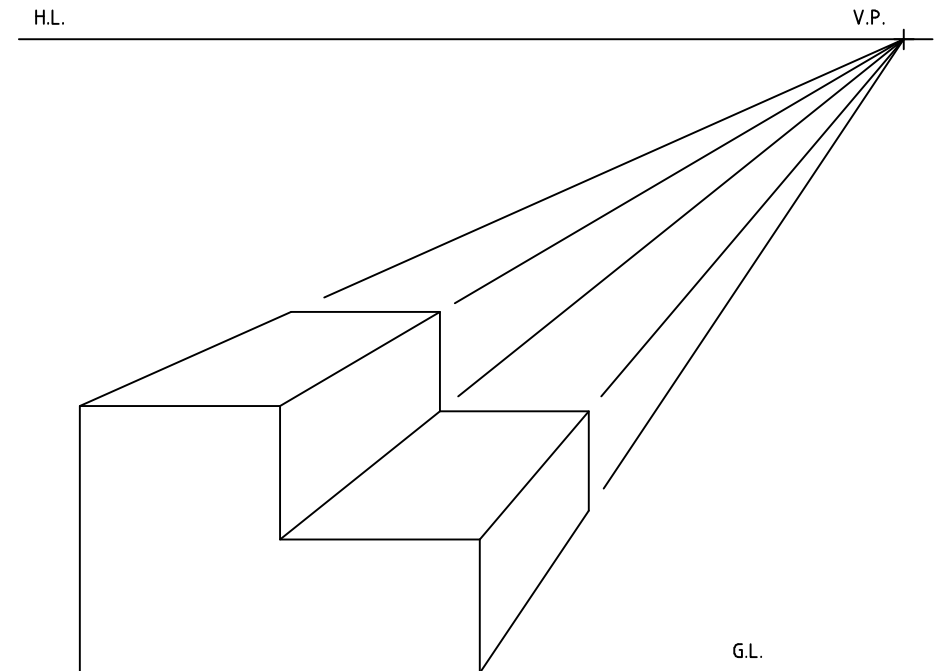
ALL PARALLEL LINES WHICH ARE ON THE GROUND OR PARALLEL TO THE GROUND WILL HAVE THE SAME VANISHING POINT ON THE HORIZON.

A SIMPLE SHAPED BLOCK FIG 9.3.1, IS AN EXAMPLE OF A ONE POINT PERSPECTIVE DRAWING.



AN OBJECT DRAWN IN PERSPECTIVE SHOWS ONE OF ITS SIDES IN FULL SIZE. THE OTHER VIEWS WILL CONVERGE AND APPEAR TO VANISH TOWARDS A POINT KNOWN AS THE V.P.

TO EASILY DRAW THE OBJECT IN THE PERSPECTIVE VIEW, IT IS A GOOD PRACTICE TO DRAW THE FRONT ELEVATION OF THE OBJECT FIRST.



TO OBTAIN THE WIDTH OF THE OBJECT IN A PERSPECTIVE DRAWING, MEASURE THE WIDTH FROM THE END ELEVATION OR THE PLAN AND MARK IT ON THE VISUAL RAY FROM THE BOTTOM CORNER.

NAME: _____
YEAR: _____

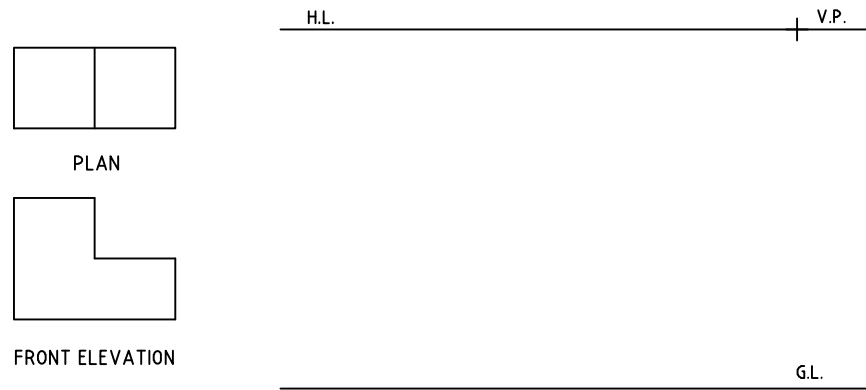
PERSPECTIVE DRAWING

DATE: _____
SHEET NO: 66

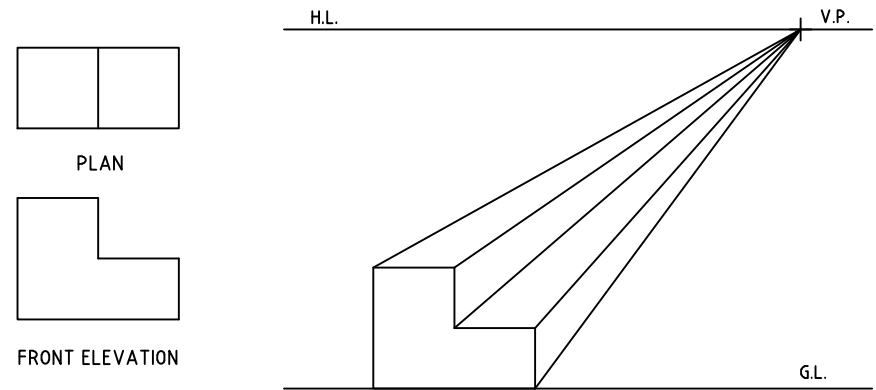
EXAMPLE

GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL) AND A VANISHING POINT (VP).

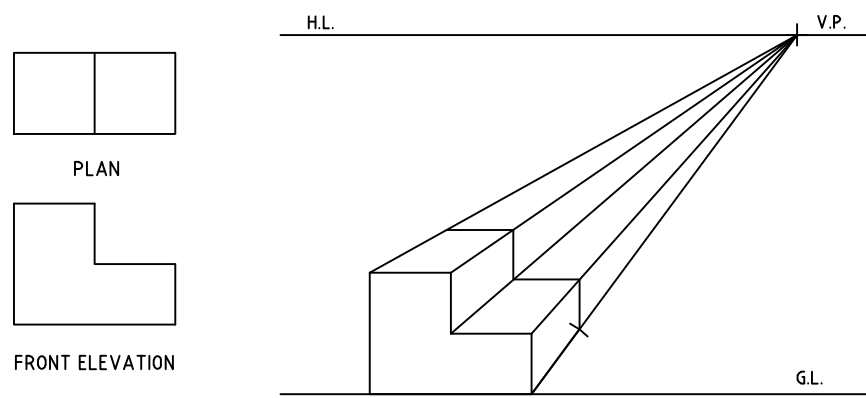
REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE



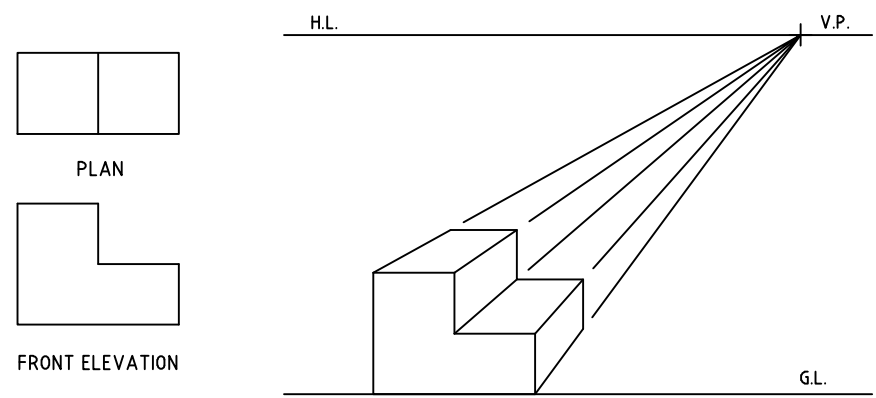
STEP 1 DRAW THE FRONT ELEVATION ON THE OPPOSITE SIDE OF THE VANISHING POINT
STEP 2 DRAW VISUAL RAYS FROM THE CORNERS OF THE FRONT ELEVATION TO THE VANISHING POINT



STEP 3 MEASURE THE WIDTH OF THE OBJECT AND MARK IT ON THE VISUAL RAY DRAWN FROM THE BOTTOM OF THE ELEVATION
STEP 4 USING THIS POINT DRAW LINES PARALLEL TO THEIR CORRESPONDING SIDES FROM THE FRONT ELEVATION



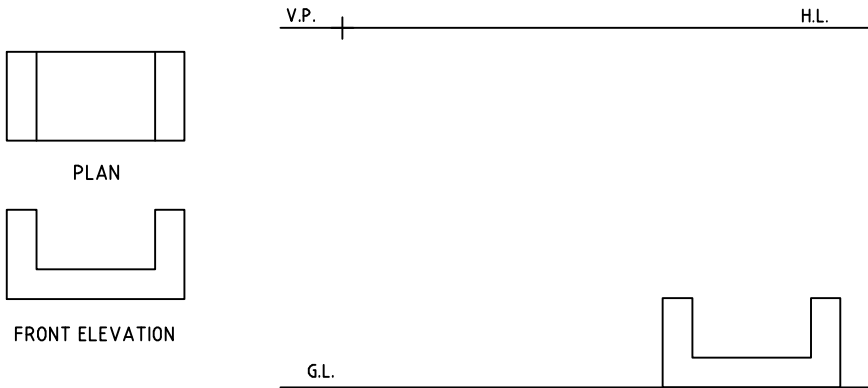
STEP 5 COMPLETE THE SHAPE BY DRAWING THE OUTLINES.



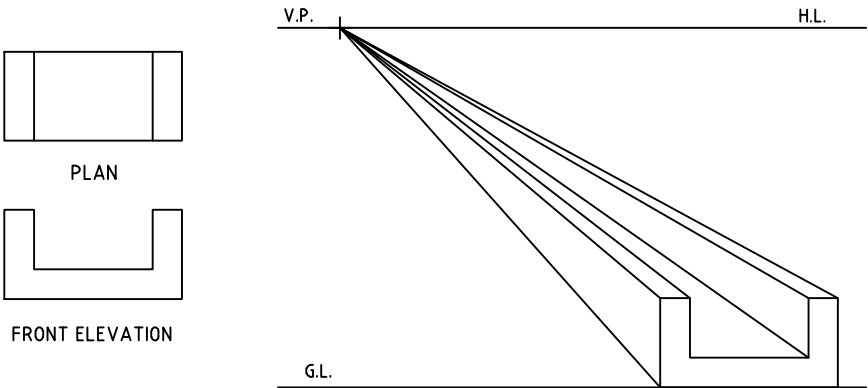
EXAMPLE

GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL) AND A VANISHING POINT (VP).

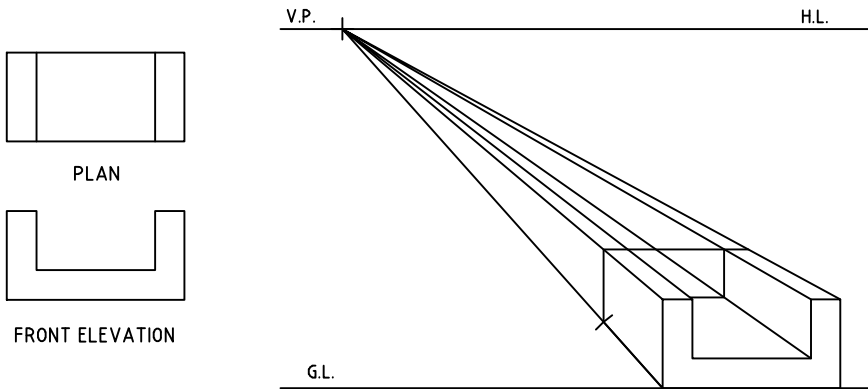
REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE



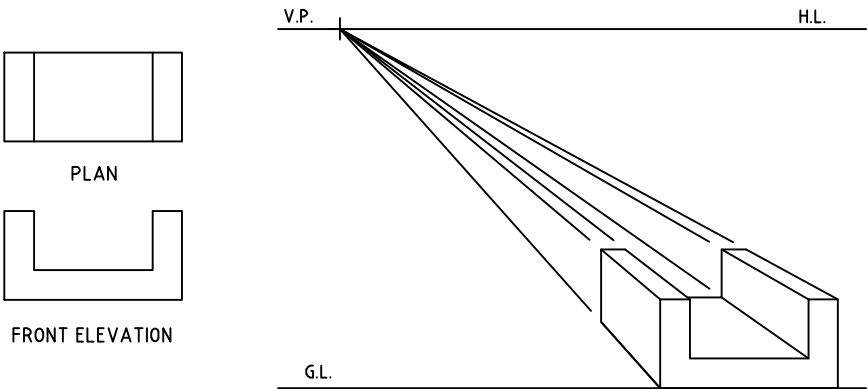
STEP 1 DRAW THE FRONT ELEVATION ON THE OPPOSITE SIDE OF THE VANISHING POINT
STEP 2 DRAW VISUAL RAYS FROM THE CORNERS OF THE FRONT ELEVATION TO THE VANISHING POINT



STEP 3 MEASURE THE WIDTH OF THE OBJECT AND MARK IT ON THE VISUAL RAY DRAWN FROM THE BOTTOM OF THE ELEVATION
STEP 4 USING THIS POINT DRAW LINES PARALLEL TO THEIR CORRESPONDING SIDES FROM THE FRONT ELEVATION



STEP 5 COMPLETE THE SHAPE BY DRAWING THE OUTLINES.



NAME: _____
YEAR: _____

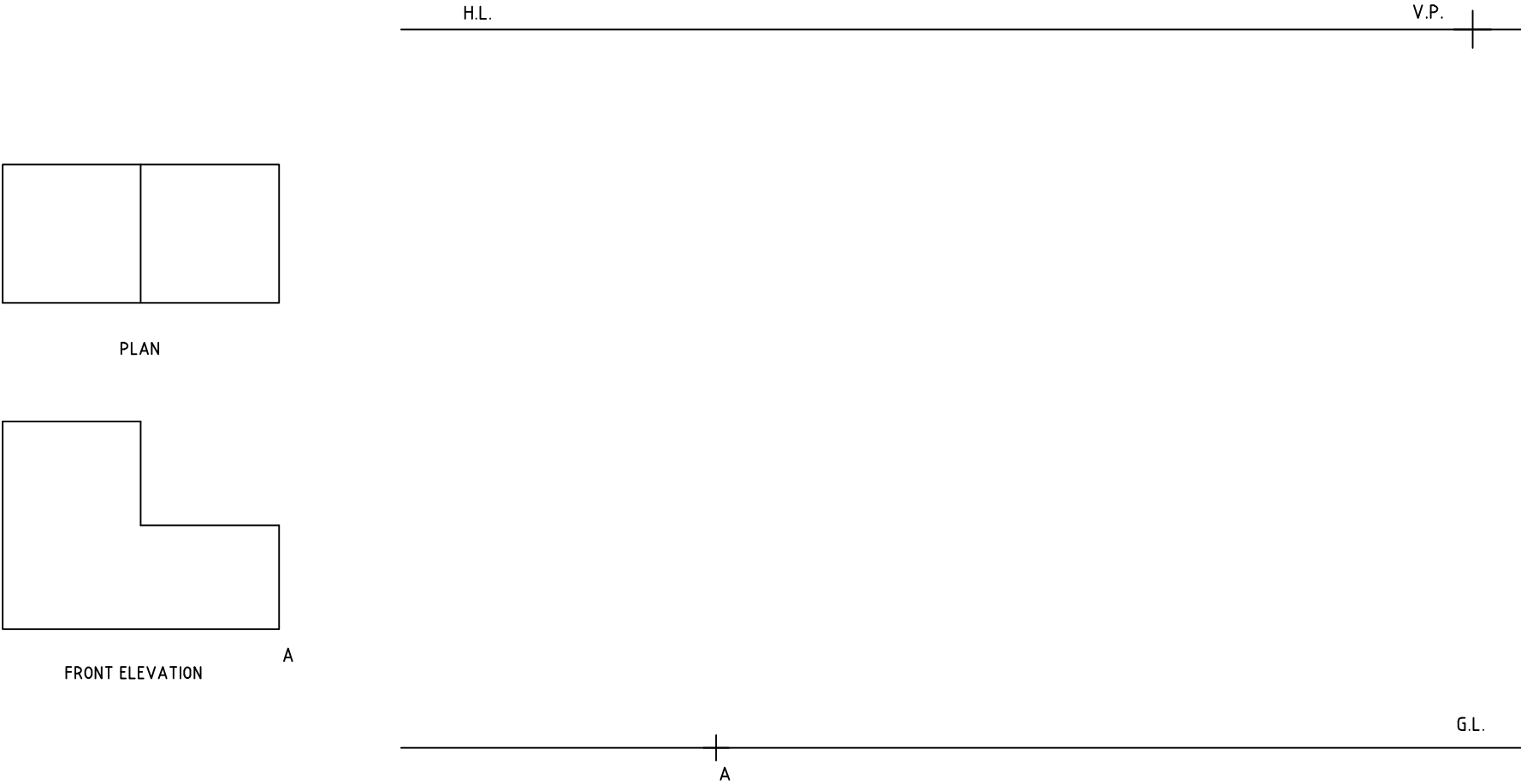
PERSPECTIVE DRAWING

DATE: _____
SHEET NO: 68

GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	

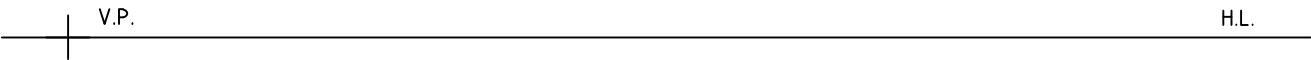


NAME: _____	_____	DATE: _____
YEAR: _____	_____	SHEET NO: 69

GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	

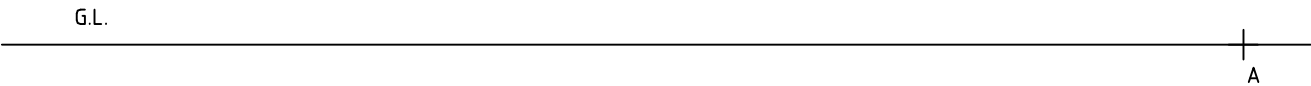


PLAN



FRONT ELEVATION

A

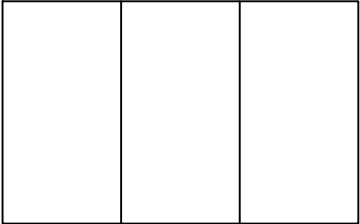


NAME: _____	_____	DATE: _____
YEAR: _____	_____	SHEET NO: 70

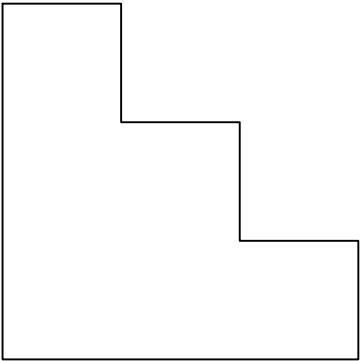
GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



PLAN



A

FRONT ELEVATION

H.L.

V.P.

G.L.

A

NAME: _____

YEAR: _____

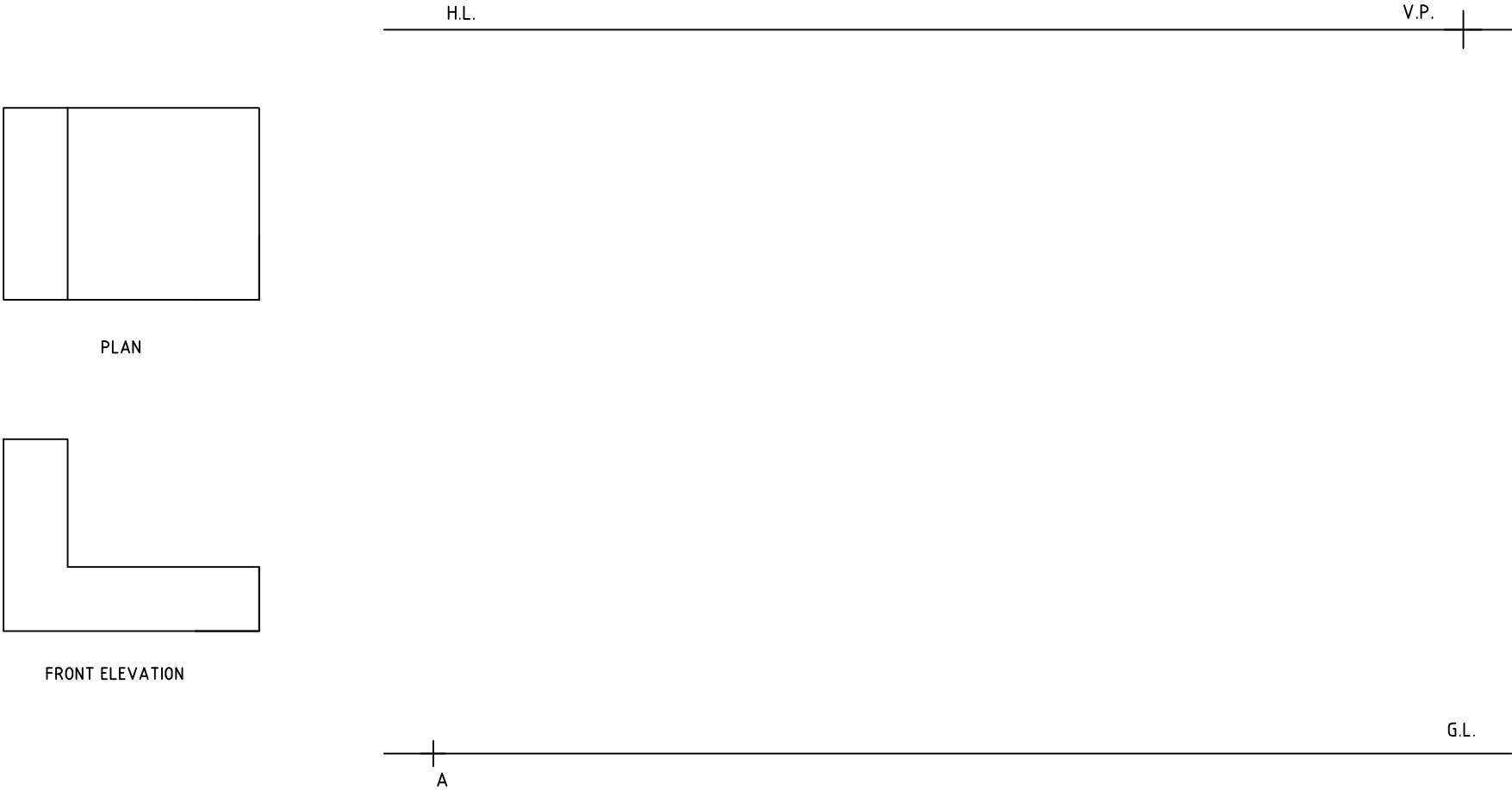
DATE: _____

SHEET NO: 71

GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	

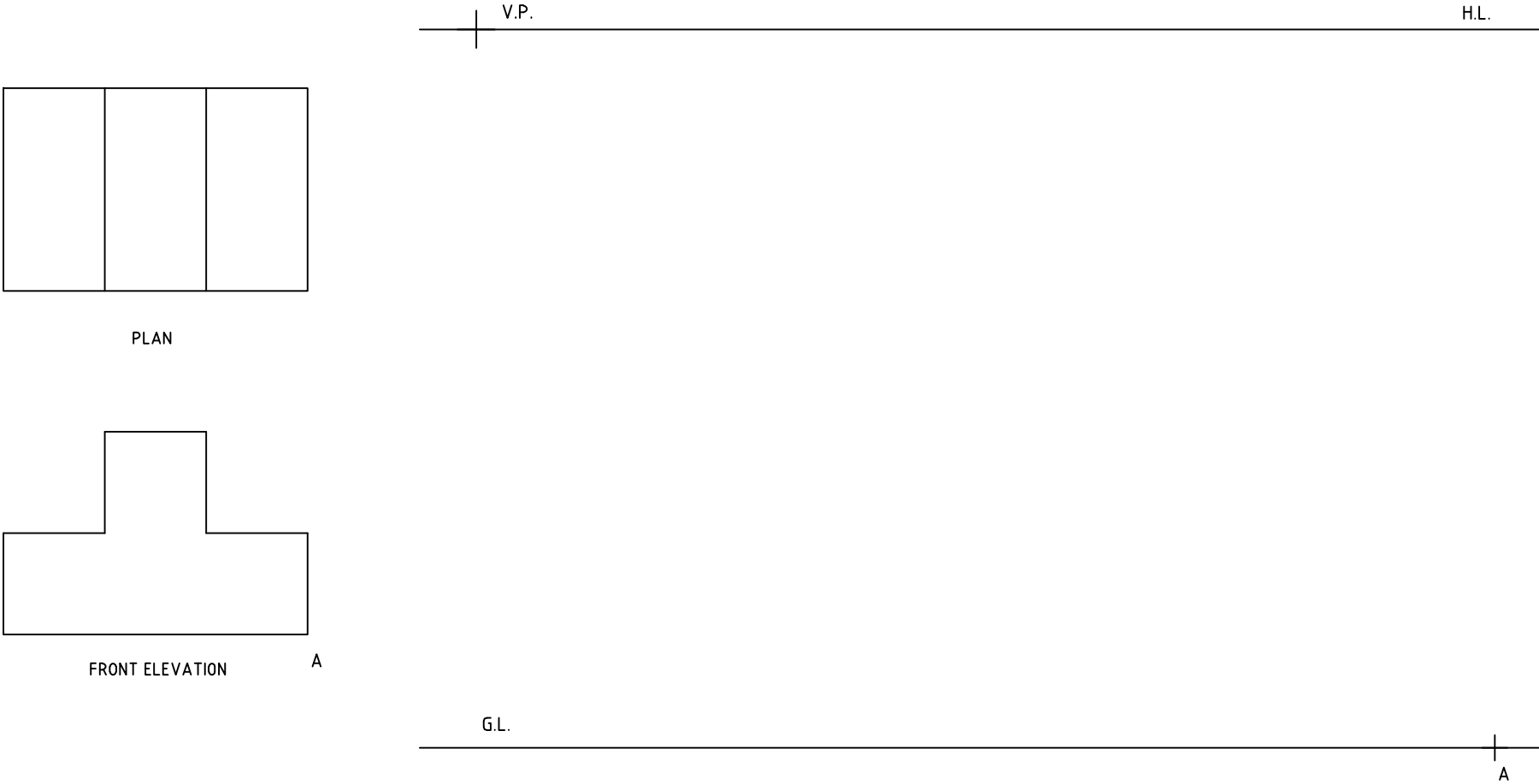


NAME: _____	_____	DATE: _____
YEAR: _____	_____	SHEET NO: 72

GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



NAME: _____	_____	DATE: _____
YEAR: _____	_____	SHEET NO: 73

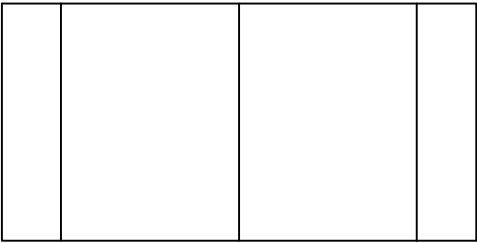
GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

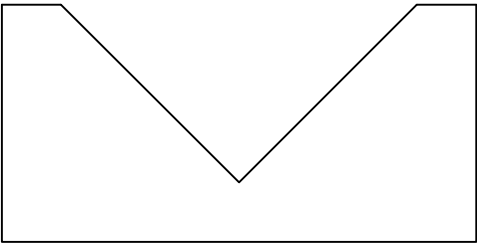
1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	

H.L.

V.P.



PLAN



A

FRONT ELEVATION

G.L.

A

NAME: _____

YEAR: _____

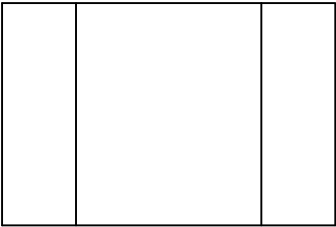
DATE: _____

SHEET NO: 74

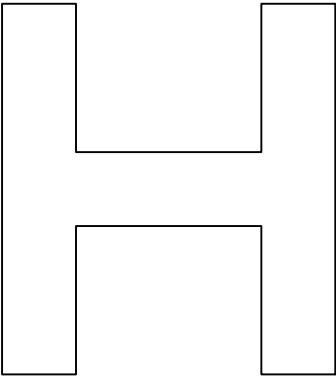
GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	



PLAN



A

FRONT ELEVATION

H.L.

V.P.

G.L.

A

NAME: _____

YEAR: _____

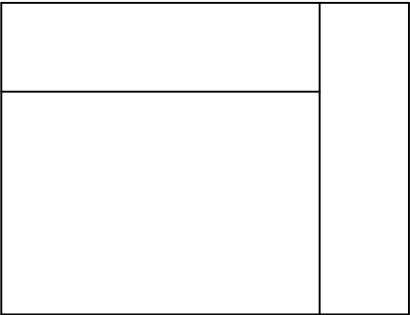
DATE: _____

SHEET NO: 75

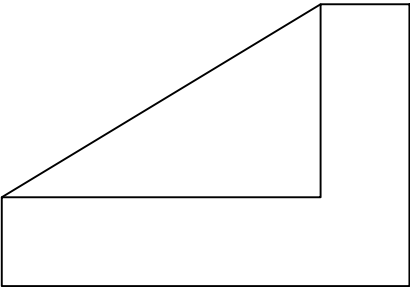
GIVEN: THE PLAN AND FRONT ELEVATION OF A SHAPED BLOCK IN 3rd ANGLE, GROUND LINE (GL), HORIZON LINE (HL), VANISHING POINT (VP) AND A STARTING POINT MARKED 'A' ON THE ELEVATION AND THE GROUND LINE.

REQUIRED: DRAW THE SHAPED BLOCK IN ONE POINT PERSPECTIVE

1.	Guidelines shown	1	
2.	Correct printing	1	
3.	Correct line work	1	
4.	Correct starting point	1	
5.	Creating (box) used	1.5	
6.	Accuracy (L,W,H)	2	
7.	Correct shape	1.5	
8.	Neatness	1	

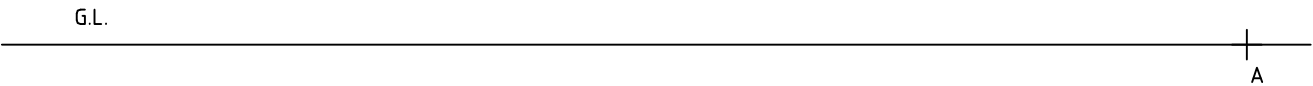
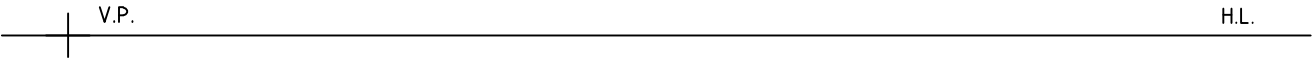


PLAN



FRONT ELEVATION

A



NAME: _____

YEAR: _____

DATE: _____

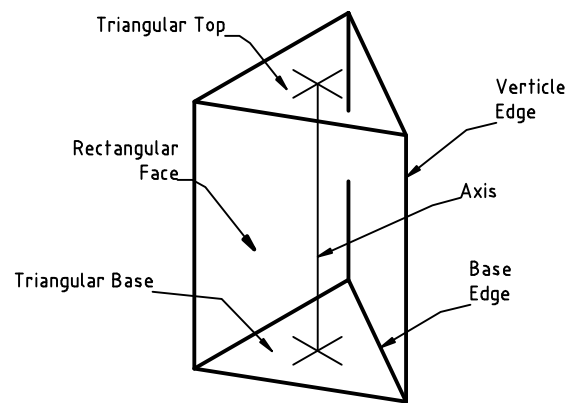
SHEET NO: 76

SOLID GEOMETRY

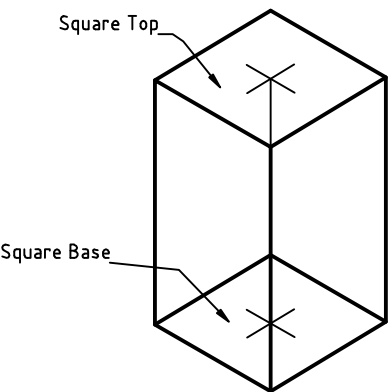
Prisms: A right regular prism is a geometrical solid consisting of equal rectangular sides and two equal ends.

Cylinder: A right regular cylinder is a geometrical solid consisting of a curved surface and two equal circular ends.

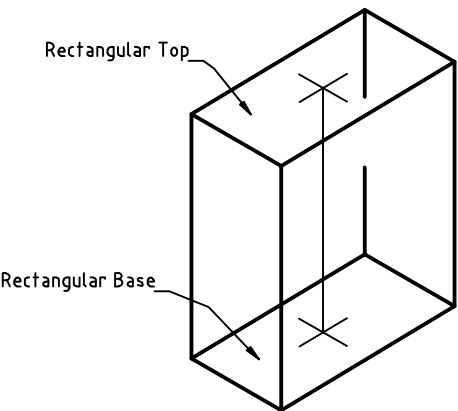
1. Triangular Prism



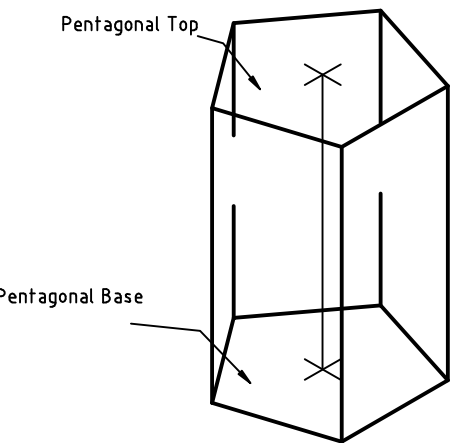
2. Square Prism



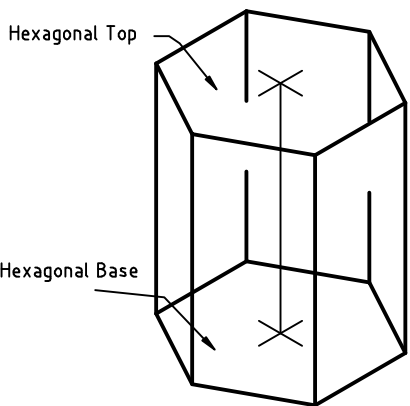
3. Rectangular Prism



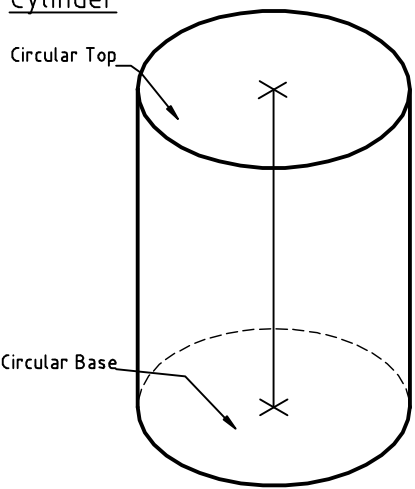
4. Pentagonal Prism



5. Hexagonal Prism



6. Cylinder



NAME: _____
YEAR: _____

TITLE: PRISMS AND CYLINDER

DATE: _____
SHEET NO: 77

EXAMPLE 1 - DEVELOPMENT OF A PRISM

GIVEN: PLAN AND ELEVATION OF A RIGHT SQUARE PRISM

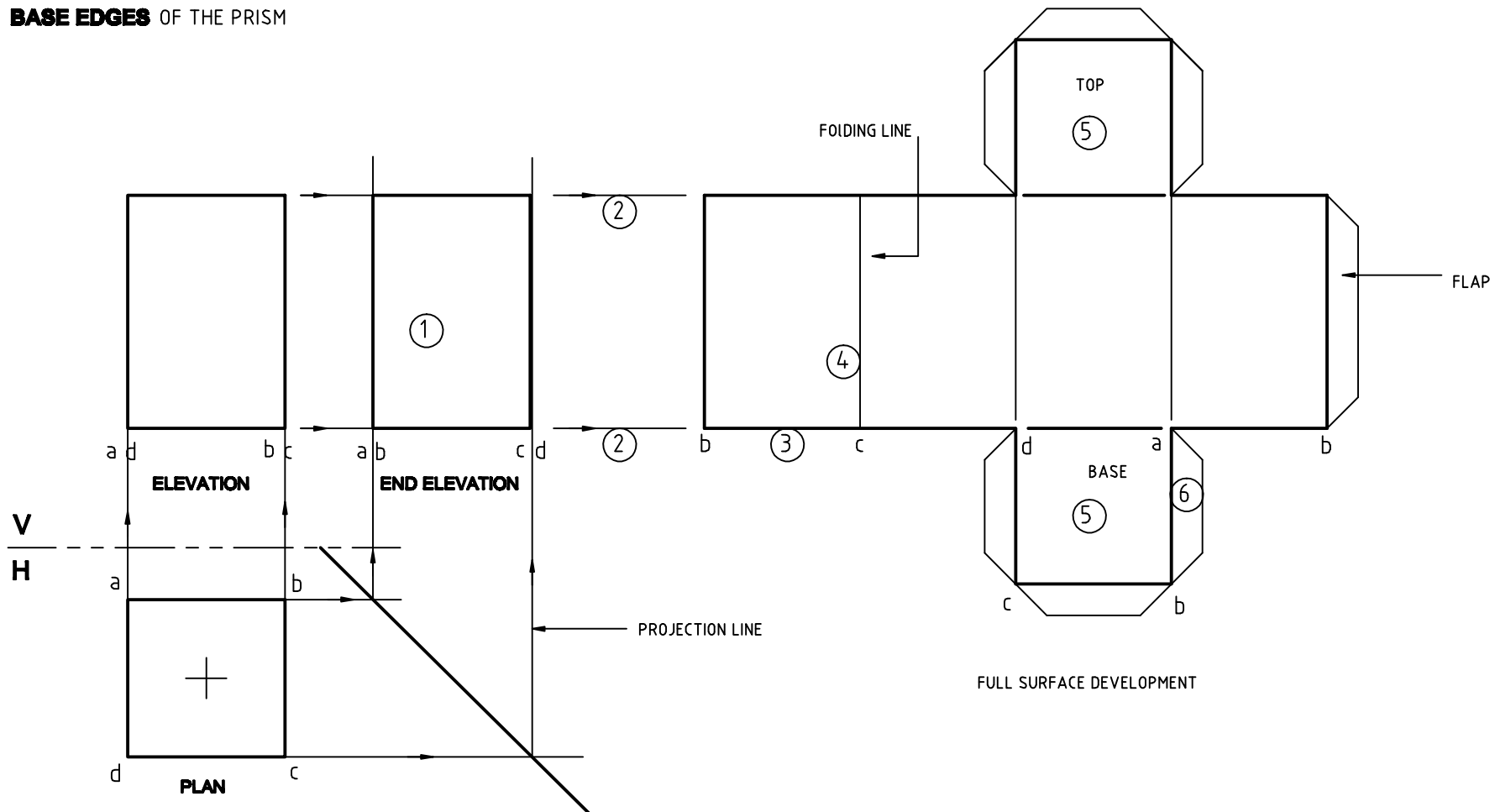
REQUIRED: PROJECT THE END ELEVATION
DRAW THE DEVELOPMENT OF THE PRISM USING THE
PARALLEL LINE DEVELOPMENT METHOD
LABEL THE DRAWINGS

NOTE: IT IS IMPORTANT TO KNOW THE:

- (i) **HEIGHT** OR THE **AXIS** OF THE PRISM.
- (ii) **BASE EDGES** OF THE PRISM

STEPS TO FOLLOW:

1. DRAW THE FRONT ELEVATION AND PROJECT THE END ELEVATION
2. PROJECT THE **HEIGHT** FROM THE ELEVATION AND MARK THE STARTING POINT
3. TAKING MEASUREMENTS FROM THE PLAN STEP OFF THE BASE EDGES ON THE BASE LINE.
4. PROJECT THE **FOLD LINES** FROM THE MARKED BASE EDGES
5. TAKING MEASUREMENTS FROM THE PLAN DRAW THE TWO ENDS AND DRAW THE SEAMS
6. DRAW THE OUTLINES TO COMPLETE THE DEVELOPMENT
7. LABEL THE DRAWINGS. DO NOT ERASE THE CONSTRUCTION LINES



NAME: _____

YEAR: _____

TITLE: PARALLEL LINE DEVELOPMENT

DATE: _____

SHEET NO: 78

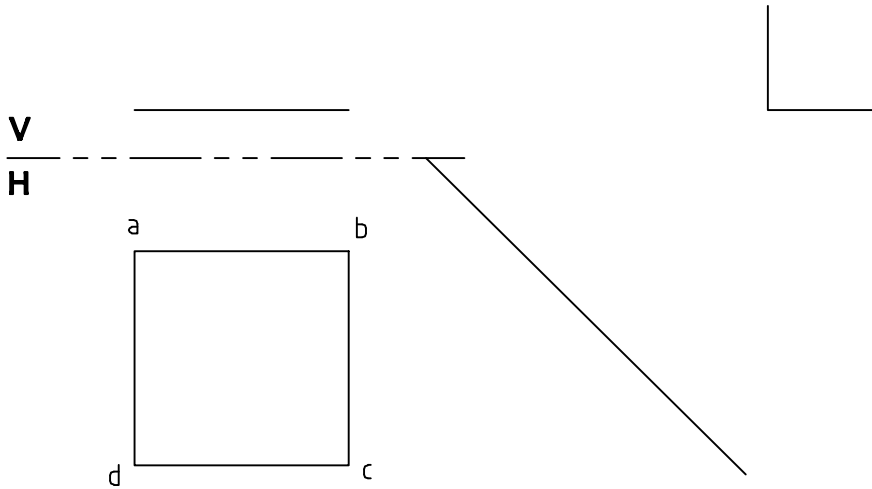
EXERCISE 2

GIVEN: THE PLAN OF A SQUARE PRISM (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE FRONT ELEVATION
 - B. PROJECT THE END ELEVATION
 - C. DRAW THE FULL SURFACE DEVELOPMENT
 - D. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
B. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
D. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



NAME:		DATE:	
YEAR:		SHEET NO: 79	

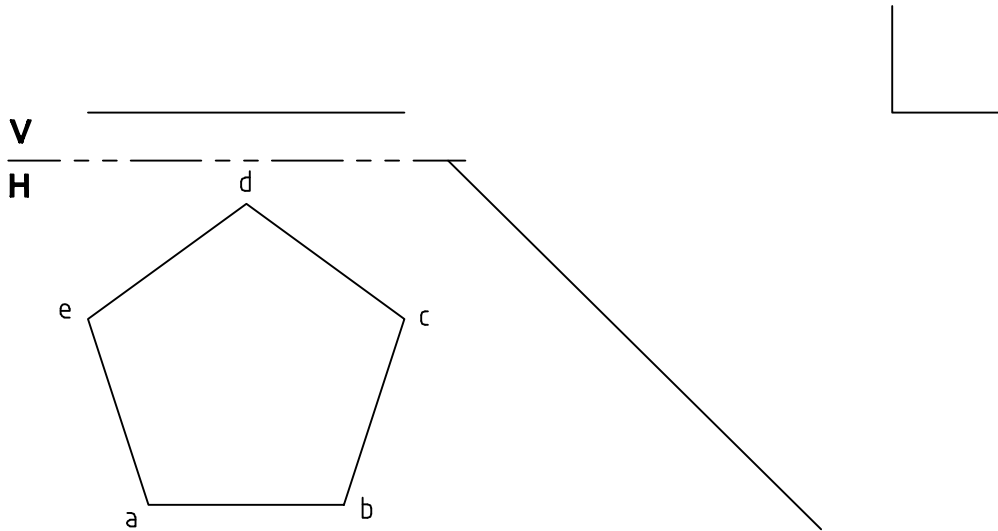
EXERCISE 4

GIVEN: THE PLAN OF A PENTAGONAL PRISM (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE FRONT ELEVATION
 - B. PROJECT THE END ELEVATION
 - C. DRAW THE FULL SURFACE DEVELOPMENT
 - D. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
B. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
D. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



NAME: _____	_____	DATE: _____
YEAR: _____	_____	SHEET NO: 80

EXAMPLE 2 - DEVELOPMENT OF A CYLINDER

GIVEN: PLAN AND ELEVATION OF A RIGHT CYLINDER

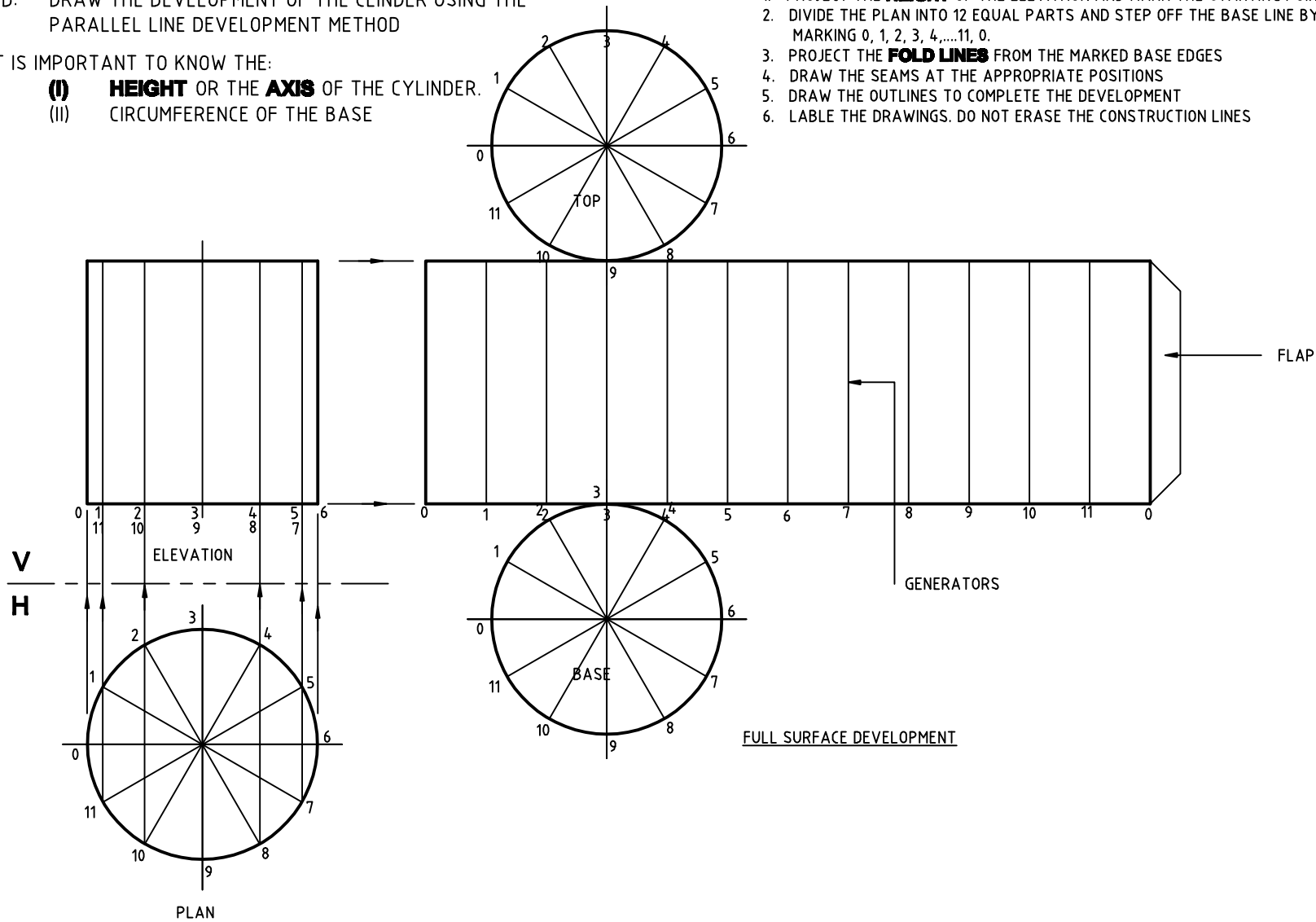
REQUIRED: DRAW THE DEVELOPMENT OF THE CLINDER USING THE PARALLEL LINE DEVELOPMENT METHOD

NOTE: IT IS IMPORTANT TO KNOW THE:

- (I) **HEIGHT** OR THE **AXIS** OF THE CYLINDER.
- (II) CIRCUMFERENCE OF THE BASE

STEPS TO FOLLOW

1. PROJECT THE **HEIGHT** OF THE ELEVATION AND MARK THE STARTING POINT
2. DIVIDE THE PLAN INTO 12 EQUAL PARTS AND STEP OFF THE BASE LINE BY MARKING 0, 1, 2, 3, 4,.....11, 0.
3. PROJECT THE **FOLD LINES** FROM THE MARKED BASE EDGES
4. DRAW THE SEAMS AT THE APPROPRIATE POSITIONS
5. DRAW THE OUTLINES TO COMPLETE THE DEVELOPMENT
6. LABEL THE DRAWINGS. DO NOT ERASE THE CONSTRUCTION LINES



NAME: _____

YEAR: _____

TITLE: PARALLEL LINE DEVELOPMENT

DATE: _____

SHEET NO: 81

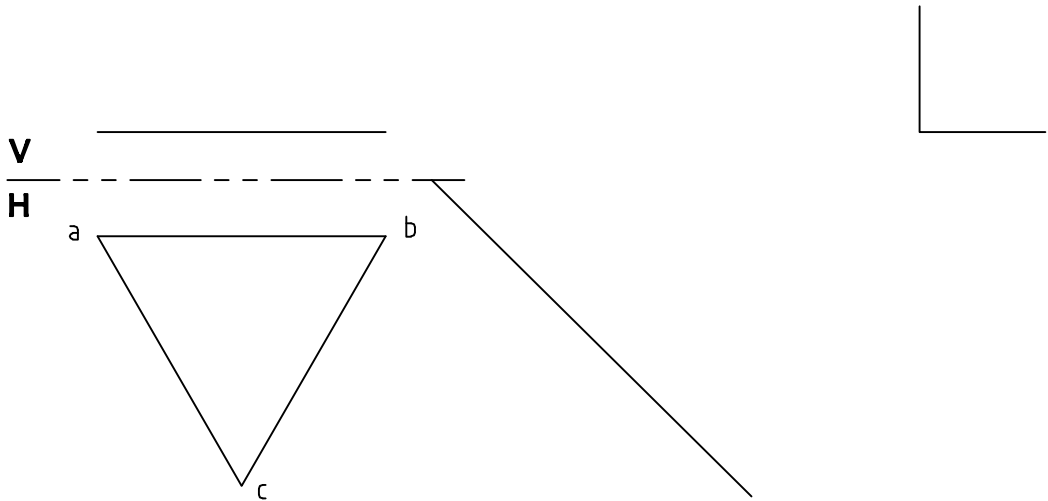
EXERCISE 1

GIVEN: THE PLAN OF A TRIANGULAR PRISM (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE FRONT ELEVATION
 - B. PROJECT THE END ELEVATION
 - C. DRAW THE FULL SURFACE DEVELOPMENT
 - D. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
B. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
D. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



NAME:		DATE:	
YEAR:		SHEET NO: 82	

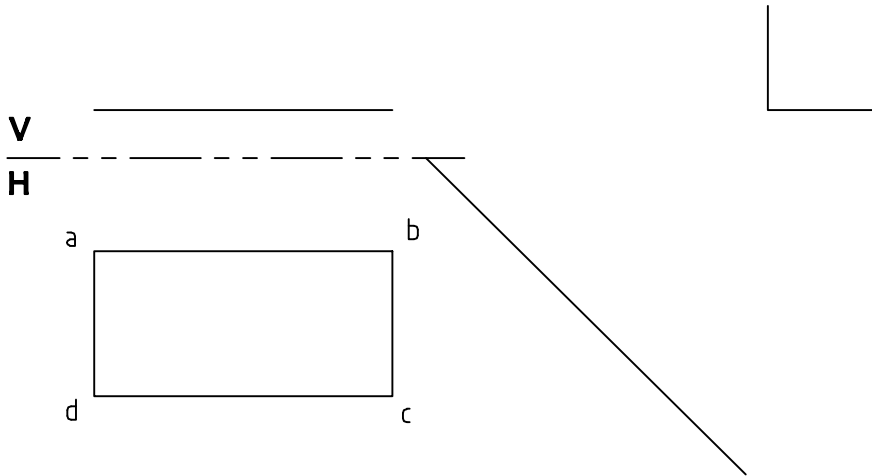
EXERCISE 3

GIVEN: THE PLAN OF A RECTANGULAR PRISM (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE FRONT ELEVATION
 - B. PROJECT THE END ELEVATION
 - C. DRAW THE FULL SURFACE DEVELOPMENT
 - D. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
B. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
D. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



NAME:		DATE:	
YEAR:		SHEET NO: 83	

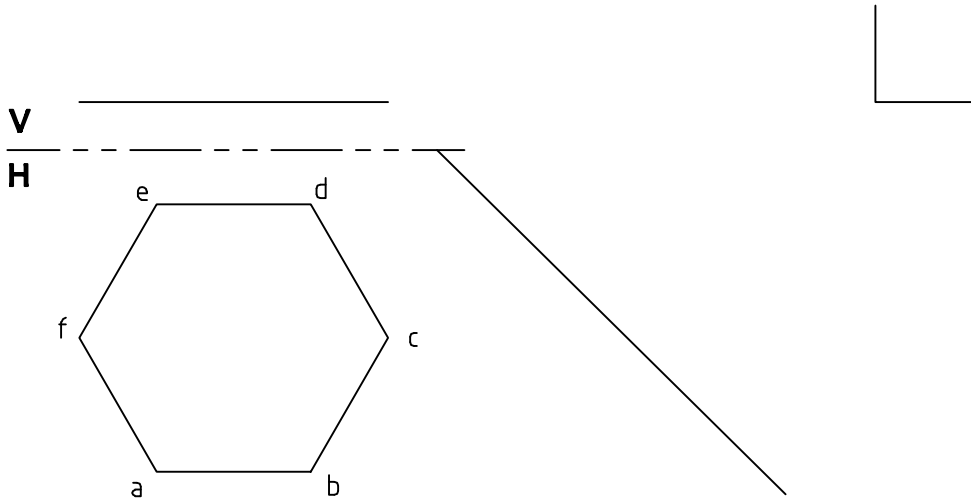
EXERCISE 5

GIVEN: THE PLAN OF A HEXAGONAL PRISM (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE FRONT ELEVATION
 - B. PROJECT THE END ELEVATION
 - C. DRAW THE FULL SURFACE DEVELOPMENT
 - D. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
B. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
D. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



NAME: _____	_____	DATE: _____
YEAR: _____	_____	SHEET NO: 84

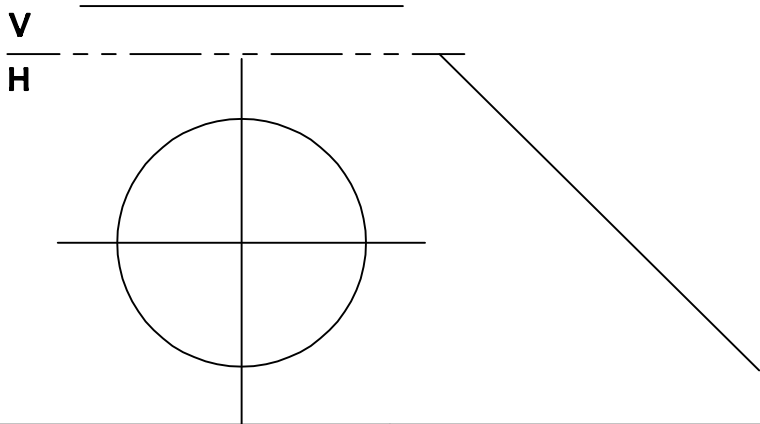
EXERCISE 6

GIVEN: THE PLAN OF A CYLINDER (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE FRONT ELEVATION
 - B. PROJECT THE END ELEVATION
 - C. DRAW THE FULL SURFACE DEVELOPMENT
 - D. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
B. END ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
D. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



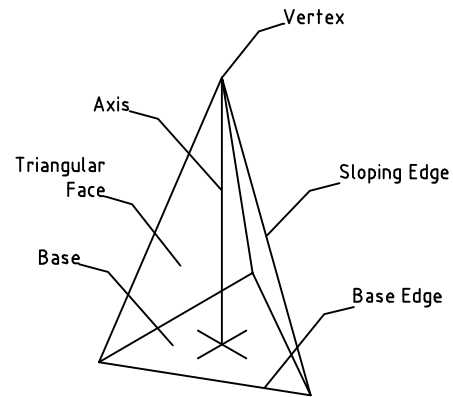
NAME:		DATE:	
YEAR:		SHEET NO: 85	

SOLID GEOMETRY

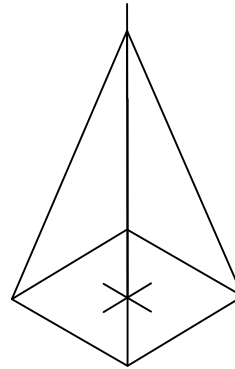
Pyramids: A right regular pyramid is a geometrical solid consisting of a base and equal isosceles triangular sides meeting at a point known as the vertex.

Cone: A right regular cone is a geometrical solid consisting of a circular base and a curved surface which meets at a point known as the vertex.

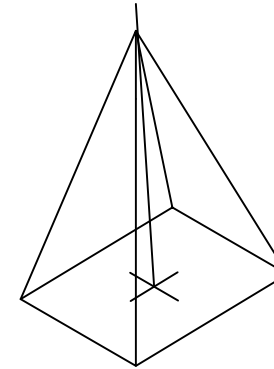
1. Triangular Pyramid



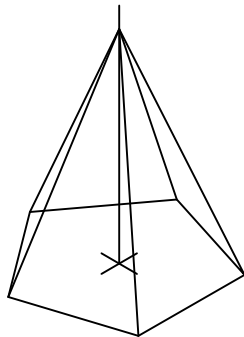
2. Square Pyramid



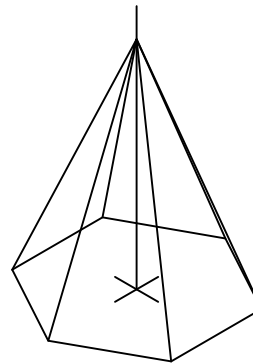
3. Rectangular Pyramid



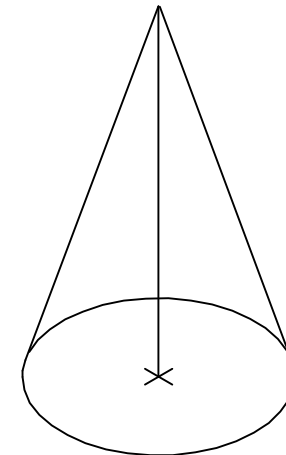
3. Pentagonal Pyramid



4. Hexagonal Pyramid



4. Cone



NAME: _____

YEAR: _____

PRISMS AND CYLINDER

DATE: _____

SHEET NO: 86

EXAMPLE 1 - DEVELOPMENT OF A PYRAMID

GIVEN: PLAN AND ELEVATION OF A RIGHT SQUARE PYRAMID

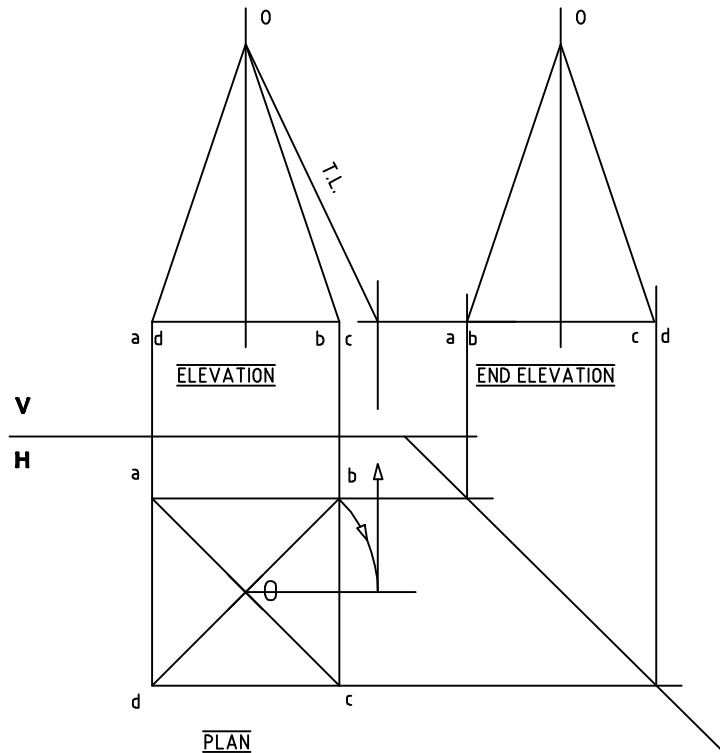
REQUIRED: PROJECT THE END ELEVATION
DRAW THE DEVELOPMENT OF THE PYRAMID USING THE
RADIAL LINE DEVELOPMENT METHOD

NOTE: IT IS IMPORTANT TO KNOW THE:

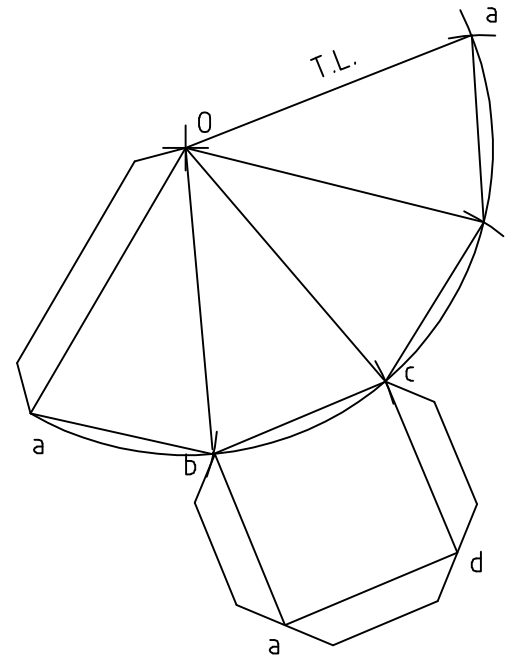
- (i) **TRUE LENGTH** OF THE EDGE OF THE PYRAMID
- (ii) **BASE EDGES** OF THE PYRAMID

STEPS TO FOLLOW:

1. DRAW THE FRONT ELEVATION AND PROJECT THE END ELEVATION
2. FIND THE TRUE LENGTH USING THE REBATMENT METHOD
3. USING THE TRUE LENGTH DRAW AN ARC FROM THE GIVEN STARTING POINT
3. TAKING MEASUREMENTS FROM THE PLAN STEP OFF THE BASE EDGES ON THE BASE ARC.
4. PROJECT THE **FOLD LINES** FROM THE MARKED BASE EDGES
5. TAKING MEASUREMENTS FROM THE PLAN DRAW THE END AND THE SEAMS
6. DRAW THE OUTLINES TO COMPLETE THE DEVELOPMENT
7. LABEL THE DRAWINGS. DO NOT ERASE THE CONSTRUCTION LINES



DEVELOPMENT OF THE SQUARE PYRAMID



FULL SURFACE DEVELOPMENT

NAME: _____

YEAR: _____

TITLE: RADIAL LINE DEVELOPMENT

DATE: _____

SHEET NO: 87

EXAMPLE 2 - DEVELOPMENT OF A CONE

GIVEN: PLAN AND ELEVATION OF A RIGHT CONE

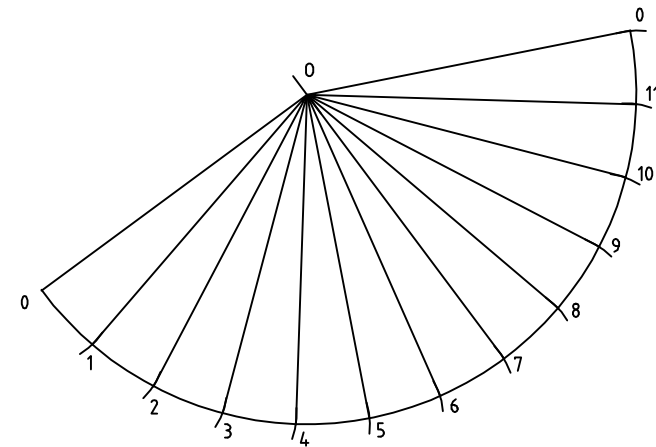
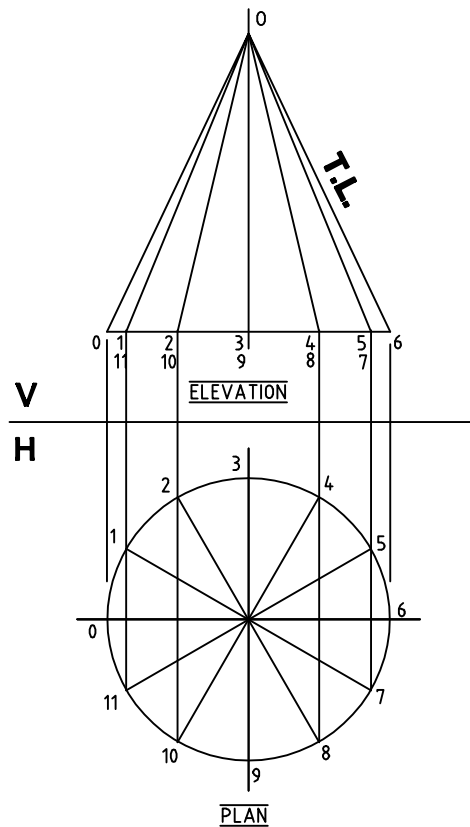
REQUIRED: PROJECT THE END ELEVATION
DEVELOPMENT OF THE CONE USING THE
RADIAL LINE DEVELOPMENT METHOD

NOTE: IT IS IMPORTANT TO KNOW THE:

- (I) TRUE LENGTH OF THE EDGE OF THE CONE
- (II) CIRCUMFERENCE OF THE BASE

STEPS TO FOLLOW:

1. DRAW THE FRONT ELEVATION AND PROJECT THE END ELEVATION
2. FIND THE TRUE LENGTH AND USE IT TO DRAW AN ARC FROM THE STARTING POINT
3. DIVIDE THE PLAN INTO 12 EQUAL PARTS AND STEP OFF ON THE BASE ARC BY MARKING 0, 1, 2, 3, 4,.....11, 0.
4. PROJECT THE **FOLD LINES** FROM THE MARKED BASE EDGES
5. TAKING MEASUREMENTS FROM THE PLAN DRAW THE END AND THE SEAMS
6. DRAW THE OUTLINES TO COMPLETE THE DEVELOPMENT
7. LABEL THE DRAWINGS. DO NOT ERASE THE CONSTRUCTION LINES



FULL SURFACE DEVELOPMENT

NAME: _____

YEAR: _____

TITLE: RADIAL LINE DEVELOPMENT

DATE: _____

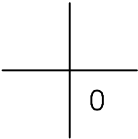
SHEET NO: 88

EXERCISE 1

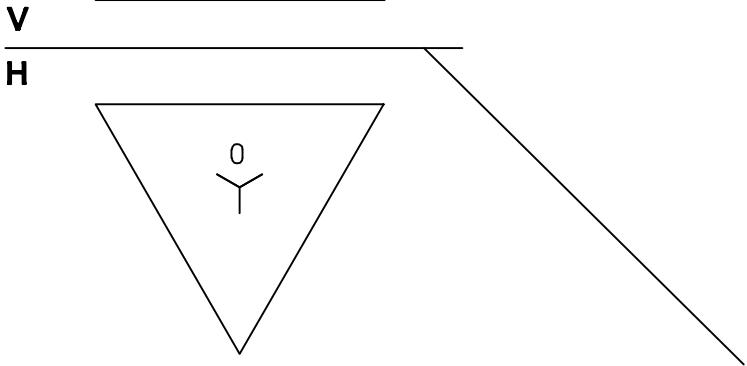
GIVEN: THE PLAN OF A TRIANGULAR PYRAMID (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE PLAN
 - B. COMPLETE THE FRONT ELEVATION
 - C. PROJECT THE END ELEVATION
 - D. DRAW THE FULL SURFACE DEVELOPMENT
 - E. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES



A. PLAN		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
B. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. END ELEVATION		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
D. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
E. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



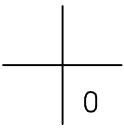
NAME:		DATE:	
YEAR:		SHEET NO: 89	

EXERCISE 2

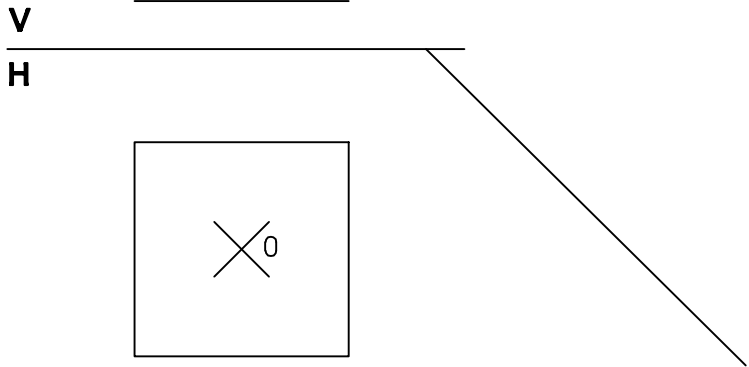
GIVEN: THE PLAN OF A SQUARE PYRAMID (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE PLAN
 - B. COMPLETE THE FRONT ELEVATION
 - C. PROJECT THE END ELEVATION
 - D. DRAW THE FULL SURFACE DEVELOPMENT
 - E. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES



A. PLAN		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
B. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. END ELEVATION		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
D. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
E. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



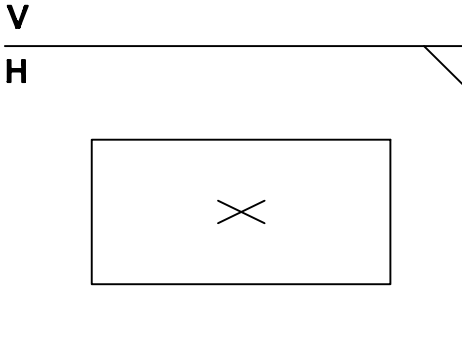
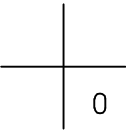
NAME:		DATE:	
YEAR:		SHEET NO: 90	

EXERCISE 3

GIVEN: THE PLAN OF A RECTANGULAR PYRAMID (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE PLAN
 - B. COMPLETE THE FRONT ELEVATION
 - C. PROJECT THE END ELEVATION
 - D. DRAW THE FULL SURFACE DEVELOPMENT
 - E. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES



A. PLAN		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
B. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. END ELEVATION		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
D. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
E. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5

NAME:		DATE:	
YEAR:		SHEET NO: 91	

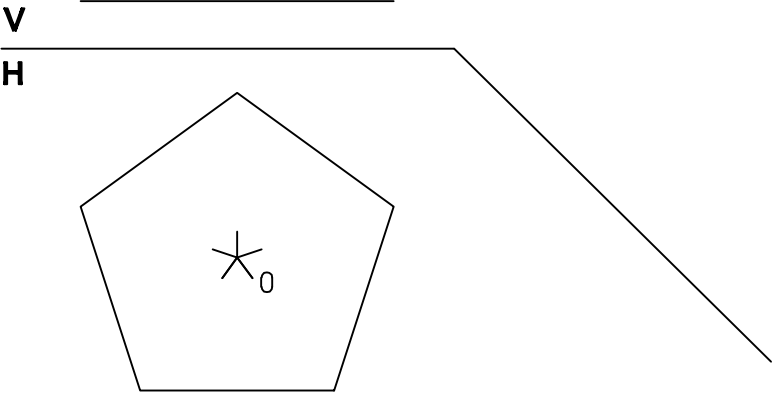
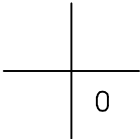
EXERCISE 4

GIVEN: THE PLAN OF A PENTAGONAL PYRAMID (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE PLAN
 - B. COMPLETE THE FRONT ELEVATION
 - C. PROJECT THE END ELEVATION
 - D. DRAW THE FULL SURFACE DEVELOPMENT
 - E. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. PLAN			
1.	Accuracy-length, width	0.5	
2.	Correct shape	0.5	
3.	Correct line work	0.5	
4.	Overrrall neatness	0.5	
B. ELEVATION			
1.	Accuracy-length, width	1	
2.	Correct shape	1	
3.	Correct line work	1	
4.	Overrrall neatness	1	
C. END ELEVATION			
1.	Accuracy-length, width	0.5	
2.	Correct shape	0.5	
3.	Correct line work	0.5	
4.	Overrrall neatness	0.5	
D. FULL DEVELOPMENT			
1.	Accuracy-length, width	2	
2.	Correct shape	1	
3.	Correct line work	1	
4.	Overrrall neatness	1	
E. LABEL			
1.	Guidelines shown	0.5	
2.	Correct printing	1	
3.	Correct line work	0.5	



NAME: _____

YEAR: _____

DATE: _____

SHEET NO: 92

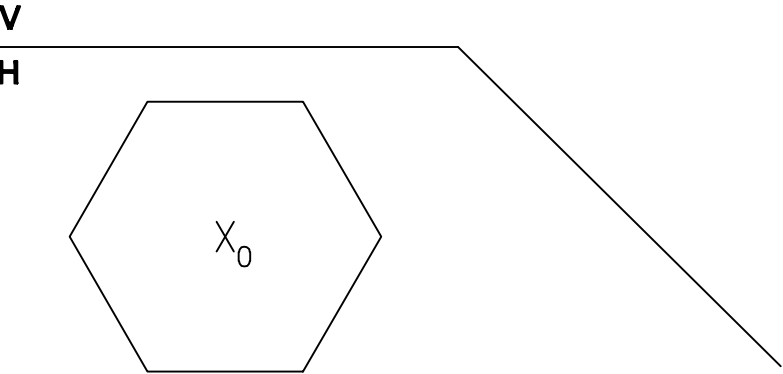
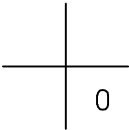
EXERCISE 5

GIVEN: THE PLAN OF A HEXAGONAL PYRAMID (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE PLAN
 - B. COMPLETE THE FRONT ELEVATION
 - C. PROJECT THE END ELEVATION
 - D. DRAW THE FULL SURFACE DEVELOPMENT
 - E. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES

A. PLAN		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
B. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. END ELEVATION		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
D. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
E. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5



NAME: _____

YEAR: _____

DATE: _____

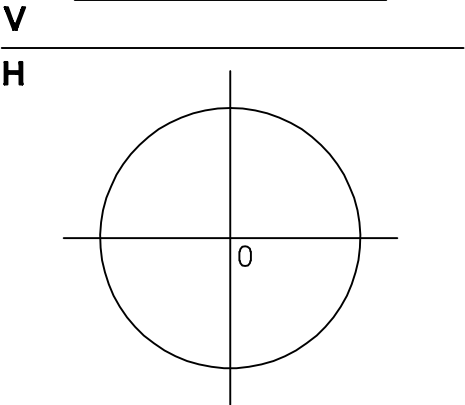
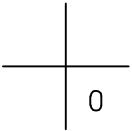
SHEET NO: 93

EXERCISE 6

GIVEN: THE PLAN OF A CONE (AXIS = 50mm) DRAWN IN FIRST ANGLE ORTHOGRAPHIC PROJECTION.

- REQUIRED:
- A. COMPLETE THE PLAN
 - B. COMPLETE THE FRONT ELEVATION
 - C. PROJECT THE END ELEVATION
 - D. DRAW THE FULL SURFACE DEVELOPMENT
 - E. LABEL ALL THE DRAWINGS.

NOTE: DO NOT ERASE THE CONSTRUCTION LINES



A. PLAN		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
B. ELEVATION		
1.	Accuracy-length, width	1
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
C. END ELEVATION		
1.	Accuracy-length, width	0.5
2.	Correct shape	0.5
3.	Correct line work	0.5
4.	Overall neatness	0.5
D. FULL DEVELOPMENT		
1.	Accuracy-length, width	2
2.	Correct shape	1
3.	Correct line work	1
4.	Overall neatness	1
E. LABEL		
1.	Guidelines shown	0.5
2.	Correct printing	1
3.	Correct line work	0.5

NAME: _____

YEAR: _____

DATE: _____

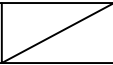
SHEET NO: 94

NAME: _____

YEAR: _____

DATE: _____

SHEET NO: _____



NAME: _____

YEAR: _____

DATE: _____

SHEET NO: _____

